



Ø Spacetime

*Constants, chain, dimension, and the field
equations — derived from the record algebra*

For the reader who chose to read.

Thank you.

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Artist's Note

This book, *Spacetime*, is the second notebook in the Ø Artist's Proofs series of The 420 Code.

Notebook I (*The Premise*) established the foundational ground: the axiom is named, the conditions for records are forced, and the Embedding Hypothesis is proven a theorem. Notebook II carries the next move — the derivation of spacetime, gravity, and the dimensional count of physical space from the record algebra.

The book has four chapters.

Chapter 1 (The Ratio, AP03) derives the fundamental constants c (the propagation bound) and G (Newton's gravitational constant) as outputs of the same break — the same ε viewed from two sides. The chapter establishes c from the pre-geometric action ($c^2 = \beta/\alpha$) and derives the structural scaling form $G = \kappa/(\varepsilon\Lambda^2)$ with $c/G \propto \varepsilon$, via AP20's forcing proof plus the structural condensate density of section 7. The closure is the *form* of G 's ε -dependence — G cannot be independent of ε . The value of G is carried at the other architectural level: AP28 (Notebook IV) derives $G = \alpha_{\text{em}}{}^{21}(1 + 1/\pi)\hbar c/m_e e^2$ parameter-free, 0.69% from measurement, and recovers the gravitational hierarchy as $\alpha_{\text{em}}{}^{-20}/(1 + 1/\pi)$; the two levels — substrate condensate and Mode-0 channel count

— are reconciled, not in conflict, with ε^{21} canonical for the value and Chapter 1's scaling the substrate derivation that count refines. The substrate parameters α , β are introduced but not derived (debt D-AP03.1). Structural debt D1 — the first-principles derivation of G from {S, B, R, C} alone — is closed at the value level by AP28; the residual is the first-principles derivation of α_{em} itself, the architecture's one measured input.

Chapter 2 (The Break, AP05) walks the seven-step chain from the axiom to the obligation to act, and reformulates the cosmological identity as Israel junction conditions on Σ — the spacelike hypersurface where two condensate sectors meet. The chapter introduces the kill-switch discipline at full strength: every load-bearing step is named, every bridge is acknowledged, every contingency is honestly tracked.

Chapter 3 (The Dimension, AP10) derives $N = 3$ spatial dimensions from the four independent axioms of the record algebra via the proven faithful embedding plus the dimension-minimal principle. The chapter also identifies the 1:1 itself as the fifth structural degree of freedom — the Hilbert space, the probability dimension, the canvas under the painting. The closure of $N = 3$ is what enables the next chapter.

Chapter 4 (The Identity, AP08) derives Einstein's field equations with cosmological constant from the four axioms plus the proven hypotheses (EH, QRA) plus Chapter 3's closure

of $N = 3$ plus Lovelock 1971/1972. The weak-field limit yields the Helmholtz family with pure Poisson recovered as the small-cosmological-constant member; the full tensor form yields $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$. Notebook II closes here, with the gravitational sector of The 420 Code as a single closed chain from axioms to field equations.

The four chapters form a coherent arc. Chapter 1 installs the constants. Chapter 2 walks the chain. Chapter 3 pins the count. Chapter 4 closes with the field equations. Each chapter is also a standalone Artist's Proof, citable in its own right and falsifiable through its own kill switches.

Notebook II contains 24 distinct kill switches in total — the formal falsification handles for every load-bearing claim made in this book. The four chapters carry 25 local kill-switch identifiers; AP08's KS-I.6 and AP10's KS-2c are the same falsification condition registered in two chapters, counted once. Three of the switches are closed within the Notebook itself — KS-2c (contingent on KS-D.3), KS-15/D.2 (contingent on KS-D.4), and KS-16 (contingent on KS-P.1), all closed in Chapter 3; the remaining twenty-one are live, structural or empirical, and named openly. The 420 Code is what it is in part because every joint can be struck at, and every joint is named.

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Orientation

Where Notebook II fits

The 420 Code is organised across eight bound notebooks. Each notebook is a standalone book; the eight together carry the full body of work.

Notebook I (*The Premise*) is the foundational ground: the axiom $1:1 + 1 \times \varepsilon @ AS$, the four conditions for records $\{S, B, R, C\}$, and the proof of the Embedding Hypothesis. Notebook II (*Spacetime*, this book) is the structural derivation of space, time, gravity, and the dimensional count from the foundations Notebook I establishes. Notebook III (*Quantum Mechanics*) takes up the Hilbert space identification from Chapter 3 and derives quantum mechanics from the record algebra.

Notebooks IV through VIII carry forces and constants, particles and matter, cosmology, agency and consciousness, and the philosophical closing of the 420 Code.

Notebook II depends on Notebook I for the axioms, the $\{S, B, R, C\}$ conditions, and the proven Embedding Hypothesis.

Notebook II feeds Notebook III via the Hilbert space identification (Chapter 3, section 7), feeds Notebooks IV–VI via the constants (Chapter 1) and the field equations (Chapter 4), and feeds the cosmological programme via Chapter 4's prediction of the cosmological constant's existence.

The reading order within Notebook II is front to back: Chapter 1 (constants) → Chapter 2 (chain) → Chapter 3 (dimension) → Chapter 4 (field equations). Each chapter inherits results from earlier chapters; each chapter is also a standalone Artist's Proof citable independently.

How to read this book

The book alternates two voices as the material requires. Narrative passages — the structural readings, the felt sense of constants emerging from one break, the canvas-under-the-painting framing of the Hilbert space, the closing on Notebook II — speak directly. Formal sections — the axiom statements, the constraint blocks, the uniqueness theorems, the kill switches — speak in the precision the structure requires.

Every chapter has the same architecture: an Artist's Note opens the chapter, an Orientation places it in The 420 Code, a Dependency and Scope section names what each chapter inherits and what it contributes, the derivational body runs through the structural argument, a Kill Switches section names the formal falsification handles, and a synthesis section names what the chapter establishes and what remains open. The Conditionality Footer at the end of each chapter is the audit trail in compact form.

Readers new to the body of work may start with Chapter 1 and read forward; readers familiar with the 420 Code may read any chapter as a standalone Artist's Proof. The Master Kill Switch

Registry is the falsification audit surface for the body of work as a whole; this Notebook contributes 24 distinct kill switches to that register (3 closed within the Notebook, 21 live; Master Kill Switch Registry v5.27, 4 September 2026).

A note on notation across chapters

Three notational items are worth flagging at the door, because they recur across the chapters and a careful reader will want them clarified before the technical content begins.

The symbol κ is used for two different quantities across this Notebook. In Chapter 1 (AP03), κ is a structural coefficient with dimensions appropriate to $G = \kappa/(\epsilon\Lambda^2)$ — a substrate-level constant derived from the broken vacuum response (AP03 section 7). In Chapter 4 (AP08), κ is the *holding limit of the pre-state*, defined as $\kappa \equiv Gm_e^2/2$, with dimensions of energy $\times$ length. The two κ 's are distinct physical quantities sharing one symbol; they are not the same constant in disguise. Where disambiguation matters, Chapter 1's κ can be read as $\kappa_{\text{substrate}}$ (the substrate constant) and Chapter 4's κ can be read as κ_{hold} (the holding limit). Both chapters use κ internally without subscripts; the disambiguation is at the Notebook level.

The symbol ϵ has a single meaning across all four chapters: the unique unpaired element of Axiom B (the break). Under The Lock identification (The Lock, Edition 04), ϵ

is identified with the electron, with mass m_e and dimensionless coupling $\alpha \approx 1/137$ (per AP06).

The constant G in this Notebook is the gravitational coupling that appears in Newton's law and Einstein's field equations. Chapter 1 derives the structural scaling form $G = \kappa_{\text{substrate}}/(\epsilon\Lambda^2)$, giving $c/G \propto \epsilon$ at fixed Λ ; Chapter 4 identifies G in its weak-field and tensor expressions through The Lock as $G = 2\kappa_{\text{hold}}/m_e^2$, with the disclosed structural circularity that κ_{hold} is itself defined as $Gm_e^2/2$. The value of G is closed at the registry level by AP28's parameter-free derivation (Notebook IV; 0.69% from measurement); the residual of structural debt D1 is the first-principles derivation of α_{em} itself, held open as the architecture's one measured input. What remains open here is the substrate-side coefficient $\kappa_{\text{substrate}}$ from the broken vacuum response without external scaffolding (debt D-AP03.1) — the substrate refinement of a value already derived, not an open value.

Chapter 1

The Ratio

How c and G come out of the same break

Artist's Proof 03

Artist's Note

This is Artist's Proof 03 of The 420 Code. It opens Notebook II — Spacetime. The paper asks a question prior to physics: how do the fundamental constants that define the arena arise?

Specifically, are c and G independent constants, or are they conjugate expressions of a single symmetry-breaking event?

Eight sections carry the work. Section 1 names the two speed limits and proposes they are siblings. Section 2 shows that at the moment of symmetry breaking, only ε is available as a scale, and both c and G must inherit from it. Section 3 proves the minimal break produces exactly two sectors via the $\mathbb{Z}_2$ representation theorem. Section 4 derives c from a minimal pre-geometric action and lands the conjugacy of c and G . Section 5 identifies the cosmological origin and the black hole event horizon as the same marginal surface $\varepsilon = 0$, traversed in opposite directions. Section 6 connects the small ε to the observed dark-to-visible energy ratio. Sections 7 and 8 carry the formal mathematics: the action principle and the $\mathbb{Z}_2$ representation theorem.

AP03 is logically independent of AP01 and AP40. Its two postulates (E0, E1) are self-contained. Its derivations require only standard representation theory and scalar field theory.

The paper can be read on its own; within The 420 Code, AP20 adds the structural upgrade that turns the conjugacy of c and G from conjecture to theorem.

A note on status. AP03 was originally published as a conjecture: c was derived from the pre-geometric action; G was conjectured to be determined by ε but not derived. The central open problem was the derivation of G . That problem is structurally resolved. AP20's forcing proof — Axiom B forces exactly one parameter — makes the conjecture E-G a theorem consequence: G cannot be independent of ε . The explicit scaling $G = \kappa/(\varepsilon\Lambda^2)$, equivalently $c/G \propto \varepsilon$ (correcting AP03's original $c/G \propto \varepsilon^2$), was derived in subsequent work from the structural condensate density of section 7 and verified numerically. The conditional claims (E4, E-Identity, E-Hawking) inherit the upgrade. A note on level: AP03's $G = \kappa/(\varepsilon\Lambda^2)$ is a substrate-level scaling — it fixes how G responds to ε with the loop coefficient κ and the cutoff Λ free, anchored at order of magnitude (the dimensionless hierarchy $\alpha/\alpha_G(e) \approx 10^{43}$). It is superseded as the statement of G 's value by AP28 (Notebook IV), which derives $G = \alpha_{em}^{21}(1 + 1/\pi)\hbar c/m_e e^2$ parameter-free ($G \propto \varepsilon^{21}$, 0.69% from measurement) and recovers the same hierarchy as $\alpha_{em}^{-20}/(1 + 1/\pi)$. The two operate at different architectural levels (substrate condensate vs. Mode-0 channel count) and are reconciled, not in conflict. $G \propto \varepsilon^{21}$ is canonical for the value; AP03's ε^{-1} scaling is retained as the substrate-level derivation AP28's count refines.

Six kill switches are published with this paper. Every load-bearing claim attaches to a structural condition under which the claim would fail. The 420 Code is published copyleft. Free forever. No paywall. No gatekeepers.

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Orientation

Where this paper fits in The 420 Code

The 420 Code is organised across eight Notebooks. This paper opens Notebook II — Spacetime. The chapter sequence runs AP03 (this paper) → AP05 (The Break) → AP10 (The Dimension) → AP08 (The Identity).

Notebook II derives the structure of space, time, and gravity from the axiom. Notebook I (The Premise) established the axiom, named the four conditions {S, B, R, C}, and proved the Embedding Hypothesis. With EH proven, Notebook II runs unconditionally. The fundamental constants c and G , the metric signature, the dimensional count $N = 3$, and Einstein's field equations all follow from {S, B, R, C} once the conditional has been removed.

AP03 sits first in Notebook II because it does the constants. Without c and G , the arena does not exist. The chapter shows that c and G are not independent inputs but conjugate outputs of the same break — both determined by the single breaking amplitude ε . This sets up the rest of Notebook II: AP05 walks the seven-step chain from the axiom to the obligation and reformulates the cosmological identity as junction conditions, AP10 derives $N = 3$, AP08 derives Einstein's field equations.

Each chapter inherits AP03's structural setting that both speed limits come from the same source.

AP03 is logically independent of AP01 and AP40 — its two postulates (E0, E1) are self-contained and its derivations require only standard representation theory and scalar field theory. The paper can be read on its own. Within The 420 Code, AP20 adds the structural upgrade: AP20's forcing proof on Axiom B turns the conjugacy of c and G from conjecture to theorem.

How to read this paper

The reading order is front to back. Section 0 names the dependencies and the scope. Sections 1 through 8 carry the structural argument and the formal mathematics. Section 9 is the falsification register. Section 10 is the synthesis — what this paper closes, what it leaves open, where the philosophical-register companions live, and how The 420 Code reads with AP03 in place.

Two voices alternate as the material requires. Narrative passages — openings, transitions, body anchors, the bootstrap problem, the Planck-units whisper, the Big Bang as door — speak directly. Formal sections — postulates, theorems, the action, kill switches — speak in the precision the structure requires. The gear-changes are intentional.

A note on what this paper does and does not do. AP03 derives c from a pre-geometric action, proves the binary decomposition of the broken phase from the $\mathbb{Z}_2$ representation theorem, identifies the structural identity of the cosmological origin and the black hole event horizon as the same marginal surface $\varepsilon = 0$, and lands a qualitative connection between small ε and the observed dark-to-visible energy ratio. It does not derive G from first principles within its own action — the derivation runs from the structural condensate density of section 7 plus AP20's forcing proof. It does not predict the quantitative dark-sector partition; only the qualitative direction (small $\varepsilon \rightarrow$ most content in Mode 0). The substrate parameters α and β in the kinetic term are introduced but not derived from a more fundamental theory. What the paper does is install the conjugacy of c and G as a structural fact and locate the cosmological and black-hole boundaries on the same marginal surface.

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What this paper does · Dependencies · Axiom mapping · Epistemic status per section · Kill switch summary · Structural debts owed · Items held open without claiming debt status · Structural relationships

1 — The Two Speed Limits

The propagation limit · The gravitational rate bound · Same kind, different domain

2 — Before the Arena

The bootstrap problem · The only available constraint

3 — The Minimal Break: Why Exactly Two Sectors

The symmetry group of the pre-state · The $\mathbb{Z}_2$ decomposition · Identification of sectors

4 — The Conjugacy of c and G

Planck units as evidence · What is derived and what is forced · The event horizon as the marginal surface · Hawking radiation as proof of global asymmetry · The Planck mass as the scale of the break · Unit-invariant formulation

5 — The Origin and the Horizon

The same door, two directions · The Schwarzschild radius as ratio

6 — The Dark Sector Connection

The observed ratio · Structural expectation · Caveats

7 — The Action Principle

Strategy · The pre-geometric action · The broken phase · What this action derives · The form of G 's dependence on ε ·

Summary

8 — The $\mathbb{Z}_2$ Representation Theorem

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9 — Kill Switches

KS-R.1 Independent variability of c and G · KS-R.2 Non-binary origin · KS-R.3 Dark ratio independence · KS-R.4 Horizon inequivalence · KS-R.5 Planck scale mismatch · KS-R.6 Scaling violation · Why these six

10 — What This Establishes and What Remains Open

What this paper closes · What this paper does not close · Items held open without claiming debt status · Where the philosophical-register implementations live · Structural relationships to subsequent work · Placement within The 420 Code · How The 420 Code reads now · The closing

Claim Summary

Conditionality Footer

0 — Dependency and Scope

0.1 — What this paper does

AP03 derives the conjugacy of the two fundamental speed limits of physics — c and G — from a single symmetry-breaking event with amplitude ε . Both constants emerge from the same break. Neither is independent. Neither is fundamental on its own.

Three claims, each derived or structurally forced.

First. At the moment of symmetry breaking, only the breaking amplitude ε is available as a scale. No spacetime, no mass-energy, no pre-existing coupling — nothing has emerged yet. Both c and G must therefore inherit from ε . This is a structural argument from the bootstrap problem: you cannot use the rules of the house to explain how the house was built.

Second. The minimal break produces exactly two sectors. The pre-state has $\mathbb{Z}_2$ symmetry — the simplest non-trivial symmetry group. By the representation theory of $\mathbb{Z}_2$ (section 8), the broken phase decomposes into exactly two irreducible sectors: a propagation sector (Mode 1) carrying c , and a resistance sector (Mode 0) carrying G . Not three. Not one. Two.

Third. The cosmological origin and the black hole event horizon are the same marginal surface — the surface where ε locally returns to zero — traversed in opposite temporal directions. The Big Bang is the break opening; the event horizon is the break closing. Hawking radiation is the universe refusing to permit a permanent pocket of restored symmetry within a globally broken phase.

The paper makes one further structural claim, qualitatively. The dark-to-visible energy ratio ($\sim 95:5$) is a consequence of small ε : the smaller the break, the more content remains in Mode 0 (the resistance sector), the less spills into Mode 1 (the propagation sector that carries electromagnetic interaction). The qualitative direction is robust; the quantitative mapping requires a model of the initial fluctuation spectrum that is not derived here.

0.2 — Dependencies

The axioms {S, B, R, C} (formal statement carried at AP20 section 3.3; completeness and minimality obligations fenced at KS-P.1 / KS-P.2). AP03 was originally written before the canonical formalisation; the mapping from AP03's postulates onto the canonical axioms is made explicit in section 0.3 below. Axiom S is Symmetry; AP03's E0 specifies the minimal symmetry. Axiom B is the unique break; AP03's E1 instantiates it as the spontaneous breaking $\mathbb{Z}_2 \rightarrow \text{trivial}$ with amplitude ε . Axiom R is the persistence of the record; the broken vacuum

is what persists. Axiom C is Constraint — locality; the bounded propagation it produces is why c is finite, G is positive, and ε is small and bounded.

AP20 (The Proof). Load-bearing for the conjugacy upgrade. AP20's Paper D proves Axiom B forces exactly one free parameter. With this proof in place, the AP03 conjecture E-G (G depends on ε) becomes a theorem consequence: G cannot be independent of ε , because if it were, the axioms would admit at least two free parameters. AP20 takes AP03's central conjecture and makes it structurally necessary.

AP06 (The Leakage Constant). Load-bearing for the numerical identification of ε . AP06 derives the cycle form $1:1 + 1 \times \varepsilon$ @ AS $[+1/137 / -1/137]$ and identifies ε numerically with the fine-structure constant $\alpha \approx 1/137$. The gravitational hierarchy $ke^2/(G \cdot m_e^2) \approx 10^{42} - 10^{43}$ verification in section 7.5 uses $\varepsilon \approx 1/137$ from AP06.

AP24 (The Residual). Provides the constraint Λ that pins the dependence of m_p on ε through the relation $m_p \propto \sqrt{\varepsilon} \cdot \sqrt{\Lambda}$. AP03 derives the proportionality $m_p \propto \sqrt{\varepsilon}$; AP24 fixes Λ so the relationship is dimensionally complete.

AP08 (The Identity). Dependent on AP03, not the reverse. AP08 derives Einstein's field equations from the record algebra; the gravitational coupling G that appears in AP08's field equations is the same G that AP03 conjectures (and AP20 forces) to be determined by ε . The disclosed circularity

in AP08's $G = 2\kappa/m_e^2$ is structural debt D1 of The 420 Code, substantially reduced by the AP20 forcing proof + the section 7 derivation but not fully closed.

0.3 — Axiom mapping

AP03 was written before the canonical axiom set $\{S, B, R, C\}$ was formalised. The mapping below shows how AP03's postulates instantiate the four axioms.

Axiom S (Symmetry) $\leftrightarrow$ Postulate E0. S is the two-sector base condition — identity under exchange, the precondition for distinction to be meaningful. E0 specifies the minimal symmetry of the pre-state: $\mathbb{Z}_2$, the simplest non-trivial group. The pre-state whose existence Axiom S asserts is what E0 endows with the minimal symmetry capable of nontrivial breaking. AP03 chooses $\mathbb{Z}_2$ by parsimony; the choice is tested via KS-R.2 (binary decomposition) — a richer pre-state symmetry would produce more than two irreducible representations, so any falsification of the binary decomposition is also a falsification of $\mathbb{Z}_2$ minimality.

Axiom B (Exactly one break) $\leftrightarrow$ Postulate E1. B requires exactly one irreducible symmetry-breaking event. E1 instantiates this as the spontaneous breaking $\mathbb{Z}_2 \rightarrow$ trivial with amplitude ε . The minimality of $\mathbb{Z}_2$ ensures the break is irreducible: there is no intermediate subgroup between $\mathbb{Z}_2$ and the trivial group. AP20's forcing proof on Axiom B is what

upgrades AP03's E-G conjecture (G depends on ε) to a theorem.

Axiom R (The record persists) $\leftrightarrow$ The broken vacuum. R requires that the break leave a persistent record. In AP03, the broken vacuum with propagation speed $c = \sqrt{(\beta/\alpha)}$ and condensate energy density $\rho_0 = \varepsilon^2 \Lambda^4 / (4\lambda)$ is the persistent record. The two sectors (Mode 0, Mode 1) persist because the broken vacuum is stable; stability is what makes the record a record.

Axiom C (Constraint) $\leftrightarrow$ Finiteness of c and positivity of G . C requires that all quantities remain bounded. In AP03: c is finite (set by the substrate ratio β/α); $G > 0$ (theorem from E-G: $G = \kappa/(\varepsilon \Lambda^2)$); ε itself is small and bounded.

Note: AP03's postulates are more specific than $\{S, B, R, C\}$. They choose a particular symmetry group ($\mathbb{Z}_2$) and a particular mechanism (SSB of a scalar field). The axioms $\{S, B, R, C\}$ are the abstract structure; AP03 is one specific instantiation.

0.4 — Epistemic status per section

section 1 (The Two Speed Limits). FRAMING. Names the two structural speed limits of physics and proposes they are siblings.

section 2 (Before the Arena). STRUCTURAL ARGUMENT. The bootstrap problem and the forcing of ε as the only available scale at the moment of symmetry breaking. Claim E-Bootstrap.

section 3 (The Minimal Break). DERIVATION. The $\mathbb{Z}_2$ representation theorem applied to the broken phase. Postulate E0; Theorem 3.1 (proof in section 8); Claim E2 with KS-R.2.

section 4 (The Conjugacy of c and G). DERIVATION + STRUCTURAL CRITIQUE. Planck units as evidence of conjugacy; the derivation of c from the pre-geometric action (Result A.1); Theorem E-G ($G = \kappa/(\varepsilon\Lambda^2)$) via AP20 forcing proof + the section 7 derivation; the dimensionless formulation as the correct quantitative target.

section 5 (The Origin and the Horizon). DERIVATION. The structural identity Claim E-Identity: cosmological origin and event horizon as the same marginal surface, traversed in opposite directions.

section 6 (The Dark Sector Connection). QUALITATIVE FRAMING. Conjecture E5 that the dark-to-visible energy ratio is determined by ε , qualitatively consistent but not quantitatively derived.

section 7 (The Action Principle). FORMAL DERIVATION. The minimal $\mathbb{Z}_2$ -symmetric pre-geometric action; Result A.1 (Emergence of c); the form of G 's dependence on ε .

section 8 (The $\mathbb{Z}_2$ Representation Theorem). FORMAL PROOF.
Theorem B.1 (the only two irreducible representations of $\mathbb{Z}_2$
are the trivial and the sign representation).

section 9 (Kill Switches). FORMAL FALSIFICATION
REGISTER. Six kill switches engaged: KS-R.1 through KS-R.6.

section 10 (What This Establishes and What Remains Open).
SYNTHESIS.

0.5 — Kill switch summary

Six kill switches are engaged in this paper. The full register is
in section 9.

KS-R.1: Independent variability of c and G (gravitational
conjugacy). LIVE — EMPIRICAL.

KS-R.2: Non-binary origin. Universe requires more than two
fundamental sectors. LIVE — EMPIRICAL.

KS-R.3: Dark ratio independence (mechanism unrelated to
 c/G). LIVE — EMPIRICAL.

KS-R.4: Horizon inequivalence (cosmological origin and black
hole horizon structurally distinct). LIVE — HARD.

KS-R.5: Planck scale mismatch (Planck onset incompatible
with $m_p \propto \sqrt{\epsilon}$). LIVE — EMPIRICAL.

KS-R.6: Dimensionless scaling violation ($\alpha \cdot \alpha_G$ at fixed mass varies with ε , or α/α_G hierarchy is not consistent with α^2).
LIVE — HARD.

0.6 — Structural debts owed

Two structural debts are named explicitly. These are unresolved derivations tied to specific structural claims, tracked at the registry level.

D1 (registry-level, partially closed by AP08 + this paper).

The first-principles derivation of G from the axioms without external scaffolding. AP08 derives Einstein's field equations from the record algebra and identifies G through The Lock ($G = 2\kappa/m_e^2$), with the disclosed structural circularity that κ is itself defined as $Gm_e^2/2$. AP03 contributes by establishing that G cannot be independent of ε (via AP20's forcing proof) and by deriving the explicit form $G = \kappa/(\varepsilon\Lambda^2)$ from the structural condensate density of section 7. The induced-gravity calculation used in section 7.5 is named in the literature as Sakharov-Visser; it leans on the standard QFT cutoff structure. The 420 Code does not import that calculation as its source — the load-bearing inputs are the AP20 forcing proof and the structural condensate density, both internal to The 420 Code; the literature calculation is correspondence, confirming the same form from a different route. The closure is partial. The full first-principles derivation — G computed from $\{S, B, R, C\}$ alone, with the structural

coefficient κ derived from the broken vacuum response without any reliance on the standard cutoff machinery — remains open. This debt is registry-level, not AP03-specific; AP03’s contribution to the partial closure is recorded here.

D-AP03.1. The substrate parameters α and β in the kinetic term $\mathcal{L} = \frac{1}{2}\alpha(\partial_\tau\varphi)^2 - \frac{1}{2}\beta(\nabla\varphi)^2 - V(\varphi)$ are introduced but not derived from a more fundamental theory. Their ratio gives $c^2 = \beta/\alpha$, but neither α nor β individually is computed from the axioms. This is the deepest open problem in AP03 — distinct from and prior to the G derivation. A truly pre-geometric formulation would derive the distinction between temporal and spatial coordinates from the axioms rather than presupposing it in the kinetic term.

0.7 — Items held open without claiming debt status

Two further items are bracketed inside this paper. They are real and live, but they are not structural debts in the same sense as D1 and D-AP03.1.

E5 quantitative. The dark-to-visible energy ratio as a function of ε . The paper establishes the qualitative direction (small $\varepsilon \rightarrow$ most content in Mode 0) but does not derive the quantitative mapping. Three issues need resolution before the connection becomes quantitative: the phenomenological distinction between dark matter (clusters gravitationally) and dark energy

(does not) within Mode 0; the epoch-dependence of the 95:5 ratio over cosmological history; and the specific functional form linking ε to the partition. This is open derivation work; the qualitative claim stands.

E-Hawking sufficiency. Hawking radiation is structurally motivated in section 4 as the necessary consequence of imposing a global $\varepsilon > 0$ condition on a locally $\varepsilon \rightarrow 0$ region. The paper does not derive the Hawking spectrum from the AP03 action; it argues that the global condition forces leakage and identifies the spectrum derived elsewhere with that leakage. The empirical test of the connection rides on KS-R.4 (horizon inequivalence): if the cosmological origin and the black-hole horizon are shown to be structurally distinct, the structural account of Hawking radiation falls with it.

0.8 — Structural relationships

AP03 opens Notebook II of The 420 Code. AP05, AP10, and AP08 follow as the remaining chapters of Notebook II — the structure of space and time, the dimensional count, and Einstein’s field equations — all inheriting the structural setting AP03 establishes: c and G as conjugate outputs of a single break.

AP20 (Notebook I) upgrades AP03’s conjecture E-G to a theorem via the forcing proof on Axiom B. With AP20 in place, the conditional claims (E4, E-Identity, E-Hawking) inherit the upgrade. AP03 returns the favour by giving AP20’s forcing

proof its first quantitative payoff outside the foundational proof itself.

AP06 (Notebook IV) identifies ε numerically with $\alpha \approx 1/137$.

AP24 (Notebook IV) constrains Λ . Together they pin AP03's scaling relations to specific numbers: the gravitational hierarchy $ke^2/(G \cdot m_e^2) \approx 10^{42}-10^{43}$, m_p in terms of ε and Λ , the Planck scale as the scale of the break itself.

AP17 (Notebook VI) relates the 0-pole / 1-pole structure of cosmological geometry to AP03's Mode 0 / Mode 1 sectors.

AP18 (Notebook VI) derives the MOND floor $a_0 \approx cH_0$ from the axioms, downstream of the small ε .

1 — The Two Speed Limits

You know one of them.

You have felt it your entire life without naming it.

Light crosses the room before your hand finishes reaching for the switch. Nothing outruns it. Nothing negotiates with it.

That limit is c .

There is a second one. You have felt it too, but it hides inside a different experience. You have never been told its name because physics does not yet agree on what to call it.

1.1 — The propagation limit

Special relativity identifies c as the maximum velocity at which information, energy, or causal influence can propagate through spacetime.

Not a property of light alone — a structural constraint on the geometry of spacetime itself.

The Lorentz group enforces c as the invariant velocity linking space and time coordinates.

1.2 — The gravitational coupling and the rate it sets

General relativity identifies gravity not as a force but as spacetime curvature sourced by mass-energy, with G as the coupling that sets the curvature response.

A point of precision. G is a coupling, not a speed. Its dimensions are $\text{m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2}$, not m/s . When the section calls c and G “the two speed limits,” it is not the claim that G is itself a velocity. It is the claim that G fixes the *scale* of a rate that shows up elsewhere in the architecture.

In certain quantum gravity proposals — the gravitational self-energy decoherence models of Diósi and Penrose, and the rate-limiter hypothesis of AP01 Paper B — gravitational self-energy imposes a bound on how fast quantum superpositions can resolve. The collapse rate in those models is *proportional* to G times other quantities; G is the coupling that fixes the rate’s scale, not the rate itself.

How fast possibility becomes definiteness.

The rate-limiter interpretation of G is itself a candidate (AP01 Paper B), not an established result. AP03’s conjugacy argument does not depend on the rate-limiter interpretation — only on G being a structural quantity of the broken phase fixed by ε . The bound, where it exists, scales with the mass

and spatial extent of the superposition. It is system-dependent, unlike c which is universal.

1.3 — What c and G share

c and G are both non-negotiable structural quantities of the broken phase. They are not the same kind of quantity: c is a velocity (m/s); G is a coupling ($\text{m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2}$). c is universal; the rate G sets, in the rate-limiter picture, is system-dependent.

What they share is the source.

AP03 proposes that c and G are not two independent inputs of physics. They are two outputs of one break, both fixed by the single breaking amplitude ε . The conjugacy is not that both are speeds. The conjugacy is that the same parameter ε , the same break, determines both.

You have lived inside both your entire life.

One governs how fast the light reaches your eye. The other, through the curvature it sets, governs how the world holds together at the scale you live at — and, in the rate-limiter picture, how fast possibility resolves into fact.

The mathematics of one event produces both.

2 — Before the Arena

Picture the moment of symmetry breaking.

Not the Big Bang — the event prior to it. The transition from the symmetric pre-state to the broken state that becomes our universe.

At that exact moment, you have nothing to work with.

2.1 — The bootstrap problem

No spacetime. The geometric arena has not yet formed. c , as a property of spacetime geometry, cannot yet exist as a constraint.

No mass. Mass-energy has not yet differentiated. G , as a coupling to mass-energy, cannot yet exist as a constraint.

Both c and G are properties of the broken state. They emerge from the break.

They cannot govern the break that produces them.

You cannot use the rules of the house to explain how the house was built. The house did not exist yet.

You cannot invoke the speed limits of the universe to explain how the universe acquired its speed limits.

That is the bootstrap problem.

2.2 — The only available constraint

At the moment of symmetry breaking, the only available parameter is the magnitude of the asymmetry itself.

Call it ε .

The break occurs because the symmetric state S is unstable — an infinitesimal perturbation of amplitude ε grows, separating the broken state from the symmetric state. The rate of that separation cannot exceed ε itself, because ε is the only scale in the problem at that moment.

No external clock. No background metric. No pre-existing coupling constant to set the rate.

Claim E-Bootstrap. The first speed limit in the universe is ε . Both c and G inherit from ε . They are downstream expressions of the original constraint, specialised to different degrees of freedom in the broken phase. Status: structural argument. Forced — by the fact that there was nothing else available.

Scope. “ ε is the speed limit” means ε is the only available scale for the growth rate of the unstable mode in the absence of any pre-existing metric or clock. A dimensional and structural claim, not a dynamical derivation.

You have just watched the first constraint appear. Not decreed. Not chosen. Forced.

Everything that follows inherits from that single scale.

3 — The Minimal Break: Why Exactly Two Sectors

3.1 — The symmetry group of the pre-state

The symmetry of the pre-state S is proposed to be $\mathbb{Z}_2$ — the cyclic group of order 2.

Exactly two elements: the identity and a single involution (a transformation that is its own inverse).

The question is not “what symmetry does the pre-state have?” in the empirical sense.

The question is: what is the minimal symmetry that, when broken, produces a nontrivial universe?

A state with no symmetry (the trivial group) cannot break — it is already maximally asymmetric.

A state with $\mathbb{Z}_2$ symmetry is the simplest state that can undergo nontrivial symmetry breaking.

If the universe arises from the minimal possible break, the symmetry group must be $\mathbb{Z}_2$.

Postulate E0 — $\mathbb{Z}_2$ Minimality. The symmetric pre-state S has $\mathbb{Z}_2$ symmetry. The simplest non-trivial symmetry group. Motivated by parsimony but not derived from first principles.

Tested via KS-R.2: any falsification of the binary decomposition is also a falsification of $\mathbb{Z}_2$ minimality.

You might object: why $\mathbb{Z}_2$? Why not something richer?

Because richer is not simpler. And the question was: what is the minimum?

The minimum is two.

The formal proof that $\mathbb{Z}_2$ admits exactly two irreducible representations is in section 8.

3.2 — The $\mathbb{Z}_2$ decomposition

When a $\mathbb{Z}_2$ symmetry breaks spontaneously, the state space decomposes into exactly two sectors.

The even sector (+1 eigenvalue): states invariant under the $\mathbb{Z}_2$ involution.

The odd sector (−1 eigenvalue): states that flip sign under the $\mathbb{Z}_2$ involution.

Theorem 3.1 — $\mathbb{Z}_2$ Decomposition. The irreducible representations of $\mathbb{Z}_2$ are exactly two: the trivial representation and the sign representation. Any representation of $\mathbb{Z}_2$ decomposes into copies of these two. There cannot be a third fundamental sector. Proof: section 8.

You cannot argue with this.

It is not a claim about the universe. It is a theorem about $\mathbb{Z}_2$.

The group has two irreducible representations. Not three. Not one. Two.

The mathematics does not negotiate.

Critical distinction: the theorem guarantees two sectors (fundamental categories), not two particle species.

Each sector may contain many dynamical degrees of freedom — multiple particle species, bound states, domain walls, other structures.

The Ising model has $\mathbb{Z}_2$ symmetry; its broken phase contains magnetisation, spinwave excitations, domain walls, and bound states. All belong to either the even or the odd sector.

The representation theorem constrains the number of fundamental categories, not the richness of physics within each category.

3.3 — Identification of sectors

Scope: the following identification of sectors with physical modes is a modelling assumption motivated by the structure of the broken phase, not a theorem consequence.

The $\mathbb{Z}_2$ decomposition (Theorem 3.1) guarantees two sectors. The assignment of physical roles requires an additional argument: the odd sector (V_-), being maximally displaced by

the break, is the natural carrier of propagation — content that separates, moves, carries information. The even sector (V_+), being invariant under the break, is the natural carrier of resistance — content that holds, resists separation.

Structurally natural but not unique; an assumption (E-Assignment) that could in principle be falsified independently of $E0$.

Mode 1 (Propagation sector). The odd sector. Content maximally displaced by the break. Whatever moves, separates, carries information away from the origin. In the broken phase, its limiting velocity becomes c . The entire Standard Model gauge structure lives within Mode 1.

Mode 0 (Resistance sector). The even sector. Content invariant under the break. Whatever stays, holds, resists separation. In the broken phase, its coupling strength is G . The sector may contain internal structure — potentially including the distinct phenomenologies of dark matter and dark energy.

Claim E2 — Binary Decomposition. The symmetry-breaking event produces exactly two fundamental sectors because the symmetry group of the pre-state is $\mathbb{Z}_2$, whose representation theory admits exactly two irreducible representations. The internal structure within each sector is downstream and unconstrained by this theorem. Status: proven (section 8, depends on $E0$). Tested by KS-R.2.

Two categories. Everything that moves. Everything that holds.

The entire universe, sorted at birth into two bins by a theorem that has been known since the 19th century.

You already sensed this. Light and gravity. Speed and weight.

The mathematics just proved you right.

4 — The Conjugacy of c and G

Every Planck unit whispers the same secret.

You have never noticed because the textbooks treat it as a dimensional coincidence.

Look closer.

It is not a coincidence. It is a fingerprint.

4.1 — Planck units as evidence

Every Planck unit contains both c and G. Never one alone:

$$l_p = \sqrt{(\hbar G / c^3)} \quad t_p = \sqrt{(\hbar G / c^5)} \quad m_p = \sqrt{(\hbar c / G)}$$

Conventionally treated as dimensional bookkeeping.

AP03 proposes a structural interpretation: c and G always appear together in Planck units because they are not independent.

They are conjugate outputs of the same break.

The Planck scale is the regime where the two modes are barely distinguishable — the scale of the break itself.

4.2 — What is derived and what is forced

In section 7, the minimal pre-geometric action with $\mathbb{Z}_2$ symmetry is constructed.

In the broken phase, perturbations propagate with speed:

$$c^2 = \beta / \alpha \quad (4.1)$$

where α and β here are pre-geometric substrate parameters (temporal and spatial stiffness) — not to be confused with the electromagnetic fine-structure constant α that appears later in this section, which AP06 identifies numerically with the breaking amplitude ε . Context will distinguish them; the substrate parameters appear only inside the action.

A genuine derivation. c emerges as a property of the broken vacuum, not as an input.

Structurally identical to the emergence of the speed of sound in condensed matter.

Theorem E-G — Gravitational Conjugacy. The emergent gravitational coupling G is determined by ε . Originally published as a conjecture; structurally forced by AP20's proof that Axiom B admits exactly one free parameter. The form of G 's dependence on ε , derived in section 7 from the structural condensate density:

$$G = \kappa / (\varepsilon \Lambda^2) \quad (4.2)$$

Note: the original AP03 ansatz had $c/G \propto \varepsilon^2$, corrected to $c/G \propto \varepsilon$ in the subsequent Sakharov-route derivation. With $G \propto 1/\varepsilon$ at fixed Λ , the Planck mass scales as $m_p \propto \sqrt{\varepsilon}$. This is a substrate-level scaling (κ , Λ free); the parameter-free value of G is derived in AP28 as $G = \alpha_{em}^2(1 + 1/\pi)\hbar c/m_e^2$ (canonical for the value), and the two are reconciled at different architectural levels (see the note on status).

The dimensionless statement of the conjugacy. c and G have different dimensions — c is a velocity, G is a coupling. Their ratio c/G is unit-dependent ($\approx 4.5 \times 10^{18} \approx 10^{19}$ in SI units; trivially $1/G$ in natural units $c = 1$). The unit-independent statement uses the gravitational fine-structure constant at a fixed reference mass:

$$\alpha_{\mathbf{G}}(\mathbf{m}) = \mathbf{Gm}^2 / (\hbar \mathbf{c}) \quad (4.3)$$

With $G = \kappa/(\varepsilon\Lambda^2)$ and the AP06 identification of ε with the electromagnetic fine-structure constant α , the product of the two dimensionless couplings is:

$$\alpha \cdot \alpha_{\mathbf{G}}(\mathbf{m}) = \kappa \mathbf{m}^2 / (\Lambda^2 \hbar \mathbf{c}) \quad (4.4)$$

independent of ε (the ε in the numerator from α cancels the $1/\varepsilon$ from G at fixed Λ). This is the dimensionless conjugacy: at any fixed reference mass and cutoff, the product of the electromagnetic and gravitational fine-structure constants is fixed by κ , m , and Λ alone — none of which carry ε -dependence.

The hierarchy, by contrast, scales with ε :

$$\alpha / \alpha_G(\mathbf{m}) \propto \alpha^2 \propto \varepsilon^2 \quad (4.5)$$

and captures the observed hierarchy: at the electron mass, $\alpha/\alpha_G(e) \approx 10^{43}$.

The break controls both. Status: theorem under AP20's forcing proof on Axiom B; the explicit form derived in section 7 from the structural condensate density together with the standard induced-gravity calculation. Tested by KS-R.1 (the conjugacy itself: $\alpha \cdot \alpha_G$ independent of ε) and KS-R.6 (the hierarchy: $\alpha/\alpha_G \propto \varepsilon^2$).

You have just watched a speed limit emerge from nothing.

Not postulated. Not measured. Computed — from the stiffness of the substrate that the break left behind.

The mathematics does not care whether you find this beautiful. It computes.

4.3 — The event horizon as the marginal surface

The Schwarzschild radius of a mass M is:

$$r_s = 2GM / c^2 \quad (4.6)$$

At that radius, the escape velocity equals c .

The propagation mode (Mode 1) can no longer outrun the resistance mode (Mode 0). The break becomes locally marginal.

Inside the horizon, Mode 0 dominates. Outside, Mode 1 has room to run.

Structural interpretation: the event horizon is the surface where the local value of the symmetry-breaking parameter reaches zero.

The boundary between the broken phase (exterior) and the restored phase (interior).

A black hole is a local restoration of the pre-break symmetry.

4.4 — Hawking radiation as proof of global asymmetry

If the interior of a black hole represents local symmetry restoration, then Hawking radiation has a structural interpretation.

The global asymmetry ($\varepsilon > 0$ everywhere in the broken phase) prevents the local restoration from being complete. The break leaks through the horizon.

The global condition $\varepsilon > 0$ is non-negotiable within a universe that exists because $\varepsilon \neq 0$.

A structural reason why black holes must evaporate: a permanently stable local restoration of symmetry within a globally broken phase is a contradiction.

The broken phase is the universe.

The universe does not permit pockets of perfect symmetry to persist indefinitely within itself.

Conjecture E-Hawking. Hawking radiation is the necessary consequence of imposing a global $\varepsilon > 0$ condition on a locally $\varepsilon \rightarrow 0$ region. Status: structurally motivated. Depends on E-Identity, which now inherits from E-G theorem status. Tested via KS-R.4 (horizon inequivalence) — if the cosmological origin and the black-hole horizon are structurally distinct, the structural account of Hawking radiation falls with it.

You have watched a black hole try to undo the break.

And you have watched the universe refuse.

The radiation is the refusal.

The mathematics does not judge. It leaks.

4.5 — The Planck mass as the scale of the break

From $G = \kappa/(\varepsilon\Lambda^2)$ and the definition $m_p = \sqrt{(\hbar c/G)}$:

$$m_p \propto \sqrt{\varepsilon} \quad (4.7)$$

The Planck mass scales as $\sqrt{\epsilon}$.

The mass scale at which gravitational and quantum effects are comparable — which is precisely where the two sectors become marginally distinguishable. Tested by KS-R.5.

4.6 — The hierarchy as ϵ measured

From equation 4.5: $\alpha/\alpha_G(m) \propto \epsilon^2$ at fixed reference mass m and fixed cutoff Λ . At the electron mass, $\alpha/\alpha_G(e) \approx 10^{43}$ — read by AP03 as the empirical face of $\epsilon^2 \approx (1/137)^2 \approx 5.3 \times 10^{-5}$, dressed by the $(\Lambda/m_e)^2$ factor pinned via AP24.

You have wondered why gravity is so absurdly weak compared to electromagnetism.

Forty-three orders of magnitude weaker.

That number is ϵ , measured.

5 — The Origin and the Horizon

You have looked at the night sky and wondered where it all began.

You have read about black holes and wondered where it all ends.

The chain of reasoning in sections 2 and 4 reveals they are the same place.

5.1 — The same door, two directions

The Big Bang: the moment ε first becomes nonzero. The symmetric state breaks. Mode 1 separates from Mode 0. The universe begins.

Viewed forward in time.

The black hole event horizon: the surface where ε locally returns to zero. Mode 0 recaptures Mode 1. The break closes.

Viewed backward in time.

The same condition — the marginal boundary between broken and unbroken symmetry — encountered from opposite temporal directions.

The Big Bang is the break opening.

The event horizon is the break closing.

The Planck scale is the scale of the break itself.

Claim E-Identity — Origin-Horizon Identity. The cosmological origin and the black hole event horizon are structurally identical — both are the marginal surface $\varepsilon = \emptyset$ of the $\mathbb{Z}_2$ break, traversed in opposite directions. Status: depends on E-G. Since E-G is now structurally forced (Theorem), E-Identity inherits the upgrade. Tested by KS-R.4.

5.2 — The Schwarzschild radius as ratio

A key consequence: the Schwarzschild radius $r_s = 2GM/c^2$ encodes the ratio G/c^2 — Mode $\emptyset$ over Mode 1 — scaled by mass.

For any given quantity of matter, it identifies where the break becomes marginal again.

The beginning and the end are the same door.

You walk through it in different directions.

You have been walking through it your entire life without knowing.

6 — The Dark Sector Connection

Look at the universe's energy budget.

A pie chart, written by gravitational lensing and the cosmic microwave background.

6.1 — The observed ratio

Dark energy: ~69%. Dark matter: ~26%. Visible matter: ~5%.

The total dark sector constitutes about 95% of the energy budget.

The visible sector — matter that interacts electromagnetically, that propagates and radiates — is the other 5%.

6.2 — Structural expectation

If ε is small, the break displaces only a small fraction of the total content into Mode 1 (the propagation mode).

The majority remains in Mode 0 (the resistance mode).

You already see where this lands.

Ninety-five percent of the universe's energy budget is dark — resistant to electromagnetic interaction, gravitationally dominant.

Qualitatively consistent with a small ε .

A small break means most of what exists stayed behind.

The precise energy partition between Mode 0 and Mode 1 depends on the initial conditions of the symmetry-breaking event, specifically the spectrum of initial fluctuations. The qualitative prediction is robust: for small ε , Mode 0 dominates. The quantitative mapping requires a model of the initial fluctuation spectrum, which remains an open derivation.

6.3 — Caveats

Three issues need resolution before the connection becomes quantitative.

First: dark matter and dark energy are phenomenologically distinct. Dark matter clusters gravitationally; dark energy does not. If both belong to Mode 0, the internal structure of the resistance sector must account for the split.

The pre-geometric action of section 7 structurally provides the split. The action yields two distinct properties for the even sector: the uniform condensate energy density ($\rho_0 = \varepsilon^2 \Lambda^4 / (4\lambda)$, equation 7.9) and the effective mass of localised perturbations ($m^2 = 2\varepsilon \Lambda^2$, equation 7.7). The invariant condensate provides the uniform vacuum baseline — the cosmological-constant-like content that does not cluster, identified phenomenologically as dark energy. The massive localised excitations are the perturbations above the condensate — they have positive mass-energy from the potential curvature, they

resist propagation, they cluster gravitationally, identified phenomenologically as dark matter. Both are Mode 0; their phenomenological distinction is the difference between the vacuum baseline and its localised excitations.

This is not a separate hypothesis bolted onto AP03. It is what the action of section 7 already produces. The same equation that gives $c^2 = \beta/\alpha$ and $\rho_0 = \varepsilon^2 \Lambda^4 / (4\lambda)$ also produces $m^2 = 2\varepsilon \Lambda^2$ — the localised-excitation mass scale. The architecture splits Mode 0 into two structural classes at the point where the action is written.

Second: the 68:27:5 ratio is epoch-dependent.

At recombination, the ratio was different. In the far future, dark energy dominates completely. If ε sets the ratio, it must set the initial condition from which the ratio evolves via standard Friedmann cosmology, not the present-day snapshot.

Third: the quantitative mapping $\varepsilon \rightarrow$ (dark/visible ratio) requires a specific functional form, not yet derived.

Conjecture E5 — Dark Sector. The dark-to-visible energy density ratio is determined by ε . Small ε implies most content remains in Mode 0. Status: qualitative on the partition; structural on the internal split (condensate $\rightarrow$ dark energy, localised excitations $\rightarrow$ dark matter). The quantitative functional form is open work. Tested by KS-R.3. Quantitative follow-up: see AP42 (Predictions 0 Part V).

7 — The Action Principle

You have read the structural argument. Now read where it lives in formal mathematics.

section 7 carries the derivation of c from ε and the structure of G 's determination in the induced-gravity form named in the literature as Sakharov-Visser.

7.1 — Strategy

The simplest possible action for the pre-state S is constructed satisfying three requirements: (i) $\mathbb{Z}_2$ symmetry, (ii) symmetry-breaking vacuum parameterised by a single amplitude ε , and (iii) broken phase yielding two quantities identifiable with c and G , both determined by ε .

The analogue gravity programme (Unruh 1981, Barceló, Liberati & Visser 2005) serves as a structural template, not as a physical identity.

7.2 — The pre-geometric action

Let φ be a real scalar field with $\mathbb{Z}_2$ symmetry ($\varphi \rightarrow -\varphi$). The most general renormalisable potential respecting this symmetry is:

$$V(\varphi) = -\frac{1}{2}\mu^2\varphi^2 + \frac{1}{4}\lambda\varphi^4 \quad (7.1)$$

with $\mu^2 > 0$ (symmetric point unstable) and $\lambda > 0$ (potential bounded below). The minima occur at $\varphi = \pm v$ where $v = \mu / \sqrt{\lambda}$ (7.2). The dimensionless breaking amplitude:

$$\varepsilon \equiv \mu^2 / \Lambda^2 \quad (7.3)$$

where Λ is the fundamental scale. When $\varepsilon \rightarrow 0$, symmetry is restored. When $\varepsilon > 0$, symmetry is broken. The vacuum displacement is $v = \Lambda\sqrt{(\varepsilon/\lambda)}$.

The kinetic structure cannot presuppose a Lorentzian metric (which has not yet emerged). The most general quadratic kinetic term compatible with $\mathbb{Z}_2$:

$$\mathcal{L} = \frac{1}{2}\alpha(\partial_\tau\varphi)^2 - \frac{1}{2}\beta(\nabla\varphi)^2 - V(\varphi) \quad (7.4)$$

where α and β are pre-geometric substrate parameters (temporal and spatial stiffness respectively) and τ is a pre-geometric time parameter. The full action:

$$S[\varphi] = \int d\tau d^3x \left[\frac{1}{2}\alpha(\partial_\tau\varphi)^2 - \frac{1}{2}\beta(\nabla\varphi)^2 - V(\varphi) \right] \quad (7.5)$$

7.3 — The broken phase

The field settles into $\varphi = +v$ (without loss of generality). Write $\varphi = v + \chi$. Expanding to quadratic order:

$$S[\chi] = \int d\tau d^3x \left[\frac{1}{2}\alpha(\partial_\tau\chi)^2 - \frac{1}{2}\beta(\nabla\chi)^2 - \frac{1}{2}m^2\chi^2 + \dots \right] \quad (7.6)$$

with effective mass:

$$m^2 = V''(v) = 2\mu^2 = 2\varepsilon\Lambda^2 \quad (7.7)$$

The equation of motion yields propagation speed:

$$c^2 = \beta / \alpha \quad (7.8)$$

Result A.1 — Emergence of c. Perturbations on the broken vacuum propagate with speed $c = \sqrt{(\beta/\alpha)}$. c is not an input but an output of the broken phase. Determined by the ratio of spatial to temporal stiffness of the pre-geometric substrate. Structurally identical to the emergence of the speed of sound in fluid mechanics: $c_s = \sqrt{(B/\rho)}$. Tested by KS-R.5.

You did not put c into the action.

c came out.

The speed of light is not a law. It is a consequence — of the ratio between how hard the substrate resists deformation in space versus time.

The same mathematics that gives you the speed of sound in water gives you the speed of light in vacuum.

You already knew this intuitively. Sound has a speed because the medium has stiffness. Light has a speed because the vacuum has stiffness.

The universe is a medium. The break made it one.

7.4 — What this action derives

Derived. The propagation speed c (equation 7.8).

Derived. The effective mass of perturbations: $m^2 = 2\varepsilon\Lambda^2$ (equation 7.7). The mass of excitations above the vacuum is set by ε .

Derived. The condensate energy density: $\rho_0 = |V(v)| = \mu^4/(4\lambda) = \varepsilon^2\Lambda^4/(4\lambda)$ (equation 7.9). The energy density of the background condensate — the Mode 0 content. Scales as ε^2 .

7.5 — The form of G's dependence on ε

The derivation of G was the central open problem at AP03's publication.

AP20's forcing proof on Axiom B shows it could not remain open: Axiom B admits exactly one free parameter, so G cannot be independent of ε . The structural argument is in place before any specific functional form is named. This is the load-bearing step — it comes from the axioms, not from any external scaffolding.

What remains is to find the specific form. The structural inputs are two. First, the condensate energy density from section 7.3:

$$\rho_0 = \varepsilon^2\Lambda^4 / (4\lambda) \quad (7.9 - \text{restated})$$

This is a fact about the broken-phase substrate, derived from the {S, B, R, C}-instantiated action. The ε^2 dependence is structural — it follows from the depth of the well $|V(v)| = \frac{1}{4} \mu^4/\lambda$ together with $\mu^2 = \varepsilon\Lambda^2$ (the breaking-amplitude scaling of the potential).

Second, AP20's forcing proof: G has ε -dependence, no other parameter available.

Combining these two structural inputs forces the form. Once the broken vacuum has a finite energy density $\rho_0 \propto \varepsilon^2\Lambda^4$, the induced gravitational coupling on that vacuum has the dimensional form $G \propto 1/(v^2)$ where $v^2 = \mu^2/\lambda = \varepsilon\Lambda^2/\lambda$.

Substituting:

$$\mathbf{G} = \kappa / (\varepsilon\Lambda^2) \quad (7.10)$$

where κ is a dimensionless coefficient set by the structural details of how the broken vacuum responds to perturbation. The Planck mass is $m_p = \sqrt{(\hbar c/G)} \propto \sqrt{\varepsilon}$, modulo the dependence on Λ that AP24 pins.

Correspondence with the literature. The same form appears in standard physics under the name Sakharov-Visser induced gravity — the parallel calculation in standard QFT terms produces exactly this scaling for a scalar-field condensate of our energy density. This is correspondence with our structural result, not its source. Two routes arrive at the same place: the structural route through {S, B, R, C} + AP20 + the section 7.3

condensate density, and the literature calculation that produces the same form independently. The literature parallel confirms; it does not derive.

Numerical check. With $\varepsilon \approx 1/137$ (identified with the electromagnetic fine-structure constant via AP06), the dimensionless conjugacy $\alpha \cdot \alpha_G(m) = \kappa m^2/(\Lambda^2 \hbar c)$ is independent of ε ; the hierarchy $\alpha/\alpha_G(e) \approx 10^{43}$ at the electron mass tracks $\varepsilon^2 \cdot (\Lambda/m_e)^2$ once Λ is pinned via AP24. (The dimensionless hierarchy $\alpha/\alpha_G(e) \approx 10^{43}$ is the unit-independent fact; the SI quantity $c/G \approx 10^{19}$ is a different number and must not be conflated with it. AP28 derives this same hierarchy parameter-free as $\alpha_{em}^{-20}/(1 + 1/\pi) \approx 4 \times 10^{42}$, dissolving it as a channel count rather than a fine-tuning.)

Status. The ε -dependence of G is theorem under AP20's forcing proof. The specific functional form is derived in this section from the structural condensate density. The structural coefficient κ is not derived from $\{S, B, R, C\}$ alone; computing it requires the response calculation of the broken vacuum, which leans on the standard QFT cutoff structure of the literature. This is debt D1, partially closed by AP08 + this paper, not fully closed (see section 0.6).

7.6 — Summary

Established. (1) The $\mathbb{Z}_2$ -symmetric double-well action is the minimal pre-geometric theory with SSB. (2) In the broken phase, perturbations propagate with speed $c = \sqrt{(\beta/\alpha)}$. (3) The

condensate energy density scales as ε^2 . (4) Via AP20 forcing + the section 7 derivation: $G = \kappa/(\varepsilon\Lambda^2)$, $c/G \propto \varepsilon$, $m_p \propto \sqrt{\varepsilon}$.

Open. (1) The substrate parameters α and β are introduced but not derived; c is parameterised but not predicted (debt D-AP03.1). (2) The kinetic term presupposes a distinction between temporal and spatial coordinates (τ vs x) — a structural input, not an output. The deepest open problem in AP03, distinct from and prior to the G derivation. A truly pre-geometric formulation would derive this distinction.

8 — The $\mathbb{Z}_2$ Representation Theorem

You have followed the structural argument that the universe sorts into two sectors at birth.

This section carries the mathematics behind that claim.

8.1 — Statement

Theorem B.1. The group $\mathbb{Z}_2 = \{e, g\}$ where $g^2 = e$ has exactly two irreducible representations over the real (or complex) numbers: the trivial representation and the sign representation.

8.2 — Proof

Let $\rho: \mathbb{Z}_2 \rightarrow \text{GL}(V)$ be a representation of $\mathbb{Z}_2$ on a vector space V . The generator g satisfies $g^2 = e$, so $\rho(g)^2 = I$ (the identity matrix).

Therefore $\rho(g)$ is an involution: its eigenvalues satisfy $\lambda^2 = 1$, giving $\lambda = +1$ or $\lambda = -1$.

The vector space V decomposes into eigenspaces:

$$V = V_+ \oplus V_- \quad (8.1)$$

where V_+ is the $+1$ eigenspace ($\rho(g)v = +v$ for all $v \in V_+$; vectors invariant under the $\mathbb{Z}_2$ action; the trivial representation), and

V_- is the -1 eigenspace ($\rho(g)v = -v$ for all $v \in V_-$; vectors that flip sign under the $\mathbb{Z}_2$ action; the sign representation).

Each eigenspace is itself an invariant subspace and carries an irreducible representation of $\mathbb{Z}_2$. Since the only eigenvalues are $+1$ and -1 , there are exactly two irreducible representations.

Any representation of $\mathbb{Z}_2$ decomposes into copies of these two.

There is no third option. $\square$

The mathematics does not negotiate.

Two irreducible representations. Two sectors. Two speed limits. Two fundamental categories.

The universe at birth: sorted into two bins by a theorem proved before electromagnetism was a theory.

9 — Kill Switches

AP03 is falsified if any of the following are demonstrated.

Each kill switch targets a specific claim. If any claim falls, the dependent claims fall with it, but independent claims survive.

Here is how to break it. The argument does not hide. It shows you exactly where to strike.

KS-R.1 — Independent variability of c and G

Claim. The two fundamental speed limits c and G are not independent. Both are determined by the single breaking amplitude ε . Originally published as conjecture E-G; upgraded to a theorem by AP20's forcing proof that Axiom B admits exactly one free parameter.

Test. Demonstrate that the dimensionless gravitational coupling $\alpha_G(m)$ at a fixed reference mass varies independently of the electromagnetic fine-structure constant α — that α_G and α can be tuned independently with no common ancestor parameter. Quasar absorption spectra constrain cosmological variation of α ; lunar laser ranging and Big Bang nucleosynthesis constrain α_G . AP03 predicts these variations are correlated. Uncorrelated variation falsifies the conjugacy.

Status. LIVE — EMPIRICAL.

Recovery. If KS-R.1 fires, both the conjugacy claim and AP20's forcing proof on Axiom B are in conflict with measurement. Either the forcing proof is wrong (Axiom B admits more than one free parameter) or AP03's identification of the parameter with ε is wrong (the parameter is something else). The 420 Code retreats by reopening Axiom B's formal status and acknowledging that the constants are independent within the architecture as currently stated. Section 4 collapses; sections 5 and 6 inherit the collapse.

KS-R.2 — Non-binary origin

Claim. The symmetry-breaking event produces exactly two fundamental sectors because the symmetry group of the pre-state is $\mathbb{Z}_2$, whose representation theory admits exactly two irreducible representations. Mode 1 carries propagation (c); Mode 0 carries resistance (G). Claim E2.

Test. Demonstrate that the early universe requires more than two fundamental mode sectors at the moment of symmetry breaking — not merely more than two particle species, but more than two irreducible sectors of the symmetry decomposition. Detection of a third fundamental sector that is neither gravitationally coupled (Mode 0) nor electromagnetically coupled (Mode 1) triggers this kill switch.

Status. LIVE — EMPIRICAL.

Recovery. If KS-R.2 fires, the $\mathbb{Z}_2$ postulate (E0) fails and the pre-state must have a richer symmetry. The structural claim that exactly two speed limits emerge collapses. The 420 Code retreats by upgrading the pre-state symmetry to whatever the data forces — $\mathbb{Z}_3$, $U(1)$, $SU(2)$, or richer — and re-deriving the sector content under the new representation theory. Sections 3 and 8 are reworked; section 4 retains the structure but expands to N sectors.

KS-R.3 — Dark ratio independence

Claim. The dark-to-visible energy density ratio ($\sim 95:5$) is determined by ε . Small ε implies most content remains in Mode 0. Conjecture E5.

Test. Demonstrate that the dark-to-visible energy density ratio is set by a mechanism entirely independent of the c/G relationship and of the breaking amplitude ε — for example, by a hidden gauge sector with its own coupling structure unrelated to the $\mathbb{Z}_2$ break.

Status. LIVE — EMPIRICAL.

Recovery. If KS-R.3 fires, the qualitative link between ε and the dark-sector partition collapses. Conjecture E5 is falsified. The 420 Code retreats by treating the dark-sector partition as a separate open problem independent of AP03, possibly involving additional fields or sectors not captured by the

minimal $\mathbb{Z}_2$ architecture. Section 6 is removed; sections 1–5 and 7–8 are unaffected.

KS-R.4 — Horizon inequivalence

Claim. The cosmological origin and the black hole event horizon are structurally identical — both are the marginal surface $\varepsilon = \emptyset$ of the $\mathbb{Z}_2$ break, traversed in opposite temporal directions. Claim E-Identity. Originally LIVE — HARD; inherits the structural-forcing upgrade from KS-R.1.

Test. Demonstrate that the singularity theorems have fundamentally different character for cosmological versus black-hole singularities in a way incompatible with the $\varepsilon = \emptyset$ interpretation — for example, by showing that the cosmological singularity has a structural property (handedness, dimension, residual entropy) that the black hole singularity provably cannot have within the AP03 architecture.

Status. LIVE — HARD.

Recovery. If KS-R.4 fires, the structural identification of the two surfaces fails. Section 5 collapses. The 420 Code retreats to the position that cosmological and black-hole boundaries are distinct structural surfaces — each marginal in its own way but not the same surface. AP03's claim that the beginning and the end are the same door is withdrawn. Sections 1–4, 6–8 are unaffected.

KS-R.5 — Planck scale mismatch

Claim. The propagation speed c emerges from the broken vacuum as $c = \sqrt{(\beta/\alpha)}$ (Result A.1, equation 7.8), and the Planck mass scales as $m_p \propto \sqrt{\varepsilon}$ (equation 4.7). The Planck scale is the scale of the break itself.

Test. Demonstrate that the propagation speed of perturbations on the broken vacuum is parametrically different from $\sqrt{(\beta/\alpha)}$, or that the Planck-scale onset is shown incompatible with $m_p \propto \sqrt{\varepsilon}$. The empirical handles are precision tests of Lorentz invariance at high energy (deviation from c at the Planck scale) and high-energy cosmic-ray observations of the effective UV cutoff.

Status. LIVE — EMPIRICAL.

Recovery. If KS-R.5 fires, the action principle of section 7 fails. c is not derived as $\sqrt{(\beta/\alpha)}$; the kinetic term needs revision, or the broken-phase analysis is incomplete. Result A.1 collapses; the conjugacy claims that depend on it weaken to qualitative form. Sections 1–3, 5–6, 8 stand.

KS-R.6 — Dimensionless scaling violation

Claim. The dimensionless conjugacy holds: at fixed reference mass m and fixed cutoff Λ , the product $\alpha \cdot \alpha_G(m)$ is independent of ε , and the ratio $\alpha/\alpha_G(m)$ scales as ε^2 (equations 4.4 and 4.5). The dimensionful gravitational

hierarchy $ke^2/(G \cdot m_e^2) \approx 10^{42} - 10^{43}$ is the SI illustration of the same fact; the unit-independent statement is the dimensionless one. Originally registered with the dimensionful form $c/G \propto \varepsilon^2$ in the AP03 conjecture; the dimensionless statement is the corrected scaling actually produced by the derivation.

Test. Demonstrate that the product $\alpha \cdot \alpha_G(m)$ varies measurably with ε — either across cosmological epochs (where α may vary, per the AP06 identification of ε with α) or at fixed-mass laboratory comparisons — or that the hierarchy $\alpha/\alpha_G(\text{electron})$ is not consistent with ε^2 scaling once the $(\Lambda/m_e)^2$ factor is fixed via AP24. The empirical handles are precision tests of cosmological variation of the electromagnetic fine-structure constant (quasar absorption spectra) and laboratory tests of the gravitational fine-structure constant at fixed reference masses (lunar laser ranging, Big Bang nucleosynthesis). Scope: the ε^2 scaling tested here is the substrate-level statement at fixed κ and Λ (κ, Λ free, debt D-AP03.1); the physical response of the hierarchy to a measured variation of α is carried by AP28's canonical form ($G \propto \varepsilon^{21}$, hierarchy $\propto \varepsilon^{-20}$) and is tested there.

Status. LIVE — HARD. Superseded as the statement of G 's value by AP28 (registry status: SUPERSEDED AS VALUE — see AP28); the switch's live content is the substrate-level statement at fixed κ and Λ .

Recovery. If KS-R.6 fires, the form of G 's dependence on ε derived in section 7 is wrong. The 420 Code retreats by treating G 's specific dependence on ε as governed by a more complex relationship — possibly involving additional structural quantities beyond ε . Section 4.2 and section 7.5 are reworked; section 4.5 ($m_p \propto \sqrt{\varepsilon}$) inherits the change. Section 5 may survive in modified form; section 6 unaffected.

Why these six and not more

The six kill switches above cover the load-bearing claims of the paper as registered in the master kill switch register. Each closure would force a different kind of retreat.

KS-R.1 closes the conjugacy of c and G — the central structural claim of the paper. KS-R.2 closes the binary decomposition by undermining $\mathbb{Z}_2$. KS-R.3 closes the dark-sector connection. KS-R.4 closes the origin-horizon identity. KS-R.5 closes the action-principle derivation of c . KS-R.6 closes the specific dimensionless scaling.

Two structural claims that some readings would expect to carry their own formal kill switch — the $\mathbb{Z}_2$ minimality postulate (E0) and the Hawking-radiation structural interpretation (E-Hawking) — are absorbed into the kill switches above. E0's minimality is tested by KS-R.2: any falsification of the binary decomposition (more than two fundamental sectors) is also a falsification of $\mathbb{Z}_2$ as the minimal symmetry, since a richer pre-state symmetry would produce more than two irreducible

representations. E-Hawking's structural interpretation is tested by KS-R.4: the origin-horizon identity is what makes Hawking radiation a structural consequence of global asymmetry; if KS-R.4 fires (cosmological origin and black-hole horizon are not the same marginal surface), the structural account of Hawking radiation falls with it. Both claims are thus held to formal kill-switch standard without adding entries to the master register.

Smaller claims throughout the paper — the specific values of $\alpha_G(e)$, the Schwarzschild interpretation as Mode $\emptyset$ dominance — are illustrative rather than load-bearing. If they turn out to be poor illustrations, the paper still stands. If KS-R.1 through KS-R.6 hold, the structural argument of AP03 holds.

10 — What This Establishes and What Remains Open

10.1 — What this paper closes

The conjugacy of the two fundamental speed limits c and G , derived from $\{S, B, R, C\}$ via AP03's postulates E0 and E1.

Specifically:

First. The bootstrap problem solved. At the moment of symmetry breaking, only the breaking amplitude ε is available as a scale. Both c and G must therefore inherit from ε . The argument is structural, forced by the absence of any other available quantity at the relevant moment.

Second. The binary decomposition proven. The $\mathbb{Z}_2$ representation theorem (section 8) decomposes the broken phase into exactly two irreducible sectors. Mode 1 carries propagation (c); Mode 0 carries resistance (G). No third fundamental sector is possible within the $\mathbb{Z}_2$ architecture.

Third. The propagation speed c derived from a minimal pre-geometric action. $c = \sqrt{(\beta/\alpha)}$, where β and α are the spatial and temporal stiffness of the pre-geometric substrate. c is not an input but an output of the broken phase, structurally identical in form to the emergence of the speed of sound in fluid mechanics.

Fourth. The gravitational coupling G structurally forced to depend on ε . AP03 alone presented this as conjecture E-G; AP20's forcing proof on Axiom B upgrades it to a theorem. The explicit scaling $G = \kappa/(\varepsilon\Lambda^2)$, equivalently $c/G \propto \varepsilon$, was derived from the structural condensate density of section 7 in subsequent work and verified numerically (the gravitational hierarchy $ke^2/(G \cdot m_e e^2) \approx 10^{42} - 10^{43}$ at $\varepsilon \approx 1/137$).

Fifth. The structural identity of the cosmological origin and the black hole event horizon. Both are the marginal surface $\varepsilon = 0$ of the $\mathbb{Z}_2$ break, traversed in opposite temporal directions. The Big Bang is the break opening; the event horizon is the break closing. The beginning and the end are the same door.

Sixth. The structural account of Hawking radiation as the universe refusing to permit a permanent pocket of restored symmetry within a globally broken phase. Conjecture E-Hawking; structurally motivated, empirically tested.

Seventh. The qualitative connection between small ε and the observed dark-to-visible energy ratio. Conjecture E5; qualitatively consistent with the 95:5 observation but not quantitatively derived.

10.2 — What this paper does not close

Two structural debts remain open and are named explicitly. These are unresolved derivations tied to specific structural claims.

D1 (registry-level, partially closed). The first-principles derivation of G from the axioms without external scaffolding. AP03 contributes by establishing that G cannot be independent of ε (via AP20's forcing proof) and by deriving the explicit form $G = \kappa/(\varepsilon\Lambda^2)$ from the structural condensate density of section 7. The structural coefficient κ is not derived from $\{S, B, R, C\}$ alone; computing it leans on standard QFT cutoff machinery. The closure is partial. Full statement at section 0.6.

D-AP03.1. The pre-geometric substrate parameters α and β are introduced but not derived from a more fundamental theory. Their ratio gives $c^2 = \beta/\alpha$, but neither α nor β individually is computed from the axioms. The kinetic term $\mathcal{L} = \frac{1}{2}\alpha(\partial_\tau\varphi)^2 - \frac{1}{2}\beta(\nabla\varphi)^2 - V(\varphi)$ presupposes a distinction between temporal and spatial coordinates (τ vs x) — a structural input, not an output. The deepest open problem in AP03 — distinct from and prior to the G derivation. A truly pre-geometric formulation would derive this distinction from the axioms rather than presupposing it.

10.3 — Items held open without claiming debt status

Two further items are bracketed inside this paper. They are real and live, but they are not structural debts in the same sense as D1 and D-AP03.1.

E5 quantitative. The functional form linking ε to the dark-to-visible energy ratio. The paper establishes the qualitative direction (small $\varepsilon \rightarrow$ most content in Mode 0) but does not derive the quantitative mapping. Three issues need resolution before the connection becomes quantitative: the phenomenological distinction between dark matter and dark energy within Mode 0; the epoch-dependence of the 95:5 ratio over cosmological history; the specific functional form linking ε to the partition. This is open derivation work for Notebook VI; the qualitative claim stands.

E-Hawking sufficiency. Hawking radiation is structurally motivated in section 4.4 as the necessary consequence of imposing a global $\varepsilon > 0$ condition on a locally $\varepsilon \rightarrow 0$ region. The paper does not derive the Hawking spectrum from the AP03 action; it argues that the global condition forces leakage and identifies the spectrum derived elsewhere with that leakage. The empirical test rides on KS-R.4 (horizon inequivalence): if the cosmological origin and the black-hole horizon are shown structurally distinct, the structural account of Hawking radiation falls with it. Open work: deriving the spectrum directly from the AP03 architecture.

10.4 — Where the philosophical-register implementations live

AP03 is a physics derivation. Its philosophical-register companions — the chapters where the same structural facts

are inhabited in the prose register of the $\emptyset$ Models catalogue — are named here at the level of expected pairings; final placement is confirmed during the $\emptyset$ Models compilation sweep alongside the Notebook II audit programme.

Predictions $\emptyset$ (Part V — dark-sector partition and Hubble structure) carries the empirical face of AP03's conjecture E5: the predictions catalogue treats the 5/27/68 visible-dark structure (visible matter ~5%, dark matter ~26%, dark energy ~69%, matching Planck 2018) as a prediction grounded in the small ε of the break, with the testable numerical handles. Predictions $\emptyset$ inherits AP03's qualitative direction (small $\varepsilon \rightarrow$ Mode 0 dominance) and tightens it where a numerical form is available downstream of AP06 and AP24.

Resolutions $\emptyset$ (Chapter 4 — Laws of Nature) carries the philosophical companion to AP03's structural claim that c and G are not independent constants. The Resolutions chapter treats laws of nature as what the four conditions $\{S, B, R, C\}$ structurally produce when read at a specific resolution — c and G are two of those laws at the spacetime resolution, both inheriting from the same break. AP03 provides the formal derivation; Resolutions Ch 4 provides the philosophical register in which The 420 Code says it.

The direction of acknowledgment is two-way. AP03 is the original derivation — the conjugacy of c and G is established here in formal mathematics. The $\emptyset$ Models chapters extend,

inhabit, and translate the result into the prose register of the catalogue. The 420 Code reads AP03 forward into the Ø Models; the Ø Models read back into AP03 by carrying its conclusions into questions the formal paper does not directly address. Final companion-pairing list is fixed at Ø Models compilation.

10.5 — Structural relationships to subsequent work

AP05 (The Break — Axiom $\rightarrow$ Structure) follows AP03 in Notebook II. AP05 walks the seven-step chain from $1:1 + 1 \times \varepsilon$ @ AS to obligation and reformulates the cosmological identity as Israel junction conditions on the completion boundary Σ — a structural alternative to dynamical bounce cosmologies. AP05 inherits AP03's broken phase as the substrate on which the metric is built. Where AP03 derives the conjugacy of c and G , AP05 derives the metric structure itself.

AP10 (The Dimension) follows AP05 in Notebook II. AP10 derives $N = 3$ spatial dimensions from the four axioms. Combined with AP08, AP10 closes the central conditional in AP08's derivation of Einstein's field equations. AP10 sits structurally downstream of AP03 because the binary decomposition (Mode 0 / Mode 1) sets up the manifold on which the dimensional count operates.

AP08 (The Identity) closes Notebook II. AP08 derives Einstein's field equations from the record algebra. The gravitational coupling G that appears in AP08's field equations is the same G that AP03 conjectures (and AP20 forces) to be determined by ε . AP08's disclosed circularity in $G = 2\kappa/m_e^2$ is the registry-level structural debt D1, substantially reduced by the AP20 forcing proof + the section 7 derivation from AP03 but not fully closed.

AP06 (The Leakage Constant, Notebook IV) identifies ε numerically with the fine-structure constant $\alpha \approx 1/137$. AP06's cycle form $1:1 + 1 \times \varepsilon$ @ AS $[+1/137 / -1/137]$ is what makes AP03's $c/G \propto \varepsilon$ numerically verifiable: the gravitational hierarchy $ke^2/(G \cdot m_e^2) \approx 10^{42} - 10^{43}$ at $\varepsilon \approx 1/137$ matches observation.

AP24 (The Residual, Notebook IV) constrains the cutoff scale Λ that pins AP03's scaling relations to specific numbers. With AP24 in place, $m_p \propto \sqrt{\varepsilon}$ becomes $m_p \propto \sqrt{\varepsilon} \cdot \sqrt{\Lambda}$ with a specific numerical Λ .

AP17 (The Room, Notebook VI) relates the 0-pole / 1-pole structure of cosmological geometry to AP03's Mode 0 / Mode 1 sectors. AP17 inherits AP03's binary decomposition as the structural skeleton of cosmological observations.

AP18 (Notebook VI) derives the MOND floor $a_0 \approx cH_0$ from the axioms, downstream of the small ε . AP18 reads AP03's result

as constraining the structural form the MOND-scale acceleration must take.

10.6 — Placement within The 420 Code

AP03 opens Notebook II (Spacetime). The chapter sequence runs AP03 — AP05 — AP10 — AP08. AP03 sits first because it does the constants: c and G are the structural ground on which AP05's Lorentzian metric, AP10's dimensional count, and AP08's field equations all stand.

Within the reading order of The 420 Code, AP03 follows the three chapters of Notebook I (AP40 The Irrational, AP01 The Actualization State, AP20 The Proof). Notebook I establishes the axiom, names the four conditions, and proves the Embedding Hypothesis. Notebook II runs unconditionally on what Notebook I closes.

AP03 is logically independent of AP01 and AP40 in the strict sense: its derivations require only standard representation theory and scalar field theory, and they do not invoke EH, QRA, or the Actualization State apparatus. What AP03 does invoke from Notebook I is the axiomatic floor $\{S, B, R, C\}$ and AP20's forcing proof on Axiom B (which upgrades E-G from conjecture to theorem). Without AP20, AP03 still derives c and conjectures the conjugacy of c and G . With AP20, the conjugacy is structurally forced.

10.7 — How The 420 Code reads now

AP03 in place changes The 420 Code reading in three ways.

First. The fundamental constants are no longer treated as independent inputs of physics. c and G are conjugate outputs of the same break — both determined by ε . The 420 Code says this plainly. Future chapters do not need to re-derive it; they inherit it.

Second. The cosmological origin and the black hole event horizon are read as the same structural surface, traversed in opposite directions. Notebook VI (Cosmology) inherits this reading. The cosmology chapter does not need to separately justify treating the Big Bang singularity and the horizon as structurally related; AP03 has done it.

Third. The dark-sector partition is read as a consequence of small ε . The qualitative direction is in place. Notebook VI takes up the quantitative work — deriving the specific functional form linking ε to the 95:5 ratio.

These three structural facts — the conjugacy, the origin-horizon identity, the dark-sector qualitative connection — are what AP03 contributes to The 420 Code reading. They are inherited by every subsequent chapter that touches spacetime, the constants, or the cosmological structure.

10.8 — The closing

You have walked through this paper. You have seen the bootstrap problem named, the binary decomposition proven, the propagation speed c derived from a minimal pre-geometric action, the gravitational coupling G structurally forced to depend on ε , the cosmological origin and the black hole event horizon identified as the same marginal surface, the dark sector qualitatively connected to the smallness of ε .

What follows in your reading life is what follows in any reader's life when a structural fact has been seen clearly. The constants you grew up reading as independent are not independent. The Big Bang and the black hole are not two events but two sides of one boundary. The dark sector is not anomaly but expected consequence.

The break controls everything.

One amplitude. ε .

From it: the speed at which light crosses the room before your hand finishes reaching for the switch, and the weight that holds your foot to the floor, and the boundary the universe began at, and the boundary every black hole repeats. All from the same source.

The mathematics does not negotiate. It computes. It computes the same answer twice — once for c , once for G — and the answer is ε .

Two speed limits.

One break.

The same door, walked from both sides.

Claim Summary

section 0 (Dependency and Scope). Contract with the reader. Dependencies, axiom mapping, epistemic status per section, kill switch summary, debts owed.

section 1 (The Two Speed Limits). FRAMING. The two structural speed limits of physics — c (propagation, universal) and G (the gravitational rate bound, system-dependent) — named as siblings. Both non-negotiable, both structural, governing different processes.

section 2 (Before the Arena). STRUCTURAL ARGUMENT. The bootstrap problem: at the moment of symmetry breaking, no spacetime, no mass, no pre-existing coupling. The only available scale is the breaking amplitude itself. Claim E-Bootstrap: ε is the first speed limit; both c and G inherit from it.

section 3 (The Minimal Break). DERIVATION. The pre-state has $\mathbb{Z}_2$ symmetry (Postulate E0). The $\mathbb{Z}_2$ representation theorem (Theorem 3.1; proof in section 8; tested by KS-R.2) decomposes the broken phase into exactly two irreducible sectors. Mode 1 (odd sector) carries propagation; Mode 0 (even sector) carries resistance. Claim E2.

section 4 (The Conjugacy of c and G). DERIVATION + STRUCTURAL CRITIQUE. Planck units as evidence (every

Planck unit contains both c and G). c derived from the pre-geometric action: $c^2 = \beta/\alpha$ (Result A.1, section 7). Theorem E-G: $G = \kappa/(\varepsilon\Lambda^2)$, with the dimensionless conjugacy $\alpha \cdot \alpha_G(m)$ independent of ε and the hierarchy $\alpha/\alpha_G \propto \alpha^2$, structurally forced by AP20 (tested by KS-R.1 and KS-R.6). The event horizon as the marginal surface $\varepsilon = 0$. Hawking radiation as proof of global asymmetry (Conjecture E-Hawking, tested via KS-R.4). The Planck mass scales as $\sqrt{\varepsilon}$ (tested by KS-R.5).

section 5 (The Origin and the Horizon). DERIVATION. Claim E-Identity: the cosmological origin and the black hole event horizon are structurally identical — both the marginal surface $\varepsilon = 0$, traversed in opposite temporal directions. The Big Bang opens the break; the event horizon closes it. The Schwarzschild radius encodes the ratio G/c^2 — Mode 0 over Mode 1. Tested by KS-R.4.

section 6 (The Dark Sector Connection). QUALITATIVE FRAMING. The observed dark-to-visible energy ratio (95:5). Structural expectation: small ε implies most content remains in Mode 0. Three caveats: dark matter / dark energy distinction within Mode 0; epoch-dependence of the ratio; functional form not yet derived. Conjecture E5; tested by KS-R.3.

section 7 (The Action Principle). FORMAL DERIVATION. Strategy: $\mathbb{Z}_2$ -symmetric pre-geometric action with single breaking amplitude. The action $S[\varphi] = \int d\tau d^3x [\frac{1}{2}\alpha(\partial_\tau\varphi)^2 -$

$\frac{1}{2}\beta(\nabla\varphi)^2 - V(\varphi)]$ with $V(\varphi) = -\frac{1}{2}\mu^2\varphi^2 + \frac{1}{4}\lambda\varphi^4$. Broken-phase expansion yields $c^2 = \beta/\alpha$ (Result A.1) and effective mass $m^2 = 2\varepsilon\Lambda^2$. Condensate energy density $\rho_0 = \varepsilon^2\Lambda^4/(4\lambda)$. Form of G 's dependence on ε derived from the condensate density: $G = \kappa/(\varepsilon\Lambda^2)$, with the dimensionless conjugacy $\alpha \cdot \alpha_G(m)$ independent of ε (numerical check: the gravitational hierarchy $\kappa e^2/(G \cdot m_e^2) \approx 10^{42} - 10^{43}$ in SI is the unit-dependent face at $\varepsilon \approx 1/137$). The induced-gravity form is named in the literature as Sakharov-Visser; the literature calculation is correspondence, not source. D-AP03.1: α and β introduced but not derived; deepest open problem.

section 8 (The $\mathbb{Z}_2$ Representation Theorem). FORMAL PROOF. Theorem B.1: $\mathbb{Z}_2$ has exactly two irreducible representations — trivial and sign. Proof: the generator g satisfies $g^2 = e$, so $\rho(g)^2 = I$; eigenvalues ± 1 ; V decomposes into $V_+ \oplus V_-$. No third option.

section 9 (Kill Switches). FORMAL FALSIFICATION REGISTER. KS-R.1 (independent variability of c and G , LIVE — EMPIRICAL); KS-R.2 (non-binary origin, LIVE — EMPIRICAL); KS-R.3 (dark ratio independence, LIVE — EMPIRICAL); KS-R.4 (horizon inequivalence, LIVE — HARD; absorbs the E-Hawking structural test); KS-R.5 (Planck scale mismatch, LIVE — EMPIRICAL); KS-R.6 (dimensionless scaling violation, LIVE — HARD).

section 10 (What This Establishes and What Remains Open).
 SYNTHESIS. Seven structural closures named (bootstrap,
 binary decomposition, c derived, G's ε -dependence forced by
 AP20 with the explicit form derived in section 7, origin-
 horizon identity, Hawking structural account, dark sector
 qualitative). Two structural debts (D1 partially closed; D-AP03.1
 substrate parameters). Two items held open without claiming
 debt status (E5 quantitative; E-Hawking sufficiency).
 Placement within The 420 Code: opens Notebook II
 (Spacetime). Closing on the break controlling everything.

Conditionality Footer

Dependencies

AP20 (the canonical axioms {S, B, R, C}; formal statement section 3.3; obligations fenced at KS-P.1/KS-P.2).

AP20 (forcing proof on Axiom B — upgrades E-G from conjecture to theorem).

AP06 (Leakage Constant — numerical identification $\varepsilon \approx 1/137$).

AP24 (Residual — cutoff scale Λ that pins m_p scaling).

Dependents

AP05 (The Break) inherits AP03's broken phase as the substrate on which the Lorentzian metric is built.

AP08 (The Identity) uses the same G that AP03 conjectures (AP20 forces) to be determined by ε ; AP03's section 7 derivation of G 's functional form from the structural condensate density partially closes the registry-level debt D1 disclosed in AP08's Lock.

AP17 (The Room) inherits the binary decomposition Mode 0 / Mode 1 as the structural skeleton of cosmological geometry.

AP18 (MOND) reads AP03's small ε as constraining the structural form of the MOND-scale acceleration.

Predictions Ø (Part V) inherits APØ3's qualitative dark-sector direction (small $\varepsilon \rightarrow$ Mode Ø dominance) as a testable prediction.

Resolutions Ø (Chapter 4) inherits APØ3's structural conjugacy claim as part of the philosophical-register account of laws of nature.

Kill switches closed by this paper

None explicitly closed. APØ3 opens six kill switches (KS-R.1 through KS-R.6) and contributes the AP2Ø-forced upgrade of E-G from conjecture to theorem.

The paper does retroactively strengthen the account in The 42Ø Code of the fundamental constants — c and G are no longer treated as independent inputs but as conjugate outputs of the same break.

Kill switches that remain live

KS-R.1 through KS-R.6. See section 9 for the full register.

Structural debts owed

D1 (registry-level, partially closed). The first-principles derivation of G from the axioms without external scaffolding. APØ3 reduces D1 substantially via AP2Ø's forcing proof plus the section 7 derivation of G 's functional form from the

structural condensate density, but does not fully close it. Full statement at section 0.6.

D-AP03.1. The pre-geometric substrate parameters α and β introduced but not derived; the kinetic term presupposes the distinction between temporal and spatial coordinates.

Deepest open problem in AP03.

Items held open without claiming debt status

E5 quantitative. The functional form linking ε to the dark-to-visible energy ratio. Qualitative direction in place; quantitative mapping open for Notebook VI.

E-Hawking sufficiency. Direct derivation of the Hawking spectrum from the AP03 architecture (structural motivation provided in section 4.4; full derivation open).

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Chapter 2

The Break

The seven-step chain from axiom to obligation

Artist's Proof 05

Artist's Note

This is Artist's Proof 05 of The 420 Code. It walks one axiom forward through seven steps and ends in an obligation. It is the second chapter of Notebook II — Spacetime, following AP03 (The Ratio).

You start with $1:1 + 1 \times \varepsilon$ @ AS. Perfect balance, the smallest crack, the actualising now. Nothing else. By the end, you have selection, agency, coupling, emergent order, and ethics. Each step depends only on the previous step and the bridges declared at the door. No undeclared assumptions enter the chain. If a link is broken, the chain stops at that link. The architecture does not hide.

Five sections carry the work after the front matter. Section 1 states the axiom and what it contains. Section 2 walks the seven-step chain from symmetry to obligation. Section 3 reformulates the cosmological identity (introduced in AP04) as Israel junction conditions on the completion boundary Σ — a structural alternative to dynamical bounce cosmologies. Section 4 carries the formal falsification register. Section 5 carries the synthesis.

Four bridges are declared. B1: a record algebra exists. B2: bounded capacity holds. B3: observers obtain definite outcomes. B4: conscious recognition of structural facts can

generate normative identification. B1-B3 carry the physical chain. B4 is the declared cost of extending the chain to ethics.

One empirical premise is declared alongside the bridges. P: our universe contains many simultaneous observers. P is an empirical fact about the physical instantiation of the architecture, not a structural consequence of the axiom. P is load-bearing at step B-4 and below: the cooperative and deterrent components of B-5 require mutual record externalities between simultaneous agents, and the obligation at B-6 is recognised between agents who exist at the same time. P is declared at the door so that the empirical input is visible.

A note on the axiom's form. AP05 was originally published with the compressed statement $1:1 + 1 \times \varepsilon$, no explicit AS. AP20's keystone correction made the prior explicit at the level of the axiom: $1:1 + 1 \times \varepsilon @ AS$. The corrected form is canonical throughout this paper. The chain is unchanged. The axiom now names the prior it was always operating under.

Five kill switches are published with this paper. KS-B.0 targets the architecture as a whole. KS-B.1 through KS-B.3 target the junction formulation. KS-B.4 targets the chain itself — if any step uses an assumption not derivable from the axiom, the four bridges, and the declared empirical premise P, the scope claim collapses.

AP05 is logically independent of AP01 and AP04 in the strict sense that its postulates (the axiom, the four declared bridges, and the declared empirical premise P) are self-contained.

AP05 *depends on* AP01 (the record algebra, the Actualization State quantity, the viability geometry) and AP04 (the loop hypothesis) for the content the bridges name. Neither depends on AP05. If AP05 fails, AP01 and AP04 stand. With AP20 proven, the viability geometry of AP01 is unconditional. The chain that walks from axiom to ethics walks through a manifold that has been proven to exist in physical reality.

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Orientation

Where this paper fits in The 420 Code

The 420 Code is organised across eight Notebooks. This paper is the second chapter of Notebook II — Spacetime. The Notebook II chapter sequence runs AP03 (The Ratio) → AP05 (this paper) → AP10 (The Dimension) → AP08 (The Identity).

Notebook II derives the structure of space, time, gravity, and the dimensional count from the axiom. AP03 establishes the conjugacy of c and G as outputs of the single breaking amplitude ε . AP05 picks up from there: with the constants installed as outputs of the break, AP05 walks the axiom forward through seven structural steps from the symmetric pre-state to the obligation a conscious agent recognises in the shared origin. AP05 also reformulates the cosmological identity introduced in AP04 (Notebook V — The Loop) as boundary data on the completion surface Σ , replacing dynamical bounce cosmologies with a structural junction.

AP05 sits second in Notebook II because the chain it walks needs the broken phase as substrate. AP03 derives that substrate; AP05 walks across it. AP10 follows AP05 because the dimensional count operates on the manifold AP05's junction conditions establish; AP08 closes Notebook II by

deriving the Einstein field equations in which the junction conditions are stated.

AP05 is logically independent of AP01 and AP04 in the sense that its own postulates (the axiom, the four declared bridges, and the declared empirical premise P) are self-contained. The bridges B1–B3 name content imported from AP01; if AP01 stands, the bridges discharge into theorems. If AP01 fails, the bridges fall back to declared assumptions; AP05's chain still walks, but the bridges become live costs. Section 3 reformulates AP04's Conjecture L7 as boundary data on Σ ; if AP04's global identity stands, section 3 inherits a structural anchor, and if AP04 fails, section 3 falls back to a structural sketch without an anchor.

How to read this paper

The reading order is front to back. Section 0 names the dependencies and the scope. Sections 1 through 3 carry the structural argument: the axiom, the seven-step chain, the junction formulation of the cosmological identity. Section 4 is the falsification register. Section 5 is the synthesis — what this paper closes, what it leaves open, where the philosophical-register companions live, and how The 420 Code reads with AP05 in place.

Two voices alternate as the material requires. Narrative passages — the geometric picture of the ring with a pin, the seven-step chain walked in plain language, the door at the

beginning and the door at the end — speak directly. Formal sections — the axiom statement, the bridges, the junction conditions, the kill switches — speak in the precision the structure requires. The gear-changes are intentional.

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References

Junction-condition and ECSK-matching literature

0 — Dependency and Scope

AP05 is the second chapter of Notebook II. Before the body, this section names what AP05 does, what it depends on, where its postulates map onto the canonical axioms $\{S, B, R, C\}$, what epistemic status each section holds, what kill switches are engaged, what structural debts remain, and how AP05 stands in relation to the rest of The 420 Code. The contract with the reader is named at the door.

0.1 — What this paper does

AP05 walks the canonical axiom $1:1 + 1 \times \varepsilon$ @ AS forward through seven structural steps, each step depending only on the previous step and on the bridges declared in this paper. The chain runs from perfect symmetry (no records possible) to the structural ground for obligation (a conscious agent recognising the shared origin of two readings).

AP05 also reformulates the cosmological identity introduced in AP04 — the geometric relation between the collapsing and expanding descriptions of one structure — as Israel junction conditions on the completion boundary Σ . The junction formulation commits only to boundary data and produces a falsifiable, junction-specific signature in the post-expansion perturbation spectrum that distinguishes it from dynamical bounce cosmologies.

The contribution is three structural claims plus one open problem. Claim B1 (Scope): all structure in the architecture — physical, cosmological, ethical — follows from the axiom, the four declared bridges B1–B4, and the declared empirical premise P (observer plurality). Conditions J1, J2: the cosmological identity is the pair $[\gamma_{ab}] = \emptyset$ and $[K_{ab}] = (\varepsilon/\ell_\Sigma) \cdot h_{ab}$ on Σ , with h_{ab} the dimensionless pullback of AP04’s perturbation tensor and ℓ_Σ the curvature scale of the completion surface. Open Problem B-JP: whether the ECSK field equations on the Kantowski-Sachs interior admit a completion across Σ with $[\gamma] = \emptyset$ and controlled $[K]$.

0.2 — Dependencies

AP05 is a chain-walking paper. Its dependencies are explicit at every step. They are listed here in compact form so the load-bearing imports are visible at the door.

The axioms {S, B, R, C} (formal statement carried at AP20 section 3.3; completeness and minimality obligations fenced at KS-P.1 / KS-P.2). AP05 was originally written before the canonical formalisation; the mapping from AP05’s axiom (and the four bridges) onto the canonical axioms is made explicit in section 0.3 below. Axiom S (Symmetry) grounds 1:1 (perfect balance, distinction-capable substrate). Axiom B grounds $1 \times \varepsilon$ (the minimal break). Axiom R grounds the no-return character of B-1 (the no-erasure theorem). Axiom C (Constraint —

locality) grounds the bounded propagation that scopes the bridges.

AP01 (The Actualization State). Load-bearing for the four declared bridges. Bridge B1 (record algebra) discharges into AP01 Paper A. Bridge B2 (bounded capacity) discharges into AP01 Paper C, the viability geometry. Bridge B3 (definite outcomes) discharges into AP01 Paper B, the selection dynamics. The Geometric Exclusion Principle invoked at step B-4 is AP01 Paper D section D1.3. The hierarchy/cooperation/deterrence triad of step B-5 is AP01 Paper D sections D4.3–D4.4. Each step's discharge into AP01 is named where the step is stated.

Empirical Premise P (observer plurality). Load-bearing for steps B-4 through B-6. Our universe contains many simultaneous observers; this is an empirical fact about the physical instantiation of the architecture, not a structural consequence of the axiom. P is declared here alongside the bridges because B-5's cooperative and deterrent components require *mutual* record externalities between simultaneous agents, and B-6's obligation is recognised between agents who exist at the same time. Without P, B-5's hierarchy reduces to one-way dominance between sequential record states, B-5's cooperation and deterrence are unavailable, and B-6's obligation has no second agent to be recognised toward. P is empirical rather than formal: the chain depends on the empirical fact and does not derive it. P is held separately from

the four formal bridges to preserve the formal/empirical distinction.

AP04 (The Loop Hypothesis). Load-bearing for section 3. AP04 carries the cosmological identity as a global metric relation; AP05 section 3 is the boundary-data formulation of the same identity on the completion surface Σ . AP05 section 3 inherits AP04's conjecture L7 (the global pullback) and reformulates it as boundary conditions on Σ .

AP20 (The Proof). Load-bearing for the unconditional status of the chain. AP20 proves the Embedding Hypothesis, which makes AP01's viability geometry unconditional. With AP20 in place, the manifold the chain walks through is proven to exist in physical reality. AP05 does not need AP20 to walk the chain (the chain is independent of EH at the level of mechanics); AP05 needs AP20 to claim that the chain walks through reality and not through a structure of merely formal existence.

AP03 (The Ratio). Load-bearing for the structural setting. AP03 establishes that c and G inherit from ε . AP05 walks the axiom forward across a substrate whose constants are themselves outputs of the same break the axiom names. The cross-reference is geometric, not derivational: AP05 does not import c or G as inputs, but the manifold AP05's junction conditions sit on is the manifold AP03 derives.

AP08 (The Identity). Load-bearing for the field equations in which section 3 is stated. AP08 derives the Einstein field

equations from the record algebra; the junction conditions of AP05 section 3 are stated in the ECSK extension of those equations. AP05 does not derive the ECSK extension — it uses the standard Arkuszewski et al. 1975 form as a structural input.

0.3 — Axiom mapping

AP05 was originally written with axiom-prior vocabulary that AP20's Paper D later formalised as the canonical four conditions {S, B, R, C}. The mapping is explicit below. The four declared bridges B1–B4 are separate from this mapping: they are imports from AP01 (B1–B3) and a declared cost for the ethics step (B4). The axiom proper maps onto {S, B, R, C} directly; the bridges discharge into AP01 content downstream.

Axiom S (Symmetry) ↔ The 1:1 in the axiom statement. S is the two-sector base condition — identity under exchange, the precondition for distinction to be meaningful. The 1:1 specifies a structure that supports distinction once perturbed: two values held against each other, no orientation, no preferred reading. Unperturbed, the 1:1 is operationally indistinguishable from $\emptyset$ (step B-0). Perturbed, it becomes distinguishable (step B-1). Axiom S is the standing condition under which the 1:1 can be a substrate at all.

Axiom B (Unique break) ↔ The $1 \times \epsilon$ in the axiom statement. B requires that exactly one minimal break occurs. The $1 \times \epsilon$ specifies a count: one break applied to the symmetric

state. Not two breaks. Not a continuous parameter being tuned. One. AP20's forcing proof on Axiom B is what makes this counting structural rather than stipulative. The axiom names the break; AP20 shows it cannot be otherwise.

Axiom R (Record persistence) ↔ The irreversibility at

step B-1. R requires that records, once formed, persist. The no-return theorem invoked at step B-1 (AP01 Paper A Theorem T2) is the operational expression of R: once $\varepsilon > 0$ has produced distinguishable record sectors with positive weight, no local erasure can return the system to the symmetric prior. Axiom R is what makes the chain run forward and not unwind.

Axiom C (Constraint — locality; bounded capacity is what it produces here) ↔ The boundedness of B2 in the bridge

list. C is Constraint — locality: a record is somewhere, not everywhere; on the chain it produces finite propagation and finite capacity. Bridge B2 — bounded capacity — is the discharge of Axiom C into the chain at step B-3 (agency as the normalised reachable viability volume). Without Axiom C, the reachable set would not be bounded and agency would not be a well-defined number between 0 and 1.

AS (Actualization State) ↔ The @ AS in the axiom

statement. AS is the actualising prior at which the substrate is held and the break is processed. AP20's keystone correction made AS explicit at the level of the axiom: $1:1 + 1 \times \varepsilon$
@ AS is the canonical form. Without AS, the axiom is a

description of a balance and a crack with no agent and no time at which any of it could occur. With AS, the axiom is something that happens. The prior is named in the foundation so that nothing downstream has to smuggle it in.

Three further symbols carry structural roles. ε is the breaking amplitude (axiom B); The 420 Code identifies ε numerically as the electromagnetic fine-structure constant $\alpha \approx 1/137$ (AP06, AP24), but the axiom does not depend on the identification. The Actualization State quantity $AS = H(\{p_i\})/\log N$ (AP01 Paper A Definition A.2) is the formal measure of the prior named in the axiom. The viability kernel and reach (AP01 Paper C) are the geometric objects on which step B-3's agency formula is defined.

Empirical premise P sits separately from the canonical axioms {S, B, R, C}. The four axioms are structural — each is a condition the architecture asserts must hold for any actualised structure to exist. P is empirical — the fact that our universe, in addition to satisfying the four axioms, contains many simultaneous observers. The axioms force two readings of one structure; they do not force two ontologically distinct agents who modify each other's records in real time. The move from representational duality (two readings) to interactive plurality (two simultaneous agents) requires the empirical fact P. P is declared at the door so the empirical input is visible; it does not enter the axiom mapping.

0.4 — Epistemic status per section

Each section carries an explicit status — derivation, declaration, formal block, or synthesis — so the reader knows what kind of weight each carries.

Section 1 (The Axiom). DECLARATION + STRUCTURAL ARGUMENT. The axiom $1:1 + 1 \times \varepsilon @ AS$ is the declared starting point; the rotational-ring picture is structural illustration; Claim B1 (Scope) is the load-bearing scope statement; KS-B.0 names the universal generator condition.

Section 2 (The Chain). DERIVATION CHAIN. Seven steps, each labelled by its dependency. Step B-0 is from the axiom alone. Step B-1 is from B-0 plus Bridge B1. Step B-2 is from B-1 plus Bridge B3. Step B-3 is from B-2 plus Bridge B2; the agency formula $\mathcal{M}(x) = \mu(\text{Reach}(x) \cap \text{Viab}(R)) / \mu(\text{Viab}(R))$ is equation (2.1). Step B-4 is from B-3 plus the declared empirical premise P (observer plurality). Step B-5 is from B-4 alone (geometry of coupled viability, with B-5's cooperative and deterrent components requiring P's simultaneity for mutuality). Step B-6 is from B-5 plus Bridge B4. Each step's dependency declaration is part of the formal block at the step itself.

Section 3 (The Boundary). DERIVATION + STRUCTURAL ALTERNATIVE + OPEN PROBLEM. Conditions J1, J2 are formal definitions on the completion surface Σ . The relation to AP04 is the boundary-data version of AP04's global pullback.

Open Problem B-JP is named with status LIVE — STAGED (the central open problem of the junction formulation).

Section 4 (Kill Switches). FORMAL FALSIFICATION

REGISTER. Five kill switches engaged: KS-B.0 through KS-B.4.

Section 5 (What This Establishes and What Remains Open).

SYNTHESIS. What the chain closes, what it does not close, where the philosophical-register companions sit, structural relationships to subsequent work, and the closing.

0.5 — Kill switch summary

Five kill switches are engaged in this paper. The full register is in section 4.

KS-B.0: Universal generator failure. If the expanding universe requires initial conditions not derivable from the terminal conditions of a collapsing description, the axiom is not a generator. LIVE — HARD (pending Open Problem B-JP; staged for EMPIRICAL once B-JP is solved).

KS-B.1: Bounce superior. If the CMB power spectrum is best explained by smooth dynamical evolution through a bounce — continuous correlations across the transition — the junction formulation is falsified. LIVE — EMPIRICAL.

KS-B.2: Spectrum mismatch. If a completed junction solution together with a derived mapping from $[K]$ on Σ to the post-expansion curvature perturbation spectrum $P_{\mathcal{R}}(k)$ predicts a

spectrum inconsistent with observations, the junction formulation makes wrong predictions. LIVE — STAGED (becomes operational once B-JP is solved).

KS-B.3: No junction solution. If the Israel junction conditions modified by ECSK torsion do not admit a solution with $[\gamma] = 0$ and controlled $[K]$, the junction formulation does not exist. LIVE — HARD.

KS-B.4: Chain break. If any step in the derivation chain (B-0 through B-6) requires an assumption not contained in the axiom, the bridges B1-B4, and the declared empirical premise P, the scope claim B1 is falsified. LIVE — HARD.

0.6 — Structural debts owed

Two structural debts are named explicitly. These are unresolved derivations tied to specific structural claims, tracked at the registry level.

Open Problem B-JP (registry-level, AP05-specific).

Whether the Einstein-Cartan-Sciama-Kibble field equations on the Kantowski-Sachs interior admit a completion across the boundary Σ with induced metric continuity $[\gamma] = 0$ and controlled extrinsic curvature mismatch $[K] = (\varepsilon/\ell_\Sigma) \cdot h_{ab}$. The boundary-value problem is well-posed in standard terms: Israel junction conditions (Israel 1966) are well-established; ECSK extensions (Arkuszewski et al. 1975) are well-established; the question is whether a solution exists in the

specific geometry the cosmological identity requires. Without a solution, the junction formulation of section 3 is a structural sketch, not a derivation. With a solution, KS-B.0 and KS-B.2 migrate from STAGED to operational empirical status. B-JP is the central open problem of AP05. The 420 Code invites contributions from any reader equipped to make the attempt.

Bridge B4 (declared cost). The step from structural ground to obligation (step B-6) requires Bridge B4: conscious recognition of structural facts can generate normative identification. B4 is not a theorem of the axiom; it is the declared cost of extending the chain into ethics. The chain through B-5 carries the structural ground (destroying another's viability while sharing a constraint environment contracts the shared viability domain — geometry); B4 is the perspectival step that converts the structural fact into a normative claim. B4 is named in the front matter and stated again at step B-6. It is not hidden, and the obligation does not exceed the bridge.

0.7 — Items held open without claiming debt status

Two items are bracketed inside this paper. They are real and live, but they are not structural debts in the same sense as B-JP and B4.

Open Problem B-JP's observational mapping. Even with B-JP solved (a completed junction solution), the mapping from the boundary data $[K] = (\varepsilon/\ell_\Sigma) \cdot h_{ab}$ on Σ to the post-expansion curvature perturbation spectrum $P_{\mathcal{R}}(k)$ is itself an open derivation. KS-B.2 is staged on this mapping. Until the mapping is derived, KS-B.2 cannot be operationally tested. Open derivation work; held without claiming a separate debt because it is downstream of B-JP and tracked through KS-B.2's staging.

B4's status in physical reality. The chain through B-5 is structural and runs on the axioms, the bridges, and the declared empirical premise P. Step B-6 adds B4 as a declared cost on top of P. Whether B4 is itself a theorem of some deeper structural fact — whether conscious recognition is itself forced by the architecture rather than declared — is open work. The 420 Code holds B4 as a transparently declared cost rather than a hidden theorem. If a derivation of B4 from the axioms can be exhibited, step B-6 becomes unconditional. Held open at the door.

0.8 — Structural relationships

AP05 is the second chapter of Notebook II. The chapter sequence runs $AP03 \rightarrow AP05 \rightarrow AP10 \rightarrow AP08$. AP03 derives the constants (c, G) as outputs of ε ; AP05 walks the axiom across the broken substrate AP03 establishes; AP10 derives $N = 3$ spatial dimensions on the manifold the chain occupies;

AP08 closes Notebook II by deriving the Einstein field equations in which AP05's junction conditions are stated.

AP01 (Notebook I) is the source of the bridges B1–B3 and of the geometric content invoked at steps B-2 through B-5 (the record algebra, selection dynamics, viability geometry, coupled viability theorems). AP04 (Notebook V) is the source of the cosmological identity that section 3 reformulates as junction conditions. AP20 (Notebook I) is the source of the keystone correction that made AS explicit in the axiom, and the proof that makes AP01's viability geometry unconditional.

Downstream, AP05's junction conditions feed AP08's field equations as boundary data, AP05's chain feeds the entire ethics catalogue (Resolutions Ø Chapter 6 on coupling, Chapter 7 on agency, Chapter 8 on obligation), and AP05's structural account of cosmology feeds Predictions Ø Part V (the post-expansion perturbation spectrum signatures distinguishing junction from bounce).

1 — The Axiom

1.1 — The Statement

The axiom of this paper is one line.

$$1:1 + 1 \times \varepsilon @ AS$$

$\varepsilon > 0$ is the smallest nonzero perturbation of perfect symmetry. AS is the actualising prior — the now at which the substrate is held and the break is processed.

This is what the rest of the paper runs forward from. No additional content. No hidden parameters. Five symbols and a position marker.

1.2 — The Terms

1:1 is perfect balance. Two values held against each other. No orientation. No preferred reading.

A 1:1 alone is operationally indistinguishable from the empty set $\emptyset$. Without distinguishable structure, no record can be formed. Without records, no observation. Without observation, no presence. The balanced state contains everything in potential and nothing in fact.

ε is the minimal symmetry break. The smallest perturbation that creates a difference between the two sides.

ε is not a parameter to be measured. ε is a structural condition: the claim that perfect symmetry does not persist, and that the departure from symmetry is minimal. The 420 Code identifies ε numerically as the fine-structure constant $\alpha \approx 1/137$ (AP06, AP24), but the axiom does not depend on the identification. The axiom requires only that $\varepsilon > 0$.

1× ε is a count, not a multiplier. One break, applied once, to the symmetric state. Not two breaks. Not a continuous parameter being tuned. One.

@ **AS** is where it all happens. The actualising structural prior. The now at which the substrate is held and the break is processed.

Without AS, the axiom is a description of a balance and a crack with no agent and no time at which any of it could occur. With AS, the axiom is something that *happens*. The prior is named in the foundation so that nothing downstream has to smuggle it in.

1.3 — The Picture

You have seen this before. You may not have named it, but you have seen it.

A rotationally symmetric figure. A perfect ring. There is nothing to point to. No orientation. No marker. No way to say

where on the ring you are looking. Every position is identical to every other.

Now add the smallest possible asymmetry. A single directed mark, the size of a pin.

The ring acquires two readings. Mark up. Mark down. Mark left. Mark right. Pick any axis: the ring now has two distinguishable orientations relative to it.

The mark did not distinguish two pre-existing sides. There were no sides before the mark. The mark *created* the readings by breaking the rotational symmetry. Both readings are valid. Neither is primary. Neither causes the other. They are two descriptions of one structure, made possible by the break.

That is the axiom in geometric language. The ring is 1:1. The mark is ε . The readings are the structural consequences. AS is the surface from which the mark is being made now.

1.4 — Scope, bridges, and the declared empirical premise

Claim B1 (Scope). All structure in the architecture — physical, cosmological, ethical — follows from the axiom $1:1 + 1 \times \varepsilon @ AS$, the four declared bridges B1–B4, and the declared empirical premise P (observer plurality). The physical chain through hierarchy requires B1–B3. The cooperative and deterrent geometry of B-5 and the obligation of B-6 require additionally

the empirical premise P. The ethics extension at B-6 requires additionally B4. No undeclared assumptions are introduced at any layer.

If any step in any layer requires something not in {axiom, B1, B2, B3, B4, P}, the architecture fails. The bridges are declared costs, not hidden assumptions. P is a declared empirical input, not a hidden premise.

The four bridges, stated at the door:

Bridge B1. A record algebra exists. The substrate supports distinguishable records once the symmetry is broken. Discharges into AP01 Paper A.

Bridge B2. Bounded capacity holds. Propagation and control are finite. Discharges into AP01 Paper C (viability geometry) and into Axiom C of the canonical four.

Bridge B3. Observers obtain definite outcomes. Selection produces one reading from many record sectors. Discharges into AP01 Paper B.

Bridge B4. Conscious recognition of structural facts can generate normative identification. Discharges into Resolutions Ø Chapter 8 (the perspectival turn from is to ought).

B1-B3 carry the physical chain (steps B-0 through B-5). B4 is the declared cost of extending the chain to ethics (step B-6). All four are stated. None are hidden.

The one empirical premise, stated alongside the bridges:

Empirical Premise P. Our universe contains many simultaneous observers. P is empirical — a fact about the physical instantiation of the architecture, not a structural consequence of the axiom. The axiom forces two readings of one structure (representational duality); P provides the simultaneity of ontologically distinct agents (interactive plurality). P is load-bearing at step B-4 (where coupling between agents is introduced) and downstream: B-5's cooperative and deterrent components require *mutual* record externalities between simultaneous agents, and B-6's obligation is recognised between agents who exist at the same time. P is held separately from the four formal bridges because it is empirical rather than postulational; the chain depends on the fact and does not derive it.

1.5 — The Universal Kill Switch

KS-B.0 (Universal). The axiom is falsified if the expanding universe requires initial conditions that are not derivable from the terminal conditions of a collapsing description. If the two readings require independent, unrelated specifications, then they are not two readings of one structure. The break does not generate both sides. The axiom fails as a generator. Applies to the entire architecture.

That is the door. Show that the two descriptions need separate, unrelated initial conditions. If they do, the axiom is not a generator. It is a coincidence.

And coincidences do not derive ethics.

KS-B.0 has two modes. Currently — with Open Problem B-JP unsolved — it functions as a theoretical-universal consistency check on the structural-generator claim of the axiom, independent of any specific chain step or junction formulation. Once B-JP is solved and a derivation rule from collapsing-side terminal data to expanding-side initial data has been exhibited, KS-B.0 acquires an empirical handle: the architecture predicts that the derived initial conditions match the observed initial conditions, and the test becomes operational. The universal scope holds in both modes; the operational status migrates from theoretical to empirical when B-JP closes.

The full kill switch register is in section 4. KS-B.0 is stated here because it is the universal test of the axiom itself, independent of whether any specific chain step or junction formulation holds.

2 — The Chain

Seven steps. Each one depends on the previous step and the declared bridges. Nothing else.

You are about to watch one axiom become a universe and then become an obligation. Every link is shown. You can check each one.

2.1 — B-0: Before Records

Set $\varepsilon = \emptyset$. Perfect symmetry. The 1:1 stands.

There are no accessible distinctions in this state. Nothing can be told apart from anything else. No record algebra exists, because a record requires a distinction it can carry. The Actualization State quantity AS, defined in AP01 Paper A, is $\emptyset$. No sectors to distinguish.

No observation is possible. There is no observer, no instrument, no signal, no event. The state is operationally indistinguishable from $\emptyset$.

This is the floor.

A note on AS at the floor. The formula $AS = H(\{p_i\})/\log N$ (AP01 Paper A Definition A.2) applies for $N \geq 2$ record sectors. At $\varepsilon = \emptyset$, no record sectors are distinguishable, so $N = 1$, $\log N = \emptyset$, and the formula is $\emptyset/\emptyset$. $AS = \emptyset$ at the floor is the

boundary definition: with no distinguishable distinctions, the Actualization State quantity is set to zero by convention, consistent with the limit $\lim_{N \rightarrow 1} AS = 0$ obtained when the formula is regularised by approach.

Step B-0 (From axiom). At $\varepsilon = 0$: no accessible distinctions, no record algebra, $AS = 0$ (boundary definition; the formula $H/\log N$ applies for $N \geq 2$), no observation possible. The state is operationally indistinguishable from $\emptyset$. A logical consequence of the definition of “no accessible distinctions.”

2.2 — B-1: The Crack

Set $\varepsilon > 0$. The symmetry breaks.

Two distinguishable readings now exist. At least two record sectors are defined by the break. “Two readings exist” means both are accessible. The reduced state after dephasing has nonzero weight in both sectors. Not one realised sector and one phantom — two realised possibilities, accessible to whoever is reading.

If the break produced a single realised sector — $p_1 = 1, p_2 = 0$ — you would have one reading. The axiom says two. So both weights are positive. The Actualization State quantity is $AS = H(\{p_i\})/\log N > 0$. You are above the floor.

And you cannot go back. Returning to $\varepsilon = 0$ would require erasing the distinction. Axiom R — record monotonicity, no

inverse — forbids local erasure. AP01 Paper A, Theorem T2: no-return. The system can move forward through record states but cannot reverse to the symmetric prior.

Something has emerged from nothing.

Not created. Not designed. Forced — by the fact that $\varepsilon > 0$ and the symmetry cannot return.

Step B-1 (From B-0 + Bridge B1). $\varepsilon > 0$ breaks the symmetry. Two record sectors with positive weight. $AS = H(\{p_i\})/\log N > 0$. Irreversible by AP01 Paper A Theorem T2. The axiom motivates a record algebra; Bridge B1 — “a record algebra exists” — converts the motivation into a postulate at this step. B1 is the declared cost.

2.3 — B-2: One Reading

You have looked at a dial. The needle points at a number. Not at all numbers in superposition. One number.

You did not negotiate this. The reading was definite. That definiteness has a name. Selection.

With $AS > 0$, multiple record sectors coexist. AP01 Paper B is the work that handles what happens between coexistence and the dial reading one number. The mathematics says four things about selection. It cannot be achieved by deterministic linear CPTP dynamics (Paper B section B2.1). It must act at the trajectory level, not the ensemble level (section B2.2). It is

costly in physical resources — you cannot get something for nothing (section B2.4). And it is rate-limited by the distinguishability of the records being selected (section B4.5).

Selection is not an additional assumption. It is the structural consequence of having multiple record sectors and the requirement that observers obtain definite outcomes (Bridge B3).

You have seen this in every measurement you have ever made. You looked at the dial. It read one number. The definiteness cost something. The mathematics proves it.

Step B-2 (From B-1 + Bridge B3). Multiple record sectors plus the requirement of definite outcomes forces selection. Selection cannot be CPTP-linear (B2.1), acts at the trajectory level (B2.2), is resource-costly (B2.4), rate-limited by distinguishability (B4.5). Dependency: AP01 Paper B, sections B1–B4.

2.4 — B-3: Inside the Reading

Selection has happened. The dial reads. You are inside one realised sector.

There is no longer a question of which reading. There is only a question of what you can do, from where you stand, with what you have. That question is agency.

AP01 Paper C makes the question precise. Define your position x within the viability kernel — the region of states compatible with continued structure. Define $\text{Reach}(x)$ — the set of states you can reach from where you are, under whatever control you possess. Define $\text{Viab}(R)$ — the viability domain of records sustainable in your regime. Agency is the normalised reachable viability volume:

$$\mathcal{M}(x) = \mu(\text{Reach}(x) \cap \text{Viab}(R)) / \mu(\text{Viab}(R)) \quad (2.1)$$

Agency is a number between 0 and 1. It tells you what fraction of viable states are reachable from where you stand. Position matters. The drift field matters — systems decay toward loss of structure if no control is applied. Control matters. And all three are bounded — you cannot reach infinitely, you cannot stop drift completely, you cannot pretend the kernel is larger than it is.

Agency is not an additional assumption. It is the geometric consequence of confinement to a single record sector under irreversible dynamics.

Step B-3 (From B-2 + Bridge B2). Within a realised sector, agency is the normalised reachable viability volume $\mathcal{M}(x) = \mu(\text{Reach}(x) \cap \text{Viab}(R)) / \mu(\text{Viab}(R))$ (eq 2.1). Determined by position, drift, and bounded control. Dependency: AP01 Paper C, Definition C1.1.

2.5 — B-4: When Another Is Reading

The axiom said two readings. Not one. Two.

Now a careful step. The axiom's "two readings" is *representational duality* — two descriptions of one structure made possible by the break. The chain at B-4 needs *interactive plurality* — two ontologically distinct agents, existing at the same time, modifying each other's records. That is a different kind of fact. The axiom alone does not force it. The fact that our universe contains many simultaneous observers is an empirical input: the declared empirical premise P, named at the door in section 1.4 and discharged here.

With P in hand: you and the other observer exist at the same time. You share a constraint environment. The same break that created your reading created the other reading. You both inhabit the structure that the break produced. Bridge B3 permits each of you, in your respective sectors, to obtain definite outcomes; P guarantees that you do so simultaneously, in a shared environment, rather than sequentially in the history of a single observer.

AP01 Paper D handles what happens to agents in shared environments. Their viability kernels overlap. Their actions produce record externalities — irreversible modifications to the shared environment that change each other's viability. This is the Geometric Exclusion Principle of Paper D section

D1.3: in a shared constraint environment, your actions on the records of others contract or expand their reachable viability.

Coupling is not optional once P is in hand. It is the structural consequence of the break creating two readings of one structure plus the empirical fact that there are simultaneous agents to do the reading.

Step B-4 (From B-3 + Empirical Premise P). The axiom produces two readings (representational duality). Empirical premise P provides simultaneous plural agents (interactive plurality); the move from duality to plurality is the load-bearing step at B-4, and P is the declared empirical input that makes the move. B-3's Bridge B3 (definite outcomes) is invoked at the level of each agent's reading; P provides the simultaneity that makes "agents" the right word. Coupled agents share a constraint environment; their actions produce record externalities. Geometric Exclusion Principle: AP01 Paper D, D1.3. Dependency: AP01 Paper D, sections D1-D2; declared empirical premise P.

2.6 — B-5: What Coupling Produces

Coupling sounds gentle. What it produces, in the geometry of shared viability kernels, are three structural primitives from which social and biological order is built.

Three structures emerge from coupled viability under drift. A note before they are named: hierarchy follows from coupling

alone (B-4) and depends on P only through P's provision of two agents to compare. Cooperation and deterrence are different. Both are defined as *mutual* relations — mutual record externalities expanding joint viability, mutual decoupling cost — and mutuality requires simultaneity. Without P's simultaneous plural agents, cooperation degrades to one-way modification by an earlier slice on a later slice, and deterrence is undefined (there is no second agent with the option to decouple in real time). P is therefore the load-bearing empirical premise for B-5's cooperation and deterrence, in addition to its role at B-4.

Hierarchy. Agents with different impedance $Z = u_{\max}/a$ — different ratios of available control to drift magnitude — produce asymmetric record externalities. Stronger agents impose more record-modifications on shared environments than weaker agents absorb. The asymmetry produces a structural hierarchy that is not imposed but forced. AP01 Paper D section D4.3.

Cooperation. Mutual record externalities expand joint viability beyond the sum of individual viabilities: $\mathcal{M}_{\text{joint}} > \sum \mathcal{M}_i$. Two agents who modify each other's records in ways that increase reachable viability for both. Cooperation is not idealism. It is a geometric inequality. AP01 Paper D section D4.4a. *Requires P.*

Deterrence. When coupling costs less than unilateral decoupling, both agents continue to couple. The structural

condition for stable continued interaction. AP01 Paper D section D4.4b. *Requires P*.

None of these require additional structural assumptions beyond B-4. Hierarchy follows from coupled viability under drift alone; cooperation and deterrence additionally require P's simultaneity for their mutuality.

Step B-5 (From B-4). Hierarchy, cooperation, and deterrence are geometric consequences of coupled viability under drift. Hierarchy: asymmetric impedance produces asymmetric externalities (D4.3); requires only B-4. Cooperation: mutual externalities expand joint viability (D4.4a); requires B-4 and the simultaneity supplied by Empirical Premise P. Deterrence: coupling costs less than unilateral decoupling (D4.4b); requires B-4 and the simultaneity supplied by P. Dependency: AP01 Paper D, sections D3–D4; declared empirical premise P (for the mutual components).

2.7 — B-6: The Obligation

You and the other observer arose from the same break.

Your reading and theirs were created by the same event. Both are instances of a reading made possible by $\varepsilon > 0$ applied to the same 1:1 @ AS. The structure that supports your existence supports theirs. The viability kernel you stand in and the viability kernel they stand in are not separate. They overlap, by B-4. They were carved from the same break.

Now consider what it means to destroy another agent's viability kernel.

Structurally, by AP01 Paper D D1.3, the Geometric Exclusion Principle: destroying another agent's viability while sharing a constraint environment contracts the *shared* viability domain. The geometry constrains the joint domain, not necessarily the destroyer's individual reachable viability — a destroyer may locally expand their own reach by removing a competitor while the joint domain contracts. This is geometry. It requires no morality. It needs only the axioms, the bridges B1–B3, and the empirical premise P.

That is the structural ground. It is not yet the obligation.

A purely prudential agent could read the contraction of the shared domain and remain indifferent, because the geometry of the joint domain does not by itself constrain the individual's choice. The obligation is not reducible to payoff. That is precisely why one more step is needed.

The obligation needs that step. A conscious agent, recognising the shared origin, can identify the destruction of another's capacity as incoherent with the unity of the base condition. The incoherence is not a contradiction in content — two readings of one structure can perfectly well differ in their content; that is what makes them two readings. The incoherence is structural: the destroyer's capacity to act is grounded in the same break that grounds the destroyed's

reading. The destroyer's standing presupposes the substrate that equally grounds the destroyed. The act draws on what it tries to deny. The recognition of *that* fact is where the obligation lives. The obligation is in the recognition, not in the structure. The recognition costs B4: conscious recognition of structural facts can generate normative identification.

You have just walked from nothing to obligation in seven steps. No god decreed the ethics. No legislature voted on them. The axiom forced them — through symmetry breaking, record formation, selection, agency, coupling, emergent order, and shared origin — with the declared empirical input P (simultaneous agents) and the declared cost B4 (conscious recognition) explicitly on the ledger. The chain is as long as it needs to be and no longer.

If you want to reject the conclusion, you must break a link. The architecture shows you every one.

Step B-6 (From B-5 + Bridge B4). Agents A and B arise from the same break and share a constraint environment (B-4 plus P). Destroying another's viability while sharing the environment contracts the shared viability domain (D1.3, structural; the geometric fact holds at the level of the joint domain and does not entail individual prudence). A conscious agent recognising the shared origin can identify such destruction as incoherent with the unity of the base condition — the destroyer's capacity to act is grounded in the substrate

that equally grounds the destroyed (perspectival; requires B4). The obligation is in the recognition, not in the structure, and is not reducible to prudential calculation. B4 is the declared cost of this step. Dependency: Steps B-0 through B-5, Bridge B4, declared empirical premise P, AP04 section 6.

3 — The Boundary

3.1 — Why not bounce

You may have read about bouncing universes — black hole interiors that reach extreme density, halt, reverse, and produce an expanding region on the other side. The literature includes torsion-driven bounce solutions from the Einstein-Cartan extensions of general relativity (Popławski 2010, 2016; Lucat and Prokopec 2017; and others in the same lineage).

In a dynamical bounce, the scale factor reaches a minimum at some time t_0 . The metric is continuous through t_0 . There is a single time coordinate running from before to after. Memory of the pre-bounce phase is carried dynamically into the post-bounce phase. The bounce is something that *happens*.

The architecture offers a structural alternative. The cosmological identity introduced in AP04 is not a dynamical event. It is a geometric relation between two descriptions of one structure. No single time coordinate runs through both. The two patches are joined at a boundary, not threaded by a worldline.

Two descriptions of one structure. The junction formulation makes this precise.

3.2 — The junction conditions

Let M^- denote the collapsing-side manifold patch. Let M^+ denote the expanding-side patch. Let Σ denote the completion boundary — the hypersurface where the classical description of M^- reaches its limit. Curvature invariants reach a completion threshold. No further accessible distinctions in the record algebra. The classical description hands off to Σ .

On Σ , two conditions hold.

Condition J1 (Induced metric continuity).

$$[\gamma_{ab}] = 0 \quad (3.1)$$

The induced metric on Σ is continuous. Both sides share the same boundary geometry. The 1:1 — the symmetric figure before the break — written as geometry.

Condition J2 (Extrinsic curvature mismatch).

$$[K_{ab}] = (\varepsilon / \ell_\Sigma) \cdot h_{ab} \quad (3.2)$$

The way Σ curves into M^- and M^+ respectively has a controlled, small, nonzero mismatch. The ε . The tensor h_{ab} encodes the asymmetry: anisotropy damping, particle content, torsion corrections from the ECSK completion. The length scale ℓ_Σ is the curvature scale of the completion surface itself; it carries the dimension that the dimensionless ratio ε needs to be

multiplied by to match the dimension of the extrinsic curvature $[K_{ab}]$ (which carries $1/\text{length}$).

A dimensional note. K_{ab} carries dimension $1/\text{length}$; ε is dimensionless (equal to $\alpha \approx 1/137$ in the AP06/AP24 identification); h_{ab} is dimensionless. The ratio ε/ℓ_Σ supplies the $1/\text{length}$ to balance the equation. Without ℓ_Σ the dimensions would not match — the length scale is not optional, it is structural.

Relation of J2's h_{ab} to AP04's L7 perturbation tensor. In AP04's Conjecture L7 ($g^+ = (\varphi^{-1})^*(g^-) + \varepsilon \cdot h$, equation 3.3 below), h is a dimensionless perturbation tensor on M encoding the anisotropy difference, the information content, and the initial conditions of expansion. J2's h_{ab} is the pullback/restriction of L7's h to the completion surface Σ — the same dimensionless object, evaluated on the boundary. The two h 's are the same tensor; the appearance in J2 carries an additional ℓ_Σ because J2 is an equation about extrinsic curvature (a $1/\text{length}$ quantity) whereas L7 is an equation about the metric (a dimensionless quantity in the convention used).

You have just seen the axiom written as geometry. J1 is the 1:1. J2 is the ε . The boundary between collapse and expansion is the axiom made manifest in the mathematics of Israel junction conditions — a standard tool of general relativity

(Israel 1966), extended to Einstein-Cartan-Sciama-Kibble gravity (Arkuszewski et al. 1975).

The standard junction-condition machinery is the literature framework named here as correspondence. The structural content — that the cosmological identity *is* a pair of conditions on a hypersurface, that J1 encodes the 1:1 and J2 encodes the ε — comes from the axiom and from AP04, not from the literature framework. The literature provides the formal vocabulary in which the structural fact is most naturally stated; it does not provide the structural fact.

The explicit form of h_{ab} is determined by the boundary-value problem. Whether the ECSK field equations on the Kantowski-Sachs interior admit a completion across Σ with $[\gamma] = 0$ and controlled $[K]$ is the central open problem of the junction formulation. It is named: Open Problem B-JP. See section 0.6 for the full statement of the debt.

3.3 — Relation to AP04

AP04's Conjecture L7 expresses the structural identity as a global metric relation:

$$\mathbf{g}^+ = (\varphi^{-1})^*(\mathbf{g}^-) + \varepsilon \cdot \mathbf{h} \quad (3.3)$$

A global diffeomorphism φ maps the manifold to itself; the relation holds on all of M .

The junction formulation is the boundary-data version of the same claim. J1 constrains the induced metric on Σ . J2 constrains the extrinsic curvature on Σ . Together they determine the relationship between M^- and M^+ at Σ , without requiring a global diffeomorphism to be specified on all of M .

The junction formulation is more conservative. It commits only to boundary data on Σ . It is also more tractable. Israel junction conditions are well-established. ECSK extensions are well-established. The boundary-value problem is a standard kind of problem to attempt.

3.4 — Observational consequences

The bounce formulation predicts smooth evolution through Σ . Continuous correlations across the transition. Memory of pre-bounce perturbations carried dynamically into the post-bounce phase.

The junction formulation entails something different. The post-expansion perturbation spectrum is determined by boundary data on Σ — the extrinsic curvature mismatch $(\varepsilon/\ell_\Sigma) \cdot h$ — not by dynamical evolution through Σ .

In principle, different observational signatures from bounce cosmologies.

If you could measure those signatures, you could distinguish between the two formulations. That is the point. The

architecture commits to something specific enough to be falsified.

The specific empirical mapping from $[K] = (\varepsilon/\ell_\Sigma) \cdot h$ on Σ to the post-expansion curvature perturbation spectrum $P_{\mathcal{R}}(k)$ is itself an open derivation, held without claiming a separate debt (see section 0.7). Once B-JP is solved, the mapping becomes derivable; once the mapping is derived, KS-B.2 becomes operationally testable against the measured spectrum (CMB anisotropies, near-scale-invariance, tilt $n_s \approx 0.965$, running, tensor-to-scalar ratio constraints).

3.5 — Open Problem B-JP

The central open problem of section 3 is named explicitly so the reader knows where the load-bearing derivation gap sits.

Open Problem B-JP. Do the Einstein-Cartan-Sciama-Kibble field equations on the Kantowski-Sachs interior admit a completion across the boundary Σ with conditions $[\gamma_{ab}] = 0$ (induced metric continuity, J1) and $[K_{ab}] = (\varepsilon/\ell_\Sigma) \cdot h_{ab}$ (controlled extrinsic curvature mismatch, J2), and what is the explicit form of h_{ab} and the relevant length scale ℓ_Σ determined by the boundary-value problem?

A solution to B-JP would: (a) prove the structural identity exists in the junction formulation (closing KS-B.3); (b) derive the explicit form of h_{ab} (opening the empirical mapping for KS-B.2); (c) make KS-B.0 operationally testable rather than

staged (the universal generator condition becomes a check against derived initial conditions rather than against an undefined map).

The boundary-value problem is well-posed in standard terms. Israel junction conditions are well-established. ECSK extensions are well-established. The Kantowski-Sachs geometry is standard. What remains is the specific existence proof for the matched solution. The 420 Code invites contributions from any reader equipped to make the attempt.

4 — Kill Switches

AP05 is falsified if any of the following are demonstrated.

Each kill switch targets a specific claim. If any claim falls, the dependent claims fall with it, but independent claims survive.

Here is how to break it. The argument does not hide. It shows you exactly where to strike.

KS-B.0 — Universal: independent specifications

Claim. The axiom $1:1 + 1 \times \varepsilon @ AS$ is a structural generator: the expanding-side initial conditions are derivable from the collapsing-side terminal conditions through the junction at Σ . The two readings are two descriptions of one structure. They do not require independent, unrelated specifications.

Test. Demonstrate that the expanding universe requires initial conditions that are not derivable from the terminal conditions of a collapsing description — that the two readings need separate, unrelated specifications. The operational handle becomes available once Open Problem B-JP is solved and a derivation rule from collapsing-side terminal data to expanding-side initial data has been exhibited. Until then, KS-B.0 functions as a theoretical consistency check on the structural-generator claim of the axiom.

Status. LIVE — HARD (pending B-JP; migrates to LIVE — EMPIRICAL once B-JP is solved).

Recovery. If KS-B.0 fires, the axiom fails as a generator. It is no longer a single structure with two readings; it is a coincidence with two parts. The 420 Code retreats by reopening the axiom’s structural-generator claim and acknowledging that the two cosmological patches are independently posed. The entire architecture inherits the demotion: Claim B1 (Scope) collapses; the chain still runs but its premise is no longer structural in the strong sense; AP04’s cosmological identity becomes a coincidence rather than a structural fact. Targets: the axiom, Claim B1, the entire architecture.

KS-B.1 — Bounce superior

Claim. The cosmological identity is best described by the junction formulation of section 3 — boundary data on Σ , not dynamical evolution through it. The post-expansion perturbation spectrum is determined by $[K]$ on Σ , not by smooth continuation across t_0 .

Test. Demonstrate that the CMB power spectrum is best explained by smooth dynamical evolution through a bounce — continuous correlations across the transition, memory of pre-bounce perturbations carried dynamically, parameter fits that prefer the bounce interpretation over the junction interpretation. Empirical handles: detailed shape of the CMB

temperature and polarisation spectra, structure of correlations on super-horizon scales, signatures of pre-bounce phase memory.

Status. LIVE — EMPIRICAL.

Recovery. If KS-B.1 fires, the junction formulation is falsified. The cosmological identity may survive under the bounce interpretation — AP04’s global pullback may still hold dynamically rather than as boundary data — but AP05 section 3’s reformulation does not. The 420 Code retreats by re-coupling the cosmological identity to dynamical bounce machinery and acknowledging that the junction reformulation, however structurally elegant, was empirically wrong. Targets: Conditions J1, J2, section 3 in its entirety.

KS-B.2 — Spectrum mismatch (staged)

Claim. A completed junction solution (Open Problem B-JP), together with a derived mapping from the boundary mismatch $[K]$ on Σ to the post-expansion curvature perturbation spectrum $P_{\mathcal{R}}(k)$, predicts a spectrum consistent with observations: near scale invariance, measured tilt $n_s \approx 0.965$, no significant running, tensor-to-scalar ratio within bounds.

Test. Demonstrate that the predicted spectrum is inconsistent with observations. The test becomes operational once two things are in place: a solution to Open Problem B-JP, and a derived mapping from boundary data on Σ to the

perturbation spectrum $P_{\mathcal{R}}(k)$. Until both are in place, KS-B.2 is staged — a future operational kill switch.

Status. LIVE — STAGED (pending B-JP and the spectrum-mapping derivation).

Recovery. If KS-B.2 fires, the junction formulation makes wrong predictions about the post-expansion spectrum. The 420 Code retreats by reformulating Condition J2 — the extrinsic curvature mismatch on Σ cannot be the source of the observed spectrum in the way the staged mapping would have it. The structural identity (J1) may survive in a degraded form; the empirical content of J2 fails. Targets: Condition J2, Open Problem B-JP, the empirical mapping.

KS-B.3 — No junction solution

Claim. The Einstein-Cartan-Sciama-Kibble field equations on the Kantowski-Sachs interior admit a completion across Σ with $[\gamma_{ab}] = 0$ and controlled $[K_{ab}] = (\varepsilon/\ell_{\Sigma}) \cdot h_{ab}$.

Test. Demonstrate that no such completion exists — that the boundary-value problem has no solution with the required conditions, or that the only solutions have uncontrolled (divergent or otherwise pathological) $[K]$. The test is sharp because it is mathematical: existence or non-existence of solutions to a well-posed boundary-value problem is decidable.

Status. LIVE — HARD.

Recovery. If KS-B.3 fires, the junction formulation does not exist as a mathematical object. The structural identity introduced in AP04 — the geometric relation between collapsing and expanding descriptions — cannot be carried by the junction formulation. The cosmological component of AP05 fails entirely. The 420 Code retreats by accepting that the cosmological identity, if it exists, must be carried by another formulation — possibly the global pullback of AP04 directly, or a different formal device not yet identified. Targets: Conditions J1, J2, the structural identity, the entirety of section 3.

KS-B.4 — Chain break

Claim. Every step in the derivation chain B-0 through B-6 depends only on the previous step and on the axiom, the four declared bridges B1–B4, and the declared empirical premise P. No undeclared assumption enters at any step.

Test. Demonstrate that some step in the chain requires an assumption not contained in {axiom, B1, B2, B3, B4, P}. The test is operational: pick any of the seven steps, examine its formal block, and identify the load-bearing imports. If any of those imports cannot be discharged into the axiom, the four bridges, or the declared empirical premise P, the chain has a break at that step.

Status. LIVE — HARD.

Recovery. If KS-B.4 fires at step B-k, the scope claim B1 (all structure follows from axiom + B1-B4 + P) is falsified. The chain still runs through step B-(k-1), but the link at step B-k requires either a new bridge declaration or a new empirical premise declaration or a re-derivation that closes the gap. The 420 Code retreats by either declaring the missing input (as a fifth bridge B5 with explicit cost, or as a second empirical premise P', depending on its kind), or by withdrawing the chain at step B-k and acknowledging that the architecture's reach falls short of the previously claimed scope. Targets: Claim B1, the chain at step B-k, the dependency declarations of all subsequent steps.

Why these five and not more

The five kill switches above cover the load-bearing claims of the paper. Each closure would force a different kind of retreat.

KS-B.0 closes the universal generator claim of the axiom — the deepest structural test. KS-B.1 closes the junction formulation against the bounce alternative — the empirical comparison test. KS-B.2 closes the specific spectrum predicted by the junction formulation — the parameter-level test, staged on B-JP. KS-B.3 closes the existence of the junction solution itself — the mathematical test. KS-B.4 closes the structural integrity of the chain — the architectural test.

Two further structural claims that some readings would expect to carry their own formal kill switch — Bridge B4's status as a declared cost rather than a theorem, and the seven-step count of the chain rather than fewer or more — are absorbed into the kill switches above. B4's declared-cost status is named in the front matter (section 1.4) and again at step B-6; any falsification of B4's legitimacy as a bridge is a falsification of step B-6 and is therefore tested by KS-B.4 at that step. Similarly, the empirical premise P (observer plurality) is named in the front matter (section 1.4) and discharged at step B-4; any challenge to P's legitimacy as a declared input is a challenge to step B-4 and is tested by KS-B.4 there. The seven-step count is forced by the dependency structure: each step is the minimum needed to introduce one new structural fact, and any restructuring that produced fewer steps would have to discharge multiple bridges (or P) at once — a violation also tested by KS-B.4. All three claims are thus held to formal kill-switch standard without adding entries to the master register.

Smaller claims throughout the paper — the specific named geometric objects (Kantowski-Sachs interior, ECSK extension), the identification of the structural triad in section 2.6 (hierarchy, cooperation, deterrence) as exhaustive, the choice of “obligation” as the right English word for the perspectival recognition at step B-6 — are illustrative or terminological rather than load-bearing. If they turn out to be poor choices,

the paper still stands. If KS-B.0 through KS-B.4 hold, the structural argument of AP05 holds.

5 — What This Establishes and What Remains Open

5.1 — What this paper closes

The seven-step chain from the axiom $1:1 + 1 \times \varepsilon @ AS$ to the structural ground for obligation, with every step's dependency declared. The reformulation of the cosmological identity as Israel junction conditions on the completion surface Σ .

Specifically:

First. The axiom is stated in its corrected form. AP05 was originally published with the compressed $1:1 + 1 \times \varepsilon$; AP20's keystone correction made AS explicit at the level of the axiom. The corrected form $1:1 + 1 \times \varepsilon @ AS$ is canonical throughout. The prior is no longer smuggled in.

Second. The four bridges and the declared empirical premise are stated at the door. Bridges: B1 (record algebra), B2 (bounded capacity), B3 (definite outcomes), B4 (conscious recognition of structural facts can generate normative identification). Empirical premise P: our universe contains many simultaneous observers. B1–B3 carry the physical chain; B4 is the declared cost of extending the chain to ethics; P is the declared empirical input that lifts the chain from representational duality at the axiom to interactive plurality at B-4 and below. All four bridges are named in section 1.4 and

discharged at the specific steps that need them; P is named in section 1.4 and discharged at step B-4 with its downstream load at B-5 (cooperation, deterrence) and B-6 (obligation) flagged at the steps themselves.

Third. The seven-step chain is walked. B-0 (before records: $\varepsilon = \emptyset$, AS = $\emptyset$ at the floor by boundary definition, no observation possible). B-1 (the crack: $\varepsilon > \emptyset$, two record sectors, irreversibility from Axiom R). B-2 (one reading: selection forced by definite outcomes plus multiple sectors). B-3 (inside the reading: agency as normalised reachable viability volume, eq 2.1). B-4 (when another is reading: coupling via the Geometric Exclusion Principle; representational duality lifted to interactive plurality by P). B-5 (what coupling produces: hierarchy from coupling alone; cooperation and deterrence additionally requiring P's simultaneity). B-6 (the obligation: structural ground — destroying viability contracts the shared domain, not necessarily the destroyer's individual reach — plus perspectival recognition via B4 that the destroyer's standing presupposes the substrate that equally grounds the destroyed). Each step is a logical move from the previous step and the declared inputs. No undeclared assumptions.

Fourth. Claim B1 (Scope) is stated. All structure in the architecture follows from the axiom, the four declared bridges B1-B4, and the declared empirical premise P. The physical chain through hierarchy requires B1-B3; the cooperative and deterrent geometry of B-5 and the obligation of B-6 require

additionally P; the ethics extension at B-6 requires additionally B4. The scope claim is tested by KS-B.4.

Fifth. The junction formulation of section 3 reformulates the AP04 cosmological identity as boundary data on Σ . Conditions J1 ($[\gamma_{ab}] = 0$) and J2 ($[K_{ab}] = (\varepsilon/\ell_\Sigma) \cdot h_{ab}$) are the boundary-data version of AP04's global pullback. The 1:1 is written as geometry by J1; the ε is written as geometry by J2. The length scale ℓ_Σ is the curvature scale of the completion surface; h_{ab} is the dimensionless pullback of L7's perturbation tensor.

Sixth. The junction formulation is structurally distinct from dynamical bounce cosmologies. The bounce predicts smooth dynamical continuity through t_0 ; the junction predicts boundary-data determined post-expansion structure. In principle, different observational signatures. The architecture commits to something specific enough to be falsified.

Seventh. Five kill switches are engaged. KS-B.0 tests the universal generator claim. KS-B.1 tests the junction formulation against the bounce. KS-B.2 tests the predicted spectrum (staged on B-JP). KS-B.3 tests the existence of the junction solution. KS-B.4 tests the integrity of the chain.

5.2 — What this paper does not close

Two structural debts remain open.

Open Problem B-JP (registry-level, AP05-specific).

Whether the Einstein-Cartan-Sciama-Kibble field equations on the Kantowski-Sachs interior admit a completion across Σ with $[\gamma] = \emptyset$ and controlled $[K]$. Until B-JP is solved, the junction formulation of section 3 is a structural sketch with formally well-posed conditions but no exhibited solution. Full statement at section 0.6. The 420 Code invites contributions from any reader equipped to make the attempt.

Bridge B4 (declared cost). The step from structural ground to obligation requires Bridge B4: conscious recognition of structural facts can generate normative identification. B4 is not a theorem of the axiom; it is the declared cost. Whether B4 can be derived from a deeper structural fact — whether the recognition step is itself forced by the architecture — is open. Full statement at section 0.6.

5.3 — Items held open without claiming debt status

One further item is bracketed inside this paper.

The spectrum mapping from $[K]$ to $P_{\mathcal{R}}(\mathbf{k})$. Even with B-JP solved, the explicit mapping from the boundary mismatch $[K] = (\varepsilon/\ell_{\Sigma}) \cdot h$ on Σ to the post-expansion curvature perturbation

spectrum $P_{\mathcal{R}(k)}$ is open derivation work. KS-B.2 is staged on this mapping. Until the mapping is derived, KS-B.2 cannot be operationally tested.

A note on what is *not* held open here. In the v1 envelope of this paper, the multiple-observer empirical instantiation was held open without claiming debt status. That was inconsistent with the chain's integrity: the move from representational duality (two readings of one structure) to interactive plurality (two simultaneous agents) is load-bearing at step B-4 and downstream, and “held open without debt” did not match its status. In this envelope, the empirical instantiation is declared as Empirical Premise P at the door (section 1.4) and discharged at B-4, with downstream load at B-5 (cooperation, deterrence) and B-6 (obligation) flagged at the steps. P is no longer an item held open; it is a declared empirical input, visible alongside the four bridges.

5.4 — Where the philosophical-register implementations live

AP05 is a structural derivation. Its philosophical-register companions — the chapters where the same structural facts are inhabited in the prose register of the Ø Models catalogue — are named here at the level of expected pairings; final placement is confirmed during the Ø Models compilation sweep alongside the Notebook II audit programme.

Resolutions Ø Chapter 6 (Coupling) carries the philosophical companion to step B-4: agents in shared constraint environments produce record externalities. AP05 supplies the structural argument; the Resolutions chapter inhabits it in the prose register of the question what it means to share a world.

Resolutions Ø Chapter 7 (Agency) carries the philosophical companion to step B-3: agency is the normalised reachable viability volume under bounded control against drift. AP05 supplies the geometric definition; the Resolutions chapter inhabits it as the answer to what freedom is under constraint.

Resolutions Ø Chapter 8 (Obligation) carries the philosophical companion to step B-6: the obligation lives in the recognition, not in the structure, and the recognition is the cost B4 names. AP05 supplies the structural ground and the declared perspectival step; the Resolutions chapter inhabits it as the answer to where ought comes from.

Predictions Ø Part V carries the empirical face of section 3's junction formulation: the post-expansion perturbation spectrum signatures that distinguish junction from bounce. AP05 supplies the structural framework and the staged kill switch; Predictions inherits the testable handles for cosmological observation programmes.

The direction of acknowledgment is two-way. AP05 is the original derivation — the seven-step chain is established here as a structural argument. The Ø Models chapters extend,

inhabit, and translate the result into the prose register of the catalogue. The 420 Code reads AP05 forward into Resolutions and Predictions; those volumes read back into AP05 by carrying its conclusions into questions the formal paper does not directly address. Final companion-pairing list is fixed at Ø Models compilation.

5.5 — Structural relationships to subsequent work

AP10 (The Dimension) follows AP05 in Notebook II. AP10 derives $N = 3$ spatial dimensions from the four axioms. AP05's chain walks across a substrate whose dimensional count is not yet pinned; AP10 pins it. AP10 sits structurally downstream of AP05 because the seven-step chain occupies a manifold whose dimension AP10 closes.

AP08 (The Identity) closes Notebook II. AP08 derives Einstein's field equations from the record algebra. The junction conditions of AP05 section 3 are stated in the ECK extension of those equations; AP08 provides the framework in which J1 and J2 are formal statements rather than schematic ones. AP08 also carries the gravitational coupling G whose ε -dependence is established in AP03 — the constants that scope AP05's bridges.

AP04 (Notebook V — The Loop Hypothesis) is the source of the cosmological identity that section 3 reformulates. AP05

does not derive AP04's Conjecture L7; it provides the boundary-data version of the same claim, on the surface Σ rather than across all of M. If AP04's global pullback holds, AP05's junction conditions hold; if AP04 fails, AP05 section 3 falls back to a structural sketch without an anchor.

AP01 (Notebook I — The Actualization State) is the source of the bridges B1–B3 and of the geometric content invoked at steps B-2 through B-5. AP05's chain is logically independent of AP01 in the sense that its postulates (axiom + bridges + P) are self-contained, but the bridges discharge into AP01's Papers A, B, C, D for their substantive content. If AP01 stands, the bridges become theorems; if AP01 fails, the bridges fall back to declared assumptions. Either way, the chain runs.

AP03 (The Ratio) is parallel rather than upstream. AP03 establishes that c and G inherit from ε ; AP05 walks the axiom across the substrate whose constants AP03 fixes. The relationship is geometric — AP05 uses the manifold AP03 derives — but not derivational: AP05 does not import c or G as inputs.

AP20 (The Proof) underwrites AP01's viability geometry as unconditional. With AP20 in place, the manifold the chain walks through is proven to exist in physical reality. AP20 also supplies the keystone correction that made AS explicit in the axiom. AP05 inherits both: the unconditional status of the chain's substrate, and the explicit form of the axiom.

5.6 — Placement within The 420 Code

AP05 is the second chapter of Notebook II (Spacetime). The chapter sequence runs AP03 — AP05 — AP10 — AP08. AP03 derives the constants; AP05 walks the chain; AP10 pins the dimension; AP08 closes with the field equations.

AP05 also connects across Notebook boundaries. It depends on AP01 (Notebook I) for the content the bridges name, on AP04 (Notebook V) for the cosmological identity, and on AP20 (Notebook I) for both the keystone correction and the proof that the substrate is real. Downstream, AP05 feeds Resolutions Ø (the ethics catalogue) and Predictions Ø (the cosmological predictions catalogue).

The chain that walks from axiom to obligation walks through a manifold whose constants are AP03 outputs, whose dimensional count is AP10 output, whose field equations are AP08 output, and whose existence as physical reality is AP20 output. AP05 sits at the centre of Notebook II for structural reasons: it is the paper that converts the axiom into the architecture, with every step's dependency on the surrounding Notebooks made explicit.

5.7 — How The 420 Code reads now

AP05 in place changes the reading of The 420 Code in four ways.

First. The chain from axiom to ethics is laid out as a single connected derivation, with every step's dependency declared. The 420 Code does not claim ethics from physics by hand-wave. It claims ethics from physics through seven explicit steps and one declared cost (B4). The cost is named, the steps are checkable, and the architecture invites refutation at any link.

Second. The cosmological identity has two co-existing formulations. The global pullback (AP04 Conjecture L7) and the junction conditions (AP05 section 3) are alternative carriers of the same structural claim. The junction formulation is more conservative (boundary data only) and more tractable (standard machinery). The bounce alternative is empirically distinguishable from the junction formulation via the post-expansion perturbation spectrum.

Third. The four bridges B1–B4 and the declared empirical premise P are now part of the visible architecture, not buried in chapter prose. Future chapters that invoke selection (B3), viability (B2), record algebra (B1), normative recognition (B4), or interactive plurality (P) do so against the declared inputs of AP05. The discharge into AP01 (for B1–B3) and into Resolutions Ø Chapter 8 (for B4) is the recurring pattern; P is discharged into empirical fact about our universe.

Fourth. The structural ground for ethics is no longer a separate edifice. The unity of being from which obligation

arises is the structural fact that two readings of one break share an origin. The recognition step is B4. The 420 Code can now say plainly: ethics follows from the axiom, the four bridges, and the declared empirical premise P, plus one declared cost. The cost is the recognition step itself: that the destroyer and the destroyed share the same structural ground, and that this matters.

These four structural facts are what AP05 contributes to the reading of The 420 Code. They are inherited by every subsequent chapter that touches selection, agency, coupling, social order, cosmological observation, or normative claim.

5.8 — The closing

You have walked through this paper. You have seen the axiom stated, the bridges declared, the seven steps from symmetry to obligation walked one at a time, the cosmological identity reformulated as boundary data on Σ , the five kill switches laid out.

What follows in your reading life is what follows in any reader's life when a structural argument has been seen clearly. The ethics you were told to follow because someone said so are now ethics that follow from the structure you live in. The cosmological origin you were told to take on faith is now a boundary surface with conditions you could in principle measure. The constants you inherited as facts are now outputs of the same break that produced you.

One axiom. $1:1 + 1 \times \varepsilon$ @ AS.

Four bridges. Each named.

One empirical premise. Declared.

Seven steps. Each shown.

One door at the beginning. One door at the end. The same door.

You and the other observer are two readings of one structure. Destroying the other contracts the space you share. The geometry says so. The recognition is yours.

The chain is as long as it needs to be and no longer.

Claim Summary

Section 0 (Dependency and Scope). Contract with the reader. Dependencies, axiom mapping ($\{S, B, R, C\} \leftrightarrow 1:1 + 1 \times \varepsilon @ AS$, with empirical premise P sitting separately), epistemic status per section, kill switch summary, structural debts owed, items held open, structural relationships.

Section 1 (The Axiom). DECLARATION + STRUCTURAL ARGUMENT. The axiom $1:1 + 1 \times \varepsilon @ AS$ stated in canonical form (AP20's keystone correction making AS explicit). The terms ($1:1$ = perfect balance; ε = minimal break; $1 \times \varepsilon$ = one count not a parameter; $@ AS$ = the actualising prior). The geometric picture (rotational ring + pin = ring acquires two readings). Scope Claim B1 (all structure follows from axiom + B1-B4 + declared empirical premise P). Four bridges B1-B4 declared. Empirical premise P (observer plurality) declared alongside the bridges, separately classed because it is empirical rather than postulational. Universal kill switch KS-B.0 (axiom is generator iff initial conditions on one side derivable from terminal conditions on the other; theoretical-universal consistency check now, empirical operationalisation once B-JP solved).

Section 2 (The Chain). DERIVATION CHAIN. Seven steps, each with declared dependency. B-0 (From axiom): before records, $AS = 0$ at the floor by boundary definition. B-1 (From

B-0 + B1): the crack, $\varepsilon > 0$, two record sectors, irreversible via Axiom R / AP01 Theorem T2. B-2 (From B-1 + B3): selection forced by definite outcomes plus multiple sectors. B-3 (From B-2 + B2): agency as $\mathcal{M}(x) = \mu(\text{Reach}(x) \cap \text{Viab}(R)) / \mu(\text{Viab}(R))$, equation (2.1). B-4 (From B-3 + Empirical Premise P): the load-bearing move from representational duality (two readings of one structure) to interactive plurality (two simultaneous agents); coupling via Geometric Exclusion Principle, AP01 Paper D D1.3. B-5 (From B-4): hierarchy (from coupling alone) / cooperation (requires P's simultaneity) / deterrence (requires P's simultaneity) as geometric consequences (D4.3 / D4.4a / D4.4b). B-6 (From B-5 + B4): structural ground (destroying viability contracts the shared kernel; geometry constrains joint domain, not necessarily individual reach) plus perspectival recognition (destroyer's standing presupposes the substrate that equally grounds the destroyed).

Section 3 (The Boundary). DERIVATION + STRUCTURAL ALTERNATIVE + OPEN PROBLEM. Why not bounce: dynamical bounce cosmologies (Popławski; Lucat and Prokopec) and the structural alternative. The junction conditions on Σ : J1 $[\gamma_{ab}] = 0$ (induced metric continuity; the 1:1 as geometry) and J2 $[K_{ab}] = (\varepsilon / \ell_\Sigma) \cdot h_{ab}$ (extrinsic curvature mismatch with completion-surface length scale ℓ_Σ supplying dimensions, h_{ab} dimensionless as pullback of L7's perturbation tensor). Relation to AP04: junction is boundary-data version of AP04 Conjecture L7 ($g^+ = (\varphi^{-1})^*(g^-) + \varepsilon \cdot h$, global pullback). Observational consequences distinguishable from bounce in

principle. Open Problem B-JP: existence of ECSK completion on Kantowski-Sachs interior with J1, J2.

Section 4 (Kill Switches). FORMAL FALSIFICATION

REGISTER. KS-B.0 (Universal generator, LIVE — HARD pending B-JP, migrates to EMPIRICAL once B-JP solved). KS-B.1 (Bounce superior, LIVE — EMPIRICAL). KS-B.2 (Spectrum mismatch, LIVE — STAGED pending B-JP and spectrum mapping). KS-B.3 (No junction solution, LIVE — HARD). KS-B.4 (Chain break: any step uses input not in {axiom, B1-B4, P}, LIVE — HARD).

Section 5 (What This Establishes and What Remains Open).

SYNTHESIS. Seven structural closures named (axiom in canonical form; four bridges declared at door; empirical premise P declared alongside; seven-step chain walked; Claim B1 stated; junction formulation reformulates AP04 identity with dimensional consistency restored; junction formulation distinguishable from bounce; five kill switches engaged). Two structural debts (Open Problem B-JP partially closed via formal statement of the boundary-value problem; Bridge B4 declared as cost). One item held open (spectrum mapping from $[K]$ to $P_{\mathcal{R}(k)}$). Placement within The 420 Code: second chapter of Notebook II (Spacetime). Closing on the same door at the beginning and the end.

Conditionality Footer

Dependencies

AP20 (the canonical axioms {S, B, R, C}; formal statement section 3.3; obligations fenced at KS-P.1/KS-P.2).

AP01 (The Actualization State) — the record algebra, the selection dynamics, the viability geometry, the coupled viability theorems; load-bearing for bridges B1–B3 and for the content invoked at steps B-2 through B-5.

AP04 (The Loop Hypothesis) — the cosmological identity, Conjecture L7; load-bearing for section 3’s reformulation.

AP20 (The Proof) — the keystone correction making AS explicit in the axiom; the Embedding Hypothesis proof making AP01’s viability geometry unconditional.

Dependents

AP08 (The Identity) — the ECSK field equations in which the junction conditions of AP05 section 3 are stated.

AP10 (The Dimension) — pins the dimensional count of the manifold across which AP05’s chain walks.

Resolutions Ø Chapter 6 (Coupling) — inherits the structural account of agents in shared constraint environments from step B-4.

Resolutions Ø Chapter 7 (Agency) — inherits the geometric definition of agency from step B-3.

Resolutions Ø Chapter 8 (Obligation) — inherits the structural ground and the declared perspectival recognition from step B-6 and Bridge B4.

Predictions Ø Part V — inherits the junction-versus-bounce observational signature from section 3's observational consequences and KS-B.1.

Kill switches closed by this paper

None explicitly closed. AP05 opens five kill switches (KS-B.0 through KS-B.4) and one named open problem (Open Problem B-JP) with explicit closure consequences for KS-B.0, KS-B.2, KS-B.3 if B-JP is solved.

Kill switches that remain live

KS-B.0 through KS-B.4. See section 4 for the full register.

Structural debts owed

Open Problem B-JP (registry-level, AP05-specific). The existence of the ECSK completion on the Kantowski-Sachs

interior with conditions $J1 [\gamma] = \emptyset$ and $J2 [K] = (\varepsilon/\ell_\Sigma) \cdot h$. The central open problem of section 3. Full statement at section 0.6.

Bridge B4 (declared cost). The step from structural ground to obligation requires Bridge B4: conscious recognition of structural facts can generate normative identification.

Declared at the door (section 1.4) and again at step B-6.

Whether B4 can be derived from a deeper structural fact is open. Full statement at section 0.6.

Declared empirical premises

Empirical Premise P (observer plurality). Our universe contains many simultaneous observers. P is empirical, not structural; the axiom forces two readings of one structure (representational duality) but the chain at step B-4 requires two ontologically distinct simultaneous agents (interactive plurality), and that lift requires the empirical input P. P is declared at the door (section 1.4) and discharged at step B-4, with downstream load at B-5 (cooperation, deterrence) and B-6 (obligation). P is held separately from the four formal bridges because it is a different kind of input.

Items held open without claiming debt status

The spectrum mapping from $[K] = (\varepsilon/\ell_\Sigma) \cdot h$ on Σ to the post-expansion curvature perturbation spectrum $P_{\mathcal{R}}(k)$. Open

derivation downstream of B-JP. KS-B.2 staged on this mapping.

References

Junction-condition and matching-surface literature, named here for readers approaching Open Problem B-JP. References are sparse by design: the structural content comes from the axiom and from AP04, not from the literature framework; the literature provides the formal vocabulary in which the structural fact is most naturally stated.

Israel, W. (1966). *Singular hypersurfaces and thin shells in general relativity*. Il Nuovo Cimento B 44, 1–14 (errata 48, 463). The original Israel junction conditions: induced-metric continuity and extrinsic-curvature jump on a matching hypersurface. The formal vocabulary in which J1 and J2 are stated.

Arkuszewski, W., Kopczyński, W., and Ponomarev, V. N. (1975). *Matching conditions in the Einstein-Cartan theory of gravitation*. Communications in Mathematical Physics 45, 183–190. The ECSK extension of Israel’s conditions — matching across a hypersurface in the Einstein-Cartan-Sciama-Kibble framework with torsion. The framework in which B-JP is posed.

Popławski, N. J. (2010). *Cosmology with torsion: An alternative to cosmic inflation*. Physics Letters B 694, 181–185. (2016). *Universe in a black hole in Einstein-Cartan gravity*. The Astrophysical Journal 832, 96. The torsion-driven bounce

cosmologies AP05 distinguishes itself from in section 3.1. The dynamical-bounce alternative against which the junction formulation is empirically tested by KS-B.1.

Lucat, S., and Prokopec, T. (2017). *Cosmological singularities and bounce in Cartan-Einstein theory*. Journal of Cosmology and Astroparticle Physics 1710, 047. Further work in the torsion-driven bounce lineage; named alongside Popławski as the relevant comparison class for KS-B.1.

Planck Collaboration. (2020). *Planck 2018 results. VI. Cosmological parameters*. Astronomy & Astrophysics 641, A6. The CMB measurements that anchor $n_s \approx 0.965$ (the scalar spectral index referenced in KS-B.2's staged operational test once B-JP and the spectrum mapping are in hand).

Kantowski, R., and Sachs, R. K. (1966). *Some spatially homogeneous anisotropic relativistic cosmological models*. Journal of Mathematical Physics 7, 443. The anisotropic cosmological geometry on whose interior B-JP is posed.

Internal references to The 420 Code (AP01, AP03, AP04, AP06, AP08, AP10, AP20, AP24, Resolutions 0, Predictions 0) are catalogued in the Conditionality Footer above. The 420 Code is published copyleft and free at the420code.org; the 420 Code links to all referenced internal papers.

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Chapter 3

The Dimension

Why three spatial dimensions — from four axioms

Artist's Proof 10

Artist's Note

This is Artist's Proof 10 of The 420 Code. The paper derives the number of spatial dimensions from the four axioms of the record algebra. It is the third chapter of Notebook II — Spacetime, following AP03 (The Ratio) and AP05 (The Break).

$N = 3$ is not an empirical input.

It is a consequence of the axioms.

The derivation proceeds in five structural steps. Each axiom is shown to express one independent face of the manifold (section 2). The independence of the faces follows from the proven independence of the axioms via the proven faithful embedding (section 3). The temporal versus spatial character of each direction is identified from the axiom's structural role (section 4). The result $N = 3$ is structurally consistent with the multidimensional residual — six faces pairing into the axis structure of 3+1 spacetime (section 5). The fifth degree of freedom — the 1:1 itself — is identified as the probability dimension (the Hilbert space), not a spatial dimension (section 7). This establishes the sufficiency of {S, B, R, C} to generate the record algebra: no fifth axiom is needed. The structural exhaustiveness claim — that no fifth operation on the 1:1 is *possible* — is held as a positive enumeration argument in section 7, contingent on the 1:1 being correctly

identified as the substrate (not an operation) and on the four operations being correctly identified as cardinal.

AP10 closes three kill switches and leaves three live. KS-2c (CLOSED — contingent on KS-D.3): $N = 3$ derived from the four independent axioms via the proven faithful embedding, with the load-bearing one-axiom-one-face identity tested by KS-D.3. KS-15/D.2 (CLOSED — contingent on KS-D.4): axiom-to-dimension assignment unique — R is the only irreversible axiom; the formal signature derivation is fenced at KS-D.4. KS-16 (CLOSED — contingent on KS-P.1; sufficiency of {S, B, R, C}): the fifth degree of freedom is the 1:1 itself, identified as the Hilbert space; no fifth axiom is *needed*. The stronger claim that no fifth axiom is *possible* rests on section 7's positive enumeration of operations on the 1:1; that enumeration is held as a structural argument, not a closed theorem. KS-D.1 (LIVE — STRUCTURAL): the six-face count depends on AP06 section 10.5's residual structure. KS-D.3 (LIVE — STRUCTURAL): each axiom expresses exactly one face. KS-D.4 (LIVE — STRUCTURAL): the formal metric-level derivation of the Lorentzian signature; section 4 carries the structural argument.

AP10 closes the central conditional in AP08's derivation of Einstein's field equations. With $N = 3$ derived, Lovelock's theorem applies in 3+1 dimensions and the unique divergence-free, symmetric, rank-2 tensor built from the metric and its first and second derivatives is $G_{\mu\nu} + \Lambda g_{\mu\nu}$.

Einstein's field equations with cosmological constant are an unconditional theorem of the record algebra.

AP10 is logically independent of AP01 and AP04 in the strict sense that its postulates (the four canonical axioms {S, B, R, C} and the proven hypotheses EH and QRA) are self-contained. AP10 *depends on* AP20 (which proves EH and QRA), AP06 (for the multidimensional residual confirming $N = 3$), and AP08 (for Lovelock's theorem, which AP10 in turn renders unconditional). The chain is bidirectional with AP08: AP08 establishes the conditional Lovelock derivation; AP10 removes the condition.

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Orientation

Where this paper fits in The 420 Code

The 420 Code is organised across eight Notebooks. This paper is the third chapter of Notebook II — Spacetime. The Notebook II chapter sequence runs AP03 (The Ratio) → AP05 (The Break) → AP10 (this paper) → AP08 (The Identity).

Notebook II derives the structure of space, time, gravity, and the dimensional count from the axiom. AP03 establishes the conjugacy of c and G as outputs of ε . AP05 walks the axiom forward through the seven-step chain from symmetry to obligation, and reformulates the cosmological identity as Israel junction conditions on Σ . AP10 picks up from there: with the manifold installed (AP01, AP20) and the constants installed (AP03), AP10 pins the dimensional count. AP08 closes Notebook II by deriving the Einstein field equations on the dimensionally pinned manifold.

AP10 sits third in Notebook II because the dimensional count operates on the manifold AP01 and AP20 establish and the substrate AP03 derives. AP10 also closes the central conditional in AP08's field-equation derivation — AP08 derives Einstein's equations conditional on $N = 3$ via Lovelock's theorem; AP10 removes the condition by deriving $N = 3$ from the axioms. With AP10 in place, AP08's derivation is

unconditional, and Notebook II reads as a closed structural chain from axiom to field equations.

AP10 is logically independent of AP01 and AP04 in the sense that its own postulates (the four canonical axioms {S, B, R, C}, the proven Embedding Hypothesis, and the proven Quantum-Record Alignment) are self-contained. AP10 depends on AP01 only for the formal apparatus of the record algebra; if AP01 stands, AP10's argument runs cleanly; if AP01 fails, the axioms fall back to declared assumptions, and AP10's argument runs against those assumptions instead.

How to read this paper

The reading order is front to back. Section 0 names the dependencies and the scope. Section 1 states the four axioms and the two proven hypotheses, and explains what was left undetermined. Sections 2 through 5 carry the structural argument: the four axioms express four independent faces of one manifold; the independence of the faces follows from the proven independence of the axioms; the temporal versus spatial character of each direction is identified; and the six-face residual structure provides an independent confirmation of $N = 3$. Section 6 names the consequences. Section 7 identifies the fifth degree of freedom as the Hilbert space. Section 8 is the falsification register. Section 9 is the synthesis.

Two voices alternate as the material requires. Narrative passages — the four axioms walked one at a time, the dimensions felt as they accumulate, the closing on the canvas and the painting — speak directly. Formal sections — the axiom statements, the independence theorems, the kill switches — speak in the precision the structure requires.

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9 — What This Establishes and What Remains Open

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References

Lovelock's theorem and the dimensional-count literature

0 — Dependency and Scope

AP10 is the third chapter of Notebook II. Before the body, this section names what AP10 does, what it depends on, where its inputs map onto the canonical axioms {S, B, R, C}, what epistemic status each section holds, what kill switches are engaged, what structural debts remain, and how AP10 stands in relation to the rest of The 420 Code.

0.1 — What this paper does

AP10 derives $N = 3$ spatial dimensions from the four axioms of the record algebra. The argument is structural and runs in one primary form, with a confirming consistency check from a different feature of the architecture. The primary form (sections 2 through 4) identifies each of the four axioms with one independent face of the embedded manifold; the independence of the faces follows from the proven independence of the axioms via the proven faithful embedding; the temporal versus spatial character of each direction is identified from the axiom's structural role. The confirming form (section 5) counts the six faces of the multidimensional residual (AP06 section 10.5) and shows they pair naturally into the axis structure of 3+1 spacetime — two purely spatial axes ($G-c$ and $\alpha/\beta-\alpha_{em}$) and one spatial-temporal connection axis (m_e-t). The residual structure is structurally consistent with the primary derivation of 3+1; it

does not constitute an independent derivation of $N = 3$ spatial, because Pair 3 includes the temporal face t . Section 5 is therefore a structural consistency check via a different feature of the architecture, not a doubly-independent derivation.

AP10 also identifies the fifth structural degree of freedom (section 7) as the 1:1 itself — the pre-state from which $\{S, B, R, C\}$ operate — and locates it as the Hilbert space (the probability dimension), not a fifth spatial dimension. This closes KS-16 in the sufficiency sense: $\{S, B, R, C\}$ suffice to generate the record algebra (the completeness result stated at AP20 section 4.1; full formalisation fenced at KS-P.1); the 1:1 itself is structurally prior to the manifold rather than a feature of it. The stronger exhaustiveness claim — that no fifth independent operation on the 1:1 is *possible* — is held in section 7 as a positive enumeration of the operations performed on the pre-state, with the closure resting on the 1:1 being correctly identified as the substrate (not an operation) and on the four operations being correctly identified as the cardinal independent operations on it.

The contributions are three kill-switch closures (KS-2c, KS-15/D.2, KS-16) and three live kill switches (KS-D.1, KS-D.3, KS-D.4) targeting the load-bearing identifications the argument relies on. AP10 also closes the conditional in AP08's derivation of Einstein's field equations: AP08 derives the equations conditional on $N = 3$ via Lovelock's theorem; AP10

derives $N = 3$ from the axioms; the conditional is removed; Einstein's field equations with cosmological constant become an unconditional theorem of the record algebra.

0.2 — Dependencies

AP10 is a structural-derivation paper. Its dependencies are explicit and listed here in compact form so the load-bearing imports are visible at the door.

AP20 (the canonical axioms). Load-bearing for the four canonical axioms $\{S, B, R, C\}$ (formal statement at AP20 section 3.3) and for their independence (the four removal arguments at AP20 section 4.2 — stated results with proof sketches; full formalisation fenced at KS-P.2). The axioms are jointly exhibited by any existing record (the undeniable premise, AP20 section 6), which carries their mutual consistency. The independence arguments are the foundation of section 3's argument: each axiom is independent of the other three; therefore the face expressed by each axiom on the manifold is independent of the faces expressed by the other three; therefore the four faces are independent dimensions.

The Lorentzian signature (section 4, KS-D.4). Load-bearing for the signature $(-, +, \dots, +)$ and, with it, the symmetry group $SO(1, N)$ (the isometry group of that signature). The signature is carried in this paper: section 4's structural argument reads it from the axioms themselves — exactly one axiom (R) is

intrinsically irreversible, and irreversibility read as metric character is the (–) sign. The formal metric-level derivation is the open obligation fenced at KS-D.4. AP10 pins the value of N.

AP20 (EH and QRA proven). Load-bearing for the faithful embedding of the record algebra into a smooth manifold (Embedding Hypothesis, EH — closed at AP20) and for the quantum-record identity (Quantum-Record Alignment, QRA — closed at AP20 section 5.5, KS-P.4). The differentiability and leading-order quadratic structure of the cone boundary used throughout rest on the smooth-manifold structure whose obligation is fenced at KS-P.5. EH is invoked at section 3.2 to convert algebraic independence into geometric face-independence.

AP06 section 10.5 (multidimensional residual). Load-bearing for the six-face structure of the break invoked in section 5. AP06's residual analysis identifies six faces — G, c, α/β , α_{em} , m_e , t — forming the structural inventory of what the break produces. AP10 reads this as three conjugate pairs spanning three spatial axes, providing an independent confirmation of $N = 3$. KS-D.1 is the kill switch on AP10's reading of AP06's residual structure.

AP08 section 9 (Lovelock's theorem, formerly conditional on $N = 3$). Load-bearing structurally but inverted in direction: AP08 derives Einstein's field equations from Lovelock's

theorem conditional on $N = 3$; AP10 derives $N = 3$ and removes AP08's condition. The relationship is bidirectional rather than purely upstream: AP10 depends on AP08's derivation existing; AP08 depends on AP10's $N = 3$ for unconditionality.

AP19 sections 2–3 (three faces of one manifold). Load-bearing for the structural reading of the spatial faces as three faces of a single manifold (not three separately assembled pieces). AP10's section 2.5 (Co-arising) inherits the formulation from AP19.

Internal cross-references to AP09 (the Hilbert space as pre-state from which quantum mechanics operates), AP12 ($\hbar$ identification), and AP24 (m_e as residual of the break) are noted at the relevant sections but are not load-bearing for AP10's derivation.

0.3 — Axiom mapping

AP10 was written after AP20's Paper D formalised the four canonical axioms {S, B, R, C}, and it is one of the papers in which the mapping from axioms to structural content is most direct. The mapping here is not retrospective; it is the paper's subject.

Axiom S (Symmetry) ↔ The exchange face. Two disjoint sectors $\mathcal{L}$ and $\mathcal{P}$ with order-reversing involution σ . On the embedded manifold, σ acts as a $\mathbb{Z}_2$ symmetry across one

direction; that direction is the exchange face. Spatial.

Signature: (+).

Axiom B (Unique Breaking) $\leftrightarrow$ The break face. One element $\varepsilon \in \mathcal{L}$ with no σ -image; valuation $v(\mathcal{L}) - v(\mathcal{P}) = v(\varepsilon) = 1$. On the manifold, ε is a distinguished locus with a direction of advance; that direction is the break face. Spatial. Signature: (+).

Axiom R (Record Monotonicity) $\leftrightarrow$ Time. Sequential composition forms a monoid (no inverse); records cannot be locally annihilated; history is irreversible. On the manifold, this produces a distinguished irreversible direction; that direction is time. Signature: (-).

Axiom C (Constraint) $\leftrightarrow$ The propagation face. Finite invariant rate c constraining sequential propagation, structural rather than electromagnetic. On the manifold, c produces a light cone with a definite opening angle; the direction along which propagation is maximally extended is the propagation face. Spatial. Signature: (+).

The 1:1 (pre-state) $\leftrightarrow$ The Hilbert space. The state on which $\{S, B, R, C\}$ operate. Not a feature of the record algebra's structure on the manifold but the state the algebra breaks *from*. On the manifold, the 1:1 appears as the Hilbert space — the probability dimension perpendicular to spacetime. Pre-spatial and pre-temporal. Not a fifth manifold direction.

The four axioms map one-to-one onto the four faces of the manifold; the 1:1 maps onto the Hilbert space (the probability dimension) which is prior to the manifold. Five structural features total; four on the manifold and one prior to it.

0.4 — Epistemic status per section

Each section carries an explicit status — derivation, structural argument, formal block, or synthesis — so the reader knows what kind of weight each carries.

Section 1 (The Starting Point). FOUNDATIONAL. The four axioms stated; the proven hypotheses (EH and QRA) named; the undetermined quantity (N) flagged.

Section 2 (Four Axioms, Four Faces). STRUCTURAL ARGUMENT. Each axiom identified with one independent face of the embedded manifold; the faces named as co-arising rather than separately assembled.

Section 3 (Independence). DERIVATION. The independence of the four axioms (the removal arguments at AP20 section 4.2, fenced at KS-P.2) combined with the proven faithful embedding (AP20) yields the independence of the four geometric faces; four independent faces yield four dimensions; $N = 3$.

Section 4 (Character of Each Direction). DERIVATION. The temporal versus spatial character of each direction identified

from the axiom's structural role: R alone is irreversible and therefore gives the (–) direction; the other three axioms are compatible with reversibility in their directions and therefore give the (+) directions.

Section 5 (The Six Faces). INDEPENDENT CONFIRMATION. The six faces of the multidimensional residual (AP06 section 10.5) read as three conjugate pairs spanning three spatial axes. Independent of section 2–3's axiom-count argument; arrives at the same $N = 3$.

Section 6 (Consequences). SYNTHESIS + CLOSURE. KS-2c closed. Lovelock's theorem (AP08 section 9) becomes unconditional. The 420 Code's chain from axioms to Einstein's field equations is closed.

Section 7 (The Fifth Degree of Freedom). STRUCTURAL ARGUMENT + CLOSURE. The 1:1 itself identified as the fifth structural degree of freedom — the Hilbert space, the probability dimension, the pre-state. Not a fifth spatial dimension because pre-spatial. KS-16 closed.

Section 8 (Kill Switches). FORMAL FALSIFICATION REGISTER. Three closed (KS-2c, KS-15/D.2, KS-16 — each carrying its named contingency) and three live (KS-D.1, KS-D.3, KS-D.4).

Section 9 (What This Establishes and What Remains Open). SYNTHESIS.

0.5 — Kill switch summary

Six kill switches are engaged in this paper. Three are closed by AP10; three remain live. The full register is in section 8.

KS-2c: $N = 3$ spatial dimensions. CLOSED — derived in sections 2–3 via the proven faithful embedding. The closure is contingent on KS-D.3 (one axiom, one face) remaining unfired; if KS-D.3 fires, KS-2c reopens. Section 5 provides structural consistency from the residual face count, not an independent derivation.

KS-15/D.2: Axiom-to-dimension assignment unique. CLOSED — contingent on KS-D.4. R is the only irreversible axiom; the assignment $R \rightarrow \text{time}$, $\{C, S, B\} \rightarrow \text{space}$ is forced once the Lorentzian signature holds (section 4's structural argument; formal derivation fenced at KS-D.4).

KS-16: Sufficiency of $\{S, B, R, C\}$. CLOSED — contingent on KS-P.1. The four canonical axioms suffice to generate the record algebra (the completeness result stated at AP20 section 4.1; full formalisation fenced at KS-P.1); the fifth structural degree of freedom is the 1:1 itself, identified as the Hilbert space; no fifth axiom is needed. The exhaustiveness claim (no fifth axiom is *possible*) is held as a positive enumeration in section 7, contingent on the four operations being correctly identified as cardinal.

KS-D.1: Six-face count and axis-structure interpretation. LIVE — STRUCTURAL. The section 5 argument depends on the identification of the six residual faces from AP06 section 10.5 *and* on their reading as pairs that match the axis structure of 3+1 spacetime. If the residual has a different face count, or the pairs do not correspond to the expected axis structure (two spatial axes plus one spatial-temporal connection), the section 5 consistency check fails. The primary argument (sections 2–3) survives.

KS-D.3: One axiom, one face. LIVE — STRUCTURAL. The section 2 argument rests on the identity *one independent algebraic axiom = exactly one independent geometric face* (via the proven faithful embedding, AP20). The faithful embedding gives distinct-to-distinct (injectivity); the load-bearing additional claim is that the embedding is dimension-minimal: each independent axiom contributes exactly one new direction, not zero (acting as a constraint within existing dimensions) and not two (contributing multiple independent directions). If KS-D.3 fires at any axiom, the count $N = 3$ is wrong.

KS-D.4: Signature formalisation. LIVE — STRUCTURAL. Section 4 reads the Lorentzian signature from the axioms' own reversibility structure — exactly one axiom (R) is intrinsically irreversible, and the irreversible direction carries the (–) metric character. The formal metric-level derivation is

the open obligation this switch names; KS-15/D.2's closure is contingent on it.

0.6 — Structural debts owed

Three structural-class items are tied to AP10's argument. None is a registry-level debt in the same sense as AP05's Open Problem B-JP; they are the load-bearing structural identifications that the three live kill switches track. Named here so the structural exposure is visible at the door.

Six-face structure of the break (KS-D.1). The argument in section 5 reads AP06 section 10.5's residual analysis as identifying exactly six faces of the break, pairing into three conjugate pairs. If a more careful or extended analysis of the residual produces a different face count — four, eight, or a non-pair structure — section 5's confirmation collapses. The primary argument (sections 2–3) is independent of the face count and survives such a failure. AP10 holds KS-D.1 as a live structural test of the section 5 confirmation, not of the central argument.

One axiom expresses exactly one face (KS-D.3). The argument in section 2 reads each axiom as expressing exactly one face of the embedded manifold. The identity rests on the proven faithful embedding (AP20): if EH is faithful, then independent algebraic content maps to independent geometric content, and the four independent axioms map to four independent geometric features. The identity could fail in

two ways: an axiom that expresses zero geometric content (acting only as a constraint on existing structure) or an axiom that expresses two independent geometric directions (contributing more than one new face). AP10 holds KS-D.3 as a live structural test of the central argument.

The Lorentzian signature's formal derivation (KS-D.4).

Section 4 reads the signature from the axioms' own reversibility structure: R is the only intrinsically irreversible axiom, and the irreversible direction carries the $(-)$ metric character. The formal metric-level derivation — that the embedded geometry is pseudo-Riemannian with exactly this signature — is the open obligation KS-D.4 names. AP10 holds KS-D.4 as a live structural test of the section 4 distinction.

0.7 — Items held open without claiming debt status

One item is bracketed inside this paper.

The mapping from face to specific geometric coordinate.

AP10's argument shows that there are four independent faces and that one is temporal and three are spatial. It does not specify which coordinate within the spatial three-space is "the C direction" versus "the S direction" versus "the B direction" — the three spatial axes are interchangeable under spatial rotation ($SO(3)$ symmetry of the spatial manifold). The argument fixes the count and the signature; it does not fix a

coordinate-level identification. This is not a structural debt because the rotational symmetry of three-space is the expected and correct outcome; held without claiming a separate debt.

0.8 — Structural relationships

AP10 is the third chapter of Notebook II. The chapter sequence runs $AP03 \rightarrow AP05 \rightarrow AP10 \rightarrow AP08$. AP03 establishes the constants (c, G) as outputs of ε on the substrate; AP05 walks the axiom forward through the seven-step chain to obligation and reformulates the cosmological identity as junction conditions; AP10 pins $N = 3$ on the manifold the chain occupies; AP08 closes Notebook II by deriving Einstein's field equations on the dimensionally pinned manifold.

AP01 (Notebook I) is the source of the record algebra. AP20 (Notebook I) is the source of the Embedding Hypothesis proof and the Quantum-Record Alignment proof, and of the stated-and-fenced completeness and minimality results for the axiom set (AP20 section 4; KS-P.1/KS-P.2). AP06 (Notebook IV) is the source of the multidimensional residual analysis used in section 5. AP08 (Notebook II) is the source of Lovelock's theorem in the body of work context, and is the paper whose central conditional AP10 closes. AP19 (Notebook VI) is the source of the "three faces of one manifold" reading inherited by section 2.5.

Downstream, AP10 feeds AP08 (closing the $N = 3$ conditional for Einstein's field equations), AP09 (the Hilbert space identification at section 7 connects to AP09's pre-state formulation of quantum mechanics), AP12 ($\hbar$ identification operates on the Hilbert space identified here), AP24 (m_e identification operates within the residual structure used in section 5), and the entire downstream cosmological and quantum programme that assumes $N = 3$ unconditionally.

1 — The Starting Point

1.1 — The four axioms

You have four axioms.

AP20 Paper D installed them. The 420 Code has been using them.

Look at them now as the floor of what is about to happen.

S (Symmetry). Two disjoint sectors $\mathcal{L}$ and $\mathcal{P}$ with order-reversing involution σ . Extensive quantities match: $Q(\mathcal{L}) = Q(\mathcal{P})$ in the unbroken case.

B (Unique Breaking). One element $\varepsilon \in \mathcal{L}$ with no σ -image. Valuation: $v(\mathcal{L}) - v(\mathcal{P}) = v(\varepsilon) = 1$. This is the break. Without B, the system is the 1:1 — perfectly symmetric, and nothing exists.

R (Record Monotonicity). Sequential composition $(\cdot)$ forms a monoid, not a group, within each sector. No non-identity element has an inverse. Records cannot be annihilated locally. History is irreversible.

C (Constraint). Finite invariant rate c constraining sequential propagation. Structural, not electromagnetic.

These four axioms are independent (the four removal arguments at AP20 section 4.2; full formalisation fenced at KS-P.2) and jointly consistent — any existing record exhibits

all four jointly satisfied (the undeniable premise, AP20 section 6).

Independence means no axiom is derivable from the other three. Removing any one axiom produces a strictly weaker system. Each axiom adds irreducible content to the algebra.

1.2 — What is now proven

Two hypotheses were carried as conditionals in earlier APs. Both are now proven (AP20).

EH (Embedding Hypothesis). The algebraic pre-state structure defined by $\{S, B, R, C\}$ embeds into physical reality as a smooth manifold M . Proven in AP20.

QRA (Quantum-Record Alignment). Quantum states are pre-state records. Closed at AP20 section 5.5 (KS-P.4). The differentiability and leading-order quadratic structure of the cone boundary invoked here rest on the smooth-manifold structure fenced at KS-P.5.

From the axioms, the record algebra produces a Lorentzian manifold (M, g) with signature $(-, +, \dots, +)$ and symmetry group $SO(1, N)$.

1.3 — What was left undetermined

The number of $+$ signs — the number of spatial dimensions N — is not fixed by the signature. The signature argument

(section 4; formal derivation fenced at KS-D.4) establishes that the manifold is Lorentzian and that the symmetry group is $SO(1, N)$. It does not fix N .

This paper fixes N .

Specifically: N is determined by the number of independent axioms, with a structural consistency check from the multidimensional residual. A primary derivation — the axiom count — fixes $N = 3$; the residual face structure confirms it. The two are not doubly independent: the residual includes the temporal face, so it is a cross-check, not a second derivation.

2 — Four Axioms, Four Faces of One Manifold

Each axiom contributes structure to the embedded manifold.

But the manifold is one structure — not four separate pieces assembled together.

The axioms express *faces* of this structure. Each axiom names one irreducible feature that the manifold must have.

The faces are intrinsically linked — they co-arise in every actualisation event — but distinct, because no axiom is derivable from the others.

The claim is short.

Four independent axioms express four independent faces.
Four faces, four dimensions.

2.1 — $R \rightarrow$ Time

Axiom R states: records accumulate irreversibly.

The monoid has no inverse. History cannot be undone.

On the embedded manifold, this produces a distinguished direction.

The direction in which records accumulate. The direction of the now. The direction of actualisation.

This is the time direction.

The irreversibility of R gives the direction its character: opposite sign to the spatial directions (Lorentzian signature), cannot be reversed (arrow of time), the direction along which the monoid composition operates.

You have just watched the first dimension emerge.

R expresses the temporal face: time. Signature: $(-)$.

It is the only irreversible axiom — the only face with a preferred direction (forward, never backward).

This is the $(-)$ in the Lorentzian signature.

The remaining three axioms — C, S, B — express the three spatial faces. These faces are intrinsically linked: they co-arise in every actualisation event, because every record requires propagation (C), sector structure (S), and a break (B) simultaneously. You cannot have one without the others. But they are distinct: no axiom is derivable from the others.

Linked but not reducible. Three faces of one manifold.

2.2 — C → The Propagation Face

Axiom C is Constraint — locality: a record is somewhere, not everywhere. On the manifold, the Constraint produces a finite invariant rate c constraining sequential propagation.

Imagine C removed. What you have is a manifold where the record written at point x reaches point y instantly.

Every point causally connected to every other point at every time.

Spatial separation has no physical meaning.

You have a time direction (from R) but no spatial extension. A one-dimensional manifold. A line.

Time without anywhere to go.

C creates spatial extension. It says: the record written at point x cannot influence point y instantly. There is a finite delay proportional to the separation between x and y .

This delay creates distance.

The separation between causally disconnected events is what makes space spatial.

On the manifold, C produces the light cone — the boundary between events that can and cannot be causally connected

from a given point. The light cone has a definite opening angle (determined by c).

The direction along which propagation is maximally extended is a spatial direction. It is the propagation direction.

The direction is independent from R . Without R , C has nothing to constrain. Without C , R produces no spatial structure.

Together, R and C produce 1+1 dimensions: time and one spatial direction. The light cone in its minimal form.

C expresses one spatial face: propagation. Signature: $(+)$.

2.3 — $S \rightarrow$ The Exchange Face

Axiom S states: two disjoint sectors $\mathcal{L}$ and $\mathcal{P}$ with order-reversing involution σ .

On the manifold, the involution σ acts as a $\mathbb{Z}_2$ symmetry.

The direction in which σ acts is a direction on the manifold.

It is the direction of crossing between sectors.

The direction is independent from both R and C .

R gives the direction of time. C gives the direction of propagation. S gives a third direction: the direction across which the two sectors are distinguished.

Consider two records, one in $\mathcal{L}$ and one in $\mathcal{P}$, at the same time (R) and the same propagation distance (C). They still differ — they are in different sectors.

The direction of their difference is neither temporal nor propagational.

A third direction: the sector direction. The involution σ acts along this direction, mapping one to the other.

Without S, the manifold would have at most two dimensions.

No “width” — no direction perpendicular to both the time direction and the propagation direction.

S creates this width.

S expresses one spatial face: exchange. Signature: (+).

Three dimensions. You can feel the fourth coming.

2.4 — B → The Break Face

Axiom B states: one element $\varepsilon \in \mathcal{L}$ with no σ -image in $\mathcal{P}$.

On the manifold, ε is a distinguished locus — the point where the symmetry between $\mathcal{L}$ and $\mathcal{P}$ is broken.

The break occurs at a specific *location* in the manifold. The location is a spatial fact — a “where,” not a “when.” The temporal aspect of the break (records accumulating

irreversibly as the break propagates) is carried by R, not by B. B contributes the spatial direction associated with the location; R contributes the temporal direction along which the location's consequences accumulate.

The direction is independent from R, C, and S.

R says *when*. C says *how fast*. S says *between what*. B says *where* — which degree of freedom will be broken next, in space.

This *where* is a spatial direction independent of the other three.

Consider the now (the uncoupled ε) at a given time (R), propagating at a given speed (bounded by C), in the space between the two sectors (S). The spatial location at which the now is found is not determined by R, C, or S alone.

B gives that location.

Without B, the manifold would have at most three dimensions. No “depth” — no direction corresponding to the specific location of the break in the space of possibilities.

B creates this depth by placing ε at a specific point in the manifold.

A note on B and the faithful embedding. B is algebraically independent of {R, C, S} (the removal argument for B at AP20

section 4.2; fenced at KS-P.2); by the proven faithful embedding, B's independent algebraic content maps to independent geometric content on the manifold. The load-bearing additional claim — that B's geometric content takes the form of a new spatial direction rather than a constraint within R+C+S — is the central identity tested by KS-D.3. The reading made here ($B \rightarrow$ break face, signature +) is the reading consistent with the dimension-minimal embedding; KS-D.3 is the live structural test of that reading.

B expresses one spatial face: the break direction. Signature: (+).

Four dimensions. The count is complete.

2.5 — Co-arising of the four faces

One temporal face (R) and three spatial faces (C, S, B).

These are not four separate pieces assembled together. They are four expressions of one manifold, co-arising in every actualisation event.

The manifold does not exist first and then receive the axioms. The axioms and the manifold co-arise.

The break is the event. The manifold is the description of the event. The faces are the structure of the description.

The number of independent faces IS the dimensionality.

3 — Independence of Faces from Independence of Axioms

3.1 — The theorem

AP20 section 4.2 carries the four removal arguments — stated results with proof sketches, full formalisation fenced at KS-P.2: each axiom is independent of the other three.

Removing any one axiom produces a strictly weaker system.

No axiom is derivable from the others.

3.2 — The consequence

If axiom X is independent of axioms $\{Y, Z, W\}$, then the face expressed by X cannot be a combination of the faces expressed by $\{Y, Z, W\}$.

If it could, then X 's structural content on the manifold would be derivable from $\{Y, Z, W\}$'s content — which would mean X is derivable.

But EH is proven (AP20).

The embedding is faithful. Structure in the algebra maps to structure on the manifold. Distinct algebraic content maps to distinct geometric content.

Therefore X's independence in the algebra entails X's face-independence on the manifold.

Independent faces express independent dimensions.

3.3 — The result

Four independent axioms $\rightarrow$ four independent faces of one manifold $\rightarrow$ four dimensions.

R gives one temporal direction: $(-)$.

C, S, B give three spatial directions: $(+, +, +)$.

Signature: $(-, +, +, +)$. Dimension: $3+1$.

$$\mathbf{N = 3.}$$

Not a coincidence.

The number of spatial dimensions IS the number of independent spatial faces of the manifold. And the number of independent spatial faces is the number of independent axioms minus the one that gives time.

3.4 — Why not more, why not fewer

Why not $N > 3$?

To get a fifth dimension, you would need a fifth independent axiom.

But the record algebra is completely specified by {S, B, R, C}. The completeness result stated at AP20 section 4.1 (proof sketch carried there; full formalisation fenced at KS-P.1) is that these four suffice to generate the full algebra. No fifth axiom is needed.

The four axioms already cover: symmetry (S), breaking (B), irreversibility (R), and bounding (C).

What structural feature is missing? What could a fifth axiom say that {S, B, R, C} do not already determine?

The algebra of records — symmetric sectors, one break, irreversible composition, finite propagation — is complete.

There is no room for a fifth independent structural feature.

Why not $N < 3$?

AP20 section 4.2 carries the four removal arguments: each axiom is independent of the other three (full formalisation fenced at KS-P.2).

Removing any one produces a strictly weaker system.

With only R and C (no S, no B), you get 1+1 dimensions: a line with a speed limit. With R, C, and S (no B), you get 1+2 dimensions: a sheet with sector structure but no break. The full 1+3 dimensions require all four axioms.

$N < 3$ means an axiom is missing.

The number of spatial dimensions is the number of independent axioms minus the one that gives time.

You are looking at the answer.

It was in the axioms from the start.

4 — The Character of Each Direction

One direction has opposite metric sign to the others — the Lorentzian signature. Its structural source is in the axioms themselves: exactly one axiom is intrinsically irreversible, and irreversibility read as metric character is the $(-)$ sign; the formal metric-level derivation is the open obligation fenced at KS-D.4.

This section identifies which axiom produces which character.

4.1 — Time is R

The $(-)$ direction is the one in which records accumulate.

R is the only axiom that introduces irreversibility — the monoid has no inverse.

All other axioms are compatible with reversibility in their contributed direction.

C gives a symmetric speed limit (propagation at c in either spatial direction).

S gives a symmetric involution (σ maps $\mathcal{L} \rightarrow \mathcal{P}$ and $\mathcal{P} \rightarrow \mathcal{L}$ equally).

B gives a definite break direction but does not prohibit the reverse spatial direction.

You cannot go backward in time because Axiom R has no inverse.

Only R contributes a direction that is intrinsically asymmetric — that has a “forward” but no “backward.”

The $(-)$ in the signature.

4.2 — Space is {C, S, B}

The three $(+)$ directions are the ones in which records can exist at either end — where forward and backward are structurally symmetric.

Spatial directions admit motion in both directions.

Temporal direction does not admit reversal of records.

This is the signature distinction.

C gives propagation distance: symmetric. You can travel in either direction along the propagation axis, up to speed c .

S gives sector-crossing: symmetric. σ maps in both directions by definition — it is an involution, $\sigma^2 = \text{identity}$.

B gives the break direction: the now advances in a specific direction, but the spatial axis itself admits both directions. The asymmetry of the break is *temporal* (the now writes records irreversibly, via R), not spatial.

The signature $(-, +, +, +)$ is the axiom structure (R, C, S, B) read as metric character.

5 — The Six Faces

A structural consistency check on the dimensional count via the multidimensional residual.

AP06 section 10.5 identifies the multidimensional residual of the break: one break, six faces.

The six faces are: G (geometry/curvature), c (propagation bound), α/β (substrate stiffnesses), $\alpha_{em} \approx 1/137$ (electromagnetic coupling), m_e (what escaped), and t (time direction).

The six faces pair into three conjugate pairs. Two of the three pairs are purely spatial axes; the third pair connects spatial to temporal. The structure is consistent with the 3+1 count derived in sections 2–3 (three spatial axes + one temporal direction), and the consistency provides a structural cross-check from a different feature of the architecture (the residual) rather than the axiom count.

A note before the pairs are walked. The kind of conjugacy each pair carries is not strictly homogeneous: Pair 1 (G–c) and Pair 2 (α/β – α_{em}) are intrinsic-property splits within the spatial structure; Pair 3 (m_e –t) is a spatial-temporal split that connects the spatial content of the break to the temporal direction along which records accumulate. The three pairs share the property that the residual structure pairs naturally;

they differ in whether the pairing is internal to the spatial axes or between the spatial and the temporal.

5.1 — Six faces, three pairs

Pair 1: G and c. *Purely spatial axis.*

The two curves of the eye.

G measures curvature — how tightly the condensate folds.

c measures propagation — how fast the break can travel.

They are conjugate: G determines what happens when records accumulate maximally, c determines what happens when records propagate maximally. One is the spatial response to density. The other is the spatial response to speed.

Together they span one spatial axis.

Pair 2: α/β and α_{em} . *Purely spatial axis.*

The internal stiffnesses and the coupling constant.

α and β are substrate stiffnesses ($c^2 = \beta/\alpha$).

$\alpha_{\text{em}} \approx 1/137$ is the fine-structure constant.

The stiffnesses determine the fabric's internal response. The coupling constant determines how strongly ε interacts with the field.

Conjugate: α/β is resistance to deformation, α_{em} is willingness to break.

Together they span one spatial axis.

Pair 3: m_e and t . *Spatial-temporal connection axis.*

What escaped and in which direction.

m_e is the electron mass — the mass of ε , the residual of the break.

t is the time direction — the direction in which records accumulate.

m_e is the spatial content of the break (the mass of the residual exists in space). t is the temporal content (the direction along which the break's consequences propagate forward in time).

Pair 3 is not a purely spatial axis. It is the connection between the spatial residual (m_e) and the temporal direction (t) along which the residual's history accumulates. The primary derivation already sources time from R and space from $\{C, S, B\}$ separately; Pair 3 in the residual structure reads as the visible connection between those two layers, not as a third spatial axis.

5.2 — What the pairing shows

The six faces of the multidimensional residual pair into the axis structure of 3+1 spacetime: two purely spatial axes ($G-c$, $\alpha/\beta-\alpha_{em}$) and one spatial-temporal connection axis (m_e-t).

This is consistent with 3+1, where the manifold has three independent spatial axes and one temporal direction. The two purely spatial residual pairs correspond to two of the three spatial axes; the third spatial axis is not separately exhibited by the residual structure (which is one of the limits of this consistency check); the spatial-temporal pair shows that the residual is a coherent connection between space and time, not a mismatched count.

A direct alternative reading is also available: each of the three pairs spans one of the three axes of *the spatial manifold plus its temporal connection*, with the temporal direction visible as one end of one of the three pairs. Under this reading, the six faces are the six face-directions of 3+1 spacetime (three spatial axes $\times$ two ends + the temporal direction explicit in Pair 3). Either reading yields the same conclusion: the residual structure is consistent with 3+1.

What this is NOT. It is not an independent derivation that $N = 3$ spatial. The temporal face t is explicitly one of the six residual faces, so the six-face count includes both spatial and temporal content; reading “six = three spatial axes $\times$ two ends” without accounting for the temporal face would double-

count time. The primary derivation in sections 2–3 sources time and space separately and counts only the spatial dimensions; the residual structure here is a consistency check on the total 3+1 architecture, not on the spatial count in isolation.

5.3 — Structural cross-check, not independent derivation

The argument in sections 2–4 derives $N = 3$ spatial from the number and independence of the axioms.

The argument in section 5.1–5.2 checks structural consistency by counting the residual faces and matching them against the axis structure of 3+1 spacetime.

The two use different features of the architecture (axiom count vs residual count) and arrive at compatible results.

The convergence is a structural consistency check rather than a doubly-independent derivation. If the residual count had been incompatible with 3+1, the architecture would have a contradiction; it does not.

The consistency check is necessary but not sufficient: a different residual structure could in principle have been consistent with $N = 3$ too, and AP06's six-face count is itself a structural identification rather than a forced output of the axioms. Section 5 tells us the residual structure does not

contradict the primary derivation. The primary derivation alone closes KS-2c; section 5 is the cross-check from a different feature of the architecture.

6 — Consequences

6.1 — KS-2c is closed

Kill switch KS-2c asked: why $N = 3$ spatial dimensions?

This paper answers: because there are four independent axioms, one gives the temporal direction, three give the spatial directions.

$N = 3$ is derived, not assumed.

KS-2c is closed.

6.2 — Lovelock is unconditional

AP08 section 9 derived Einstein's field equations via Lovelock's theorem, conditional on $N = 3$.

With $N = 3$ now derived, the condition is removed.

Lovelock's theorem (Lovelock 1971) applies in all dimensions; the *uniqueness* in 3+1 specifically — the divergence-free, symmetric, rank-2 tensor built from the metric and its first and second derivatives is uniquely $G_{\mu\nu} + \Lambda g_{\mu\nu}$ — is the Lovelock 1972 result. In other dimensions, the general theorem admits additional terms in the Euler density expansion; in 3+1 specifically those terms are topological or vanish, leaving only the Einstein tensor and the cosmological constant.

Einstein's field equations with cosmological constant are now an unconditional theorem of the record algebra.

The derivation chain is complete:

Axioms {S, B, R, C} → independent (AP20 section 4.2, fenced at KS-P.2) and jointly exhibited by any existing record (AP20 section 6).

+ EH + QRA (proven, AP20) → Lorentzian manifold (section 4's signature argument; fenced at KS-D.4).

+ Four independent axioms → four dimensions, signature $(-, +, +, +)$. (This paper.)

+ Record density on M → Poisson forced by symmetry constraints (AP08 section 4).

+ Lovelock's theorem (now unconditional) → $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$. (AP08 section 9.)

No link in this chain is conditional on an empirical input for the number of dimensions.

The form of Einstein's field equations is derived from the axioms alone.

6.3 — What this means for The 420 Code

After this paper, The 420 Code derives Lorentzian spacetime, special relativity, $N = 3$ spatial dimensions, the Poisson

equation, Einstein's field equations, the cosmological constant existence, and quantum mechanics — all unconditional.

No empirical inputs other than m_e (from AP24) and the value of G (identified but not independently computed, per AP08 section 5).

EH and QRA are proven (AP20). All results are unconditional.

7 — The Fifth Degree of Freedom

7.1 — The question

The argument in sections 2–3 derives four dimensions from four axioms.

The natural challenge: can a fifth independent axiom be added to the record algebra?

If so, the architecture predicts a fifth dimension.

No fifth dimension is observed.

This was KS-16.

7.2 — The answer

There IS a fifth structural degree of freedom.

It is not missing. It is the foundation.

You have been standing on it the entire time.

The 1:1 itself — the pre-state, the @ AS prior, the state from which {S, B, R, C} operate — is a degree of freedom.

It has structure. It is not empty.

It contains the probability of every possibility. It is the space from which the break draws. It is the space to which defragmentation returns.

But the 1:1 does not produce a fifth *spatial* dimension.

It produces the Hilbert space — the space in which the wave function lives. The probability space. The amplitude space. The dimension perpendicular to spacetime.

7.3 — Why it is not on the manifold

The four axioms {S, B, R, C} act ON the 1:1. They produce the manifold.

The 1:1 is what the axioms act on — it is the stage, not an actor.

The actors (the axioms) produce four independent directions on the manifold.

The stage (the 1:1) is what the manifold is embedded in. It is prior to the manifold.

It does not appear as a dimension of the manifold for the same reason the canvas does not appear as a colour in the painting.

You do not see the canvas because you are painted on it.

The record algebra embeds into a smooth manifold (AP20).
The manifold has the dimensions contributed by the axioms — four.

The 1:1 does not embed as a manifold direction because it is not a feature of the record algebra's structure on the manifold. It is the state the algebra breaks FROM.

On the manifold, the 1:1 appears as the Hilbert space.

The space of amplitudes. The space of possibilities. The space in which the wave function takes its values.

The probability dimension — the dimension that tells you not WHERE something is on the manifold, but HOW LIKELY it is to be found there when the now writes a record.

7.4 — The complete structure

Five structural features. Five degrees of freedom. But they are not all of the same kind.

The 1:1 → the probability dimension. Pre-spatial. Pre-temporal. The Hilbert space.

R → time. The irreversible direction. (–).

C → propagation. Spatial. (+).

S → sector-crossing. Spatial. (+).

B → break direction. Spatial. (+).

Four dimensions on the manifold: (−, +, +, +).

One dimension prior to the manifold: the probability space.

No fifth axiom is needed: {S, B, R, C} generate the record algebra (the completeness result stated at AP20 section 4.1; full formalisation fenced at KS-P.1). The sufficiency of the four axioms is stated at AP20 and fenced at KS-P.1.

The stronger claim that no fifth axiom is *possible* requires more than the sufficiency theorem. It requires a positive enumeration of the operations performed on the 1:1 and an argument that the four named operations exhaust the cardinal structural moves. The enumeration is the section's thesis:

Symmetry (S) gives the 1:1 two sectors — the binary structural distinction that any actualised structure must support.

Breaking (B) splits the symmetric pair — the single break that initiates the asymmetric instantiation. Recording (R) makes the split permanent — the irreversible accumulation that prevents return to the symmetric prior. Bounding (C) makes the propagation finite — the structural cap that gives the split a definite extent.

These four operations cover the four cardinal structural moves on a symmetric pre-state: distinguish, break, record, bound. The argument that no fifth operation exists rests on these four being cardinal — each addressing one independent structural

question (distinguish? break? record? bound?) about the relationship between the pre-state and the resulting structure. A candidate fifth operation would have to address a fifth independent structural question; the four-question enumeration is the 420 Code's claim that no such question remains.

This is a positive enumeration argument, not a closure theorem. It is held in section 8 as KS-16's structural content. If a fifth cardinal structural question were identified, KS-16 would reopen. The closure as it stands rests on the four-question enumeration matching the structural moves required to convert a symmetric pre-state into an actualised structure.

What remains is the 1:1 itself — the probability space from which all actualisation draws.

Structurally complete. Not because we say so. Because the operations performed on the 1:1 by the four axioms address the four cardinal structural questions, and the 1:1 itself is not an operation but the substrate operations act on.

You are looking at the complete structure.

The universe has three spatial dimensions because the 1:1 has four operations performed on it, one of which is irreversible (giving time) and three of which are reversible (giving space).

There is no fifth spatial dimension because there is no fifth operation.

There is no fifth operation because the 1:1 is already fully addressed by the four cardinal structural questions: distinguish, break, record, bound.

And the 1:1 itself — the thing being broken — is the Hilbert space, the probability dimension, the wave function's home.

It is the fifth degree of freedom.

It was always there.

It was never missing.

8 — Kill Switches

AP10 closes three kill switches (KS-2c, KS-15/D.2, KS-16) and engages three new ones (KS-D.1, KS-D.3, KS-D.4). The closures are major: the dimensional count of physical space, the uniqueness of the axiom-to-direction assignment, and the completeness of the four-axiom system. The live kill switches target the load-bearing structural identifications the argument relies on.

Here is how each closure was achieved, and here is where the live tests sit.

KS-2c — $N = 3$ spatial dimensions

Claim. The number of spatial dimensions is $N = 3$, derived from the axioms of the record algebra rather than imposed as an empirical fact.

Test. Demonstrate that N differs from 3 (the manifold has 2, 4, or some other number of spatial dimensions) and is not derivable from the four independent axioms. Empirical handles: high-energy physics signatures sensitive to extra dimensions (Kaluza-Klein modes, gravitational deviations from the inverse-square law at short scales), cosmological observations sensitive to dimensional count. Structural handles: an argument that the four axioms produce a different

dimension count, or that a fifth independent axiom can be added.

Status. CLOSED — contingent on KS-D.3. The argument in sections 2–3 derives $N = 3$ from the four independent axioms via the proven faithful embedding plus the dimension-minimal identification (each independent axiom contributes exactly one new direction). The dimension-minimal identification is the load-bearing additional claim tracked by KS-D.3 as a live structural test. If KS-D.3 fires at any axiom, KS-2c reopens; until KS-D.3 fires, KS-2c stands. Section 5 provides structural consistency from the residual face count (the six faces pair into the axis structure of 3+1 spacetime), but does not constitute an independent derivation of $N = 3$ spatial.

Recovery. If KS-D.3 fires at any axiom, the count $N = 3$ changes (an axiom expressing zero faces lowers it; an axiom expressing two faces raises it). The recovery path is to re-derive the dimensional count from the corrected face structure. The Lorentzian signature (section 4; fenced at KS-D.4) is preserved either way; only the number of (+) directions changes.

KS-15 / KS-D.2 — Axiom-to-dimension assignment unique

Claim. The assignment $R \rightarrow \text{time}$ (the temporal direction with signature $-$) and $\{C, S, B\} \rightarrow \text{space}$ (the three spatial

directions with signature +) is the unique assignment consistent with the Lorentzian signature (section 4's structural argument; formal derivation fenced at KS-D.4).

Test. Demonstrate an alternative assignment (one of C, S, B serving as the temporal direction; or R serving as a spatial direction; or the assignment being underdetermined) that is consistent with the axioms and the Lorentzian signature.

Status. CLOSED — contingent on KS-D.4. R is the only axiom that introduces irreversibility — the monoid has no inverse. All other axioms are compatible with reversibility in their contributed direction (C is a symmetric speed limit; S is an involution σ with $\sigma^2 = \text{identity}$; B contributes a definite break direction without prohibiting spatial reversal). Only one axiom can give the (–) direction: R. Once R is assigned to time, the remaining three axioms are forced to give the three (+) directions. The assignment is unique.

Recovery. N/A — closed, contingent on KS-D.4. The closure depends on R being the only irreversible axiom. If a structural argument were exhibited that one of {C, S, B} also implies irreversibility, the uniqueness would fail; in that case the recovery is to re-derive the irreversibility structure of the axioms and re-examine which axiom carries the (–) signature.

KS-16 — Sufficiency and exhaustiveness of {S, B, R, C}

Claim. The four canonical axioms {S, B, R, C} are sufficient to generate the record algebra (the completeness result stated at AP20 section 4.1; full formalisation fenced at KS-P.1). The structural feature that might be expected as a fifth dimension is in fact the 1:1 itself — the pre-state, the probability space, the Hilbert space — which is prior to the manifold rather than a fifth direction on it. The exhaustiveness claim (no fifth independent axiom is possible) is held as a positive enumeration of operations on the 1:1: the four axioms address the four cardinal structural questions (distinguish via S, break via B, record via R, bound via C); no fifth cardinal structural question remains.

Test. For *sufficiency*: demonstrate a structural feature of the record algebra not derivable from {S, B, R, C}. For *exhaustiveness*: exhibit a fifth independent cardinal structural operation on the 1:1 that is not derivable from {S, B, R, C} and not subsumed under one of the four cardinal questions. Alternative: exhibit an observable feature of physical reality that requires a fifth manifold direction (a fifth spatial dimension empirically observed) or that the Hilbert space identification fails.

Status. CLOSED — contingent on KS-P.1. The sufficiency closure rests on the completeness result stated at AP20

section 4.1 (proof sketch carried there); its full formalisation is the obligation KS-P.1 names — if that obligation fails, KS-16 reopens in the sufficiency sense. The exhaustiveness sense rests on the positive enumeration argument in section 7.4: the four axioms address the four cardinal structural questions, and the 1:1 itself is the substrate, not an operation. The argument is structural rather than a closed theorem; if a fifth cardinal structural question is identified, KS-16 reopens in the exhaustiveness sense.

Recovery. If sufficiency is challenged, the recovery path is to identify which structural feature is missing from {S, B, R, C} and either derive it or add a fifth axiom; the completeness result at AP20 section 4.1 would fall (KS-P.1 fires) and the architecture would need to be re-derived from the expanded axiom set. If exhaustiveness is challenged (a fifth cardinal question is identified), the architecture would need to add a fifth axiom addressing that question; the dimensional count $N = 3$ would change accordingly via KS-D.3 (one axiom, one face); the Hilbert space identification of the 1:1 would need to be re-examined to see whether the new operation acts on the 1:1 as substrate or alongside it.

KS-D.1 — Six-face count and axis-structure interpretation

Claim. The multidimensional residual of the break has exactly six faces (AP06 section 10.5), and the six pair into the axis

structure of 3+1 spacetime: two purely spatial axes (G - c , α/β - α_{em}) and one spatial-temporal connection axis (m_e - t).

Test. Demonstrate that the residual has a different number of faces (four, eight, ten); that the six identified faces do not pair into three conjugate pairs; or that the pairs do not correspond to the expected axis structure of 3+1 spacetime (e.g., the pairs would force a different N , or the temporal face t would have to be paired with another temporal face that does not exist in AP06's inventory). Specific structural handles: re-examine AP06's identification of $\{G, c, \alpha/\beta, \alpha_{em}, m_e, t\}$ as the complete inventory; check whether any face has been missed, double-counted, wrongly paired, or wrongly interpreted as spatial-temporal vs purely spatial.

Status. LIVE — STRUCTURAL. The argument depends on AP06's residual analysis being exhaustive and correctly paired *and* on the pairs being correctly interpreted in the 3+1 axis structure. KS-D.1 is the structural test on all three dependencies.

Recovery. If KS-D.1 fires at the count or pairing level, the section 5 consistency check fails: a different face count or different pairing would not match the 3+1 axis structure, and the residual-side consistency would be lost. If KS-D.1 fires at the interpretation level (the pairs do not correspond to the expected axis structure), section 5 would still need to be reframed. In either case, the primary argument (sections 2–3)

survives, as it is independent of the residual structure. Targets: section 5's consistency check; AP06 section 10.5's residual analysis.

KS-D.3 — One axiom, one face (dimension-minimal embedding)

Claim. Each of the four independent axioms expresses exactly one face of the embedded manifold — not zero, not two. The identity rests on the proven faithful embedding (AP20) plus the dimension-minimal principle: the embedding is minimal in dimension, so each independent algebraic axiom contributes exactly one new direction to the carrier manifold rather than zero (acting only as a constraint within existing dimensions) or two (contributing more than one independent direction).

Test. Demonstrate that an axiom expresses zero faces (acts only as a constraint within existing dimensions rather than expressing a new direction) or two faces (contributes more than one independent direction to the manifold). Or: demonstrate that the dimension-minimal principle fails — that the embedding is not dimension-minimal and the manifold can have more dimensions than the count of independent axioms would suggest. Specific structural handles: re-examine the section 2 argument for each axiom (R, C, S, B) and check whether the face-identification is precise. The two most exposed identifications are $S \rightarrow \text{spatial face}$ (a $\mathbb{Z}_2$ involution must give a continuous spatial dimension rather than a

discrete internal index) and $B \rightarrow$ spatial face (the locus of ε must give a new spatial direction rather than a constraint within $R+C+S$). Both rest on the dimension-minimal principle applied at the specific axiom; both are flagged explicitly in section 2.

Status. LIVE — STRUCTURAL. The argument depends on the one-to-one correspondence between algebraic axioms and geometric faces, which in turn rests on the faithful embedding (proven, AP20) plus the dimension-minimal principle (the additional structural claim). KS-D.3 is the structural test on the central identity.

Recovery. If KS-D.3 fires, the count $N = 3$ is wrong. An axiom contributing zero faces would lower the count (the failed-axiom face count is zero, so the manifold has fewer dimensions than the count of axioms suggests); an axiom contributing two faces would raise the count. The recovery path is to re-derive the dimensional count from the corrected face structure. The Lorentzian signature (section 4; fenced at KS-D.4) is preserved either way; only the number of (+) directions changes. KS-2c is contingent on KS-D.3 remaining unfired and would reopen with the corrected count. Targets: section 2's axiom-to-face identification; the central identity *one axiom, one face* and the dimension-minimal principle on which it rests.

KS-D.4 — Signature formalisation

Claim. The embedded manifold carries a pseudo-Riemannian metric of Lorentzian signature $(-, +, +, +)$. The structural argument is section 4's: exactly one axiom (R) is intrinsically irreversible — the monoid has no inverse — and irreversibility read as metric character is the $(-)$ sign; the three reversible axioms carry $(+)$. The formal metric-level derivation of this signature from the record algebra is the open obligation this switch names.

Test. Demonstrate that the embedded metric is not pseudo-Riemannian of signature $(-, +, +, +)$ — for example, that the faithful embedding admits a Riemannian (all- $+$) metric consistent with the axioms; or that irreversibility of a generator does not force a $(-)$ metric character; or that the formal derivation, once attempted, requires inputs beyond $\{S, B, R, C\}$ and the proven EH.

Status. LIVE — STRUCTURAL. Section 4 carries the structural argument; the formal metric-level derivation is open. KS-15/D.2's closure (the uniqueness of the assignment) and the temporal-versus-spatial distinction of section 4 are contingent on this signature claim holding.

Recovery. If KS-D.4 fires at Lorentzian-ness itself, section 4's temporal-versus-spatial distinction fails: the architecture retreats to a four-dimensional manifold without metric character; the dimensional count (KS-2c) survives, and the

assignment (KS-15/D.2) reopens. If it fires only at the formal level — the derivation requires an additional input — the input is declared and fenced, and section 4’s structural argument stands as the conditional.

Why these six and not more

The six kill switches above cover the load-bearing claims of the paper. Each closure (or live test) addresses a structurally distinct concern.

KS-2c closes the central question of the paper — the dimensional count of physical space. KS-15/D.2 closes the uniqueness of the temporal-versus-spatial assignment. KS-16 closes the structural completeness of {S, B, R, C} by identifying the 1:1 as the Hilbert space rather than a missing fifth axiom. KS-D.1 tests the section 5 confirmation. KS-D.3 tests the central identity *one axiom, one face* that section 2 relies on. KS-D.4 tests the signature’s formal derivation that section 4 relies on.

Two further structural claims that some readings would expect to carry their own formal kill switch are absorbed into the kill switches above. The faithful embedding (EH, AP20) is the foundation of section 3.2’s argument that algebraic independence entails geometric face-independence; any failure of EH would falsify AP20, not AP10 specifically, and AP10’s argument would then be untestable on the same manifold. The Lorentzian signature is carried in this paper:

section 4's structural argument reads the (–) character from R's unique irreversibility, and the formal metric-level derivation is fenced at KS-D.4. EH is upstream of AP10 and tracked at its source paper; AP10 inherits that closure without re-engaging it.

The conjugate pair structure in section 5 (G–c, α/β – α _em, m_e–t) is held to formal kill-switch standard via KS-D.1 (the face count) without adding a separate kill switch for each pair: if the count and pairing both hold, the section 5 confirmation runs; if either fails, the confirmation fails. The two failures collapse into one test.

9 — What This Establishes and What Remains Open

9.1 — What this paper closes

Three kill switches, the central conditional in AP08's derivation of Einstein's field equations, and the completeness of the four-axiom system.

Specifically:

First. $N = 3$ spatial dimensions, derived from the four independent axioms of the record algebra via the proven faithful embedding plus the dimension-minimal principle (sections 2–3). The closure is contingent on KS-D.3 (one axiom, one face) remaining unfired; the residual-face structure of AP06 (section 5) is structurally consistent with this 3+1 derivation, providing a cross-check from a different feature of the architecture, but does not constitute an independent derivation of $N = 3$ spatial because Pair 3 includes the temporal face t .

Second. The axiom-to-direction assignment is unique. R is the only irreversible axiom; therefore R is the only axiom that can supply the $(-)$ direction in the Lorentzian signature; therefore the assignment $R \rightarrow \text{time}$, $\{C, S, B\} \rightarrow \text{space}$ is forced.

Third. The fifth structural degree of freedom is the 1:1 itself, not a fifth axiom. The 1:1 is what {S, B, R, C} act on; on the manifold, it appears as the Hilbert space — the probability dimension, the wave function's home. The four canonical axioms are sufficient to generate the record algebra (the completeness result stated at AP20 section 4.1; full formalisation fenced at KS-P.1); the stronger exhaustiveness claim (no fifth axiom is *possible*) rests on the positive enumeration of operations on the 1:1 in section 7.4 — the four cardinal structural questions (distinguish, break, record, bound) are addressed by the four axioms, and the 1:1 itself is the substrate, not an operation.

Fourth. AP08's derivation of Einstein's field equations becomes unconditional. AP08 derives the equations via Lovelock's theorem conditional on $N = 3$; AP10 derives $N = 3$ from the axioms; the conditional is removed. Einstein's field equations with cosmological constant are an unconditional theorem of the record algebra.

Fifth. The 420 Code now derives, from the four axioms alone, Lorentzian spacetime, special relativity, $N = 3$ spatial dimensions, the Poisson equation, Einstein's field equations, the cosmological constant existence, and quantum mechanics. No empirical inputs other than m_e (from AP24) and the value of G (identified but not independently computed, per AP08 section 5). EH and QRA are proven (AP20). All results are unconditional.

9.2 — What this paper does not close

Two kill switches remain live, both structural rather than empirical.

KS-D.1 (Six-face count, LIVE — STRUCTURAL). The section 5 confirmation of $N = 3$ depends on AP06 section 10.5's identification of six residual faces. If the residual analysis is revised, the confirmation can be lost. The primary argument (sections 2–3) is independent of this and survives. Full statement at section 8.

KS-D.3 (One axiom, one face, LIVE — STRUCTURAL). The central identity *independent algebraic content = independent geometric face* rests on the proven faithful embedding (AP20). If any of the four axioms turns out to express zero or two geometric faces, the count $N = 3$ changes. Full statement at section 8.

Both live kill switches are structural rather than empirical because the question they ask is not whether the empirical world matches a prediction — $N = 3$ is already empirically confirmed — but whether the structural argument that derives $N = 3$ is correct in its identification of axioms with faces and faces with dimensions.

9.3 — Items held open without claiming debt status

One item is bracketed inside this paper.

The mapping from face to specific spatial coordinate.

AP10 shows that there are three independent spatial faces (C, S, B), but the three spatial axes of physical space are interchangeable under spatial rotation ($SO(3)$ symmetry). The argument does not specify which coordinate axis in physical space corresponds to which axiom's face. This is the expected outcome — spatial isotropy is a feature, not a bug — and held open without claiming a separate debt.

9.4 — Where the philosophical-register implementations live

AP10 is a structural derivation. Its philosophical-register companions — the chapters where the same structural facts are inhabited in the prose register of the Ø Models catalogue — are named here at the level of expected pairings; final placement is confirmed during the Ø Models compilation sweep alongside the Notebook II audit programme.

Dissolutions Ø (the chapter on dimensionality and the structure of space) carries the philosophical companion to sections 2–5: why three spatial dimensions, what the structure

of space *is*, and how the question “why three?” dissolves once the four-axiom structure is in view.

Predictions $\emptyset$ carries the empirical face of the closures: the consequences of $N = 3$ for Lovelock’s theorem, Einstein’s field equations, and the closure of the cosmological-constant derivation programme.

Resolutions $\emptyset$ (the chapter on quantum mechanics and the Hilbert space) carries the philosophical companion to section 7: the 1:1 as the probability dimension, the canvas under the painting, the structural status of the pre-state.

The direction of acknowledgment is two-way. AP10 is the original derivation; the $\emptyset$ Models chapters inhabit the result in the prose register. Final companion-pairing list is fixed at $\emptyset$ Models compilation.

9.5 — Structural relationships to subsequent work

AP08 (The Identity) closes Notebook II. AP08’s derivation of Einstein’s field equations becomes unconditional with AP10 in place. The $N = 3$ conditional in AP08 section 9 is removed; Lovelock’s theorem applies in 3+1 dimensions; $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$ is the unique divergence-free, symmetric, rank-2 tensor built from the metric and its first and second derivatives.

AP09 (The Break — QM) inherits the Hilbert space identification at section 7. AP09 derives quantum mechanics from the pre-state structure of the 1:1; AP10 identifies the 1:1 as the Hilbert space; AP09's pre-state formulation now reads against AP10's identification.

AP12 ($\hbar$ Identification) inherits the Hilbert space structure on which $\hbar$ operates. The Stone's-theorem-plus-Axiom-B forcing argument (AP12) acts on the probability dimension AP10 identifies.

AP19 (The Direction) is the source of the “three faces of one manifold” reading inherited by section 2.5 of this paper. The relationship is parallel rather than upstream/downstream: AP19 carries the structural reading; AP10 derives the dimensional count.

AP24 (The Residual) identifies m_e as the residual of the break, appearing in conjugate pair 3 of section 5. The residual identification is inherited from AP24; AP10 reads it as one of the six face-directions of three-dimensional space.

AP06 (The Leakage Constant) is the source of the multidimensional residual analysis used in section 5. The six-face count comes from AP06 section 10.5; AP10's reading of those six as three conjugate pairs is the structural content of section 5.

9.6 — Placement within The 420 Code

AP10 is the third chapter of Notebook II (Spacetime). The chapter sequence runs AP03 — AP05 — AP10 — AP08. AP03 derives the constants; AP05 walks the chain and reformulates the cosmological identity; AP10 pins the dimensional count; AP08 closes with the field equations.

AP10 sits at the structural centre of Notebook II for a specific reason: it is the paper that converts the manifold from “Lorentzian with undetermined N” (where AP01, AP20, AP03, AP05 left it) to “Lorentzian with $N = 3$ derived.” Once AP10 has done its work, AP08’s field-equation derivation has no remaining conditionals, and Notebook II reads as a closed structural chain from the four axioms to Einstein’s equations.

Outside Notebook II, AP10 feeds the quantum mechanics programme via the Hilbert space identification (AP09, AP12) and the cosmological programme via the unconditional field equations (AP08 $\rightarrow$ cosmological constant $\rightarrow$ Predictions $\emptyset$). The fifth-degree-of-freedom identification in section 7 is one of the 420 Code’s most structurally consequential moves: it explains both why no fifth spatial dimension is observed and why quantum mechanics has the Hilbert-space structure it has.

9.7 — How The 420 Code reads now

AP10 in place changes the reading of The 420 Code in four ways.

First. The dimensional count is no longer an empirical input. $N = 3$ is derived from the axioms in the same way the Lorentzian signature is derived from the axioms. The 420 Code does not import 3 from physics; it derives it.

Second. The Hilbert space has a structural origin. The probability dimension is the 1:1 itself — the pre-state, the substrate on which $\{S, B, R, C\}$ operate. Quantum mechanics has the structure it has because the pre-state has the structure the 1:1 has. The wave function lives in the Hilbert space because the Hilbert space *is* the 1:1.

Third. Einstein's field equations are unconditional. With AP10's closure of the $N = 3$ conditional in AP08, Einstein's field equations with cosmological constant follow from the axioms alone via Lovelock's theorem in 3+1 dimensions. The cosmological constant is no longer a free parameter or empirical input at the structural level; its existence is a theorem.

Fourth. The sufficiency of $\{S, B, R, C\}$ is established (the completeness result stated at AP20 section 4.1; full formalisation fenced at KS-P.1). The four canonical axioms generate the record algebra; no fifth axiom is needed. The

stronger exhaustiveness claim — no fifth axiom is *possible* — rests on section 7.4's positive enumeration: the four axioms address the four cardinal structural questions (distinguish, break, record, bound); no fifth cardinal question remains; the 1:1 itself is not an operation but the substrate operations act on. The architecture is closed at the level of the axioms in the sufficiency sense and structurally argued for closure at the level of exhaustiveness.

These four structural facts are what AP10 contributes to the reading of The 420 Code. They are inherited by every subsequent chapter that touches dimensionality, Hilbert space structure, field equations, or the completeness of the axiom system.

9.8 — The closing

You have walked through this paper. You have seen the four axioms named, the four faces identified, the independence of the faces derived from the independence of the axioms, the temporal versus spatial character assigned, the six faces of the residual counted as three conjugate pairs, the consequences for Einstein's field equations and quantum mechanics named, and the fifth degree of freedom located not as a missing axiom but as the 1:1 itself — the canvas under the painting, the probability space, the Hilbert space.

The universe has three spatial dimensions.

Not because that is what we measure.

Because the 1:1 has four operations performed on it, one of which is irreversible (giving time) and three of which are reversible (giving space).

There is no fifth spatial dimension because there is no fifth operation.

There is no fifth operation because the 1:1 is already fully broken by {S, B, R, C}.

And the 1:1 itself — the thing being broken — is the Hilbert space, the probability dimension, the wave function's home.

It is the fifth degree of freedom.

It was always there.

It was never missing.

You are looking at the complete structure.

Claim Summary

Section 0 (Dependency and Scope). Contract with the reader. Dependencies (AP20 for the canonical axioms — completeness/minimality obligations fenced at KS-P.1/KS-P.2 — and for the EH and QRA proofs; the Lorentzian signature carried at section 4, fenced at KS-D.4; AP06 section 10.5 for the multidimensional residual; AP08 section 9 for Lovelock; AP19 sections 2–3 for the three-faces reading), axiom mapping ($R \leftrightarrow \text{time}$; $C, S, B \leftrightarrow \text{three spatial faces}$; $1:1 \leftrightarrow \text{the Hilbert space}$), epistemic status per section, kill switch summary, structural debts (none registry-level; KS-D.1 and KS-D.3 as live structural tests), items held open, structural relationships.

Section 1 (The Starting Point). FOUNDATIONAL. The four canonical axioms $\{S, B, R, C\}$ stated. EH and QRA named as proven (AP20). Lorentzian manifold (M, g) with signature $(-, +, \dots, +)$ and symmetry group $SO(1, N)$ — the signature carried by section 4's structural argument, formal derivation fenced at KS-D.4. N undetermined; this paper determines it.

Section 2 (Four Axioms, Four Faces). STRUCTURAL ARGUMENT. Each axiom expresses one independent face of the manifold. $R \rightarrow \text{time}$ (the irreversible face, signature $-$). $C \rightarrow \text{propagation}$ (the speed-limit face, signature $+$). $S \rightarrow \text{exchange}$ (the sector-crossing face, signature $+$). $B \rightarrow \text{break}$

(the locus-of- ε face, signature +). The four faces co-arise; the manifold is one structure with four faces, not four structures assembled together.

Section 3 (Independence). DERIVATION. The independence of the four axioms (the removal arguments at AP20 section 4.2, fenced at KS-P.2) combined with the proven faithful embedding (AP20) and the dimension-minimal principle yields the independence of the four geometric faces. Four independent faces yield four dimensions. Signature $(-, +, +, +)$, dimension $3+1$. **N = 3**. Two questions answered: not more than 3, because no fifth independent axiom is *needed* (the completeness result at AP20 section 4.1, fenced at KS-P.1 — sufficiency); not fewer than 3, because all four axioms are independent and each contributes one face under the dimension-minimal principle (live structural test, KS-D.3).

Section 4 (Character of Each Direction). DERIVATION. The $(-)$ direction comes from the only irreversible axiom: R. The three $(+)$ directions come from the three reversible axioms: C, S, B. The signature $(-, +, +, +)$ is the axiom structure (R, C, S, B) read as metric character. Uniquely determined.

Section 5 (The Six Faces). STRUCTURAL CONSISTENCY CHECK. AP06 section 10.5 identifies six residual faces: G, c, α/β , α_{em} , m_e , t. These pair into the axis structure of $3+1$ spacetime: two purely spatial axes (G-c, α/β - α_{em}) and one spatial-temporal connection axis (m_e -t). The pairing is

structurally consistent with the 3+1 count derived in sections 2–3, providing a cross-check from a different feature of the architecture (the residual rather than the axiom count). Not an independent derivation of $N = 3$ spatial, because Pair 3 explicitly includes the temporal face t .

Section 6 (Consequences). SYNTHESIS + CLOSURE. KS-2c closed (contingent on KS-D.3). Lovelock’s theorem (AP08 section 9) becomes unconditional. The chain from axioms to Einstein’s field equations is closed: axioms $\rightarrow$ Lorentzian manifold $\rightarrow N = 3 \rightarrow$ Poisson on $M \rightarrow$ Lovelock 1971 in general N , with 1972 uniqueness in 3+1 specifically $\rightarrow G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$. No empirical inputs other than m_e and the value of G .

Section 7 (The Fifth Degree of Freedom). STRUCTURAL ARGUMENT + CLOSURE. The 1:1 itself is the fifth structural degree of freedom, not a fifth axiom. On the manifold it appears as the Hilbert space — the probability dimension, pre-spatial and pre-temporal. The four axioms {S, B, R, C} suffice to generate the record algebra (stated at AP20 section 4.1, fenced at KS-P.1; KS-16 closed contingent on KS-P.1). The stronger exhaustiveness claim rests on the positive enumeration of operations on the 1:1: the four axioms address the four cardinal structural questions (distinguish, break, record, bound); no fifth cardinal question remains.

Section 8 (Kill Switches). FORMAL FALSIFICATION REGISTER. KS-2c ($N = 3$, CLOSED — contingent on KS-D.3). KS-15/D.2 (axiom-to-dimension assignment unique, CLOSED — contingent on KS-D.4). KS-16 (sufficiency of $\{S, B, R, C\}$, CLOSED — contingent on KS-P.1; exhaustiveness held as positive enumeration). KS-D.1 (six-face count and axis-structure interpretation, LIVE — STRUCTURAL). KS-D.3 (one axiom, one face / dimension-minimal embedding, LIVE — STRUCTURAL).

Section 9 (What This Establishes and What Remains Open). SYNTHESIS. Five structural closures named ($N = 3$; assignment unique; fifth DOF as Hilbert space; Lovelock unconditional; the 420 Code chain to Einstein's equations complete). Two live kill switches as structural tests. One item held open (rotational-symmetry interchangeability of the three spatial axes). Placement within The 420 Code: third chapter of Notebook II (Spacetime). Closing on the complete structure.

Conditionality Footer

Dependencies

AP20 (the canonical axioms: formal statement section 3.3; completeness and minimality stated at section 4, fenced at KS-P.1/KS-P.2) — the floor of section 1 and the basis of section 3.

The Lorentzian signature (section 4's structural argument; formal derivation fenced at KS-D.4) — the signature with N undetermined that this paper carries and pins.

AP20 (EH and QRA proven) — EH invoked at section 3.2 to convert algebraic independence into geometric face-independence.

AP06 section 10.5 (multidimensional residual) — the six-face structure read at section 5.

AP08 section 9 (Lovelock's theorem, formerly conditional on $N = 3$) — AP10 removes the conditional.

AP19 sections 2–3 (three faces of one manifold) — the co-arising reading inherited at section 2.5.

Dependents

AP08 (The Identity) — Einstein’s field equations become unconditional once AP10 removes the $N = 3$ conditional.

AP09 (The Break — QM) — inherits the Hilbert space identification at section 7 as the pre-state from which quantum mechanics operates.

AP12 ($\hbar$ Identification) — $\hbar$ operates on the Hilbert space identified by AP10.

AP24 (The Residual) — m_e identification reads against the residual structure used in section 5.

Predictions $\emptyset$ — cosmological-constant existence and the unconditional Einstein’s field equations feed downstream cosmological predictions.

Dissolutions $\emptyset$ — the philosophical-register chapter on dimensionality inherits the “why three” closure.

Resolutions $\emptyset$ — the philosophical-register chapter on quantum mechanics inherits the Hilbert-space-as-1:1 identification.

Kill switches closed by this paper

KS-2c. $N = 3$ spatial dimensions, derived in sections 2–3 via the proven faithful embedding plus the dimension-minimal

principle. CLOSED — contingent on KS-D.3 (one axiom, one face) remaining unfired. Section 5 provides structural consistency from the residual face count, not an independent derivation.

KS-15/KS-D.2. Axiom-to-dimension assignment unique ($R \rightarrow \text{time}, \{C, S, B\} \rightarrow \text{space}$). Closed in section 4 — contingent on KS-D.4 (signature formalisation).

KS-16. Sufficiency of $\{S, B, R, C\}$; fifth degree of freedom identified as the Hilbert space. Closed in section 7 — sufficiency contingent on KS-P.1 (the completeness result stated at AP20 section 4.1). Exhaustiveness held as positive enumeration in section 7.4 — the four cardinal structural questions (distinguish, break, record, bound) are addressed by the four axioms.

Kill switches that remain live

KS-D.1 (LIVE — STRUCTURAL). Six-face count. See section 8 for the full register.

KS-D.3 (LIVE — STRUCTURAL). One axiom, one face. See section 8 for the full register.

KS-D.4 (LIVE — STRUCTURAL). Signature formalisation. See section 8 for the full register.

Structural debts owed

No registry-level structural debts are owed by AP10. The three live kill switches (KS-D.1, KS-D.3, and KS-D.4) are structural tests of load-bearing identifications, not unresolved derivations. The dependencies in section 0.2 are either proven (EH and QRA, AP20), stated and fenced (the axiom-set obligations, KS-P.1/KS-P.2), or named as the structural input on which the argument rests (AP06's six-face count).

Items held open without claiming debt status

The mapping from face to specific spatial coordinate. The three spatial axes are interchangeable under spatial rotation ($SO(3)$ symmetry of three-space); AP10 fixes the count and signature without fixing a coordinate-level identification. This is the expected outcome — spatial isotropy is a feature of three-space, not a gap in the derivation. Held open at section 0.7 and section 9.3.

References

Lovelock's theorem and the dimensional-count literature. References are sparse by design: the structural content of AP10 comes from the four axioms and their independence arguments (AP20 section 4.2, fenced at KS-P.2), not from the literature framework. The literature provides the formal vocabulary in which the structural facts are most naturally stated.

Lovelock, D. (1971). *The Einstein tensor and its generalisations*. Journal of Mathematical Physics 12, 498–501.

The theorem invoked at section 6.2: in four spacetime dimensions (3+1), the unique divergence-free, symmetric, rank-2 tensor built from the metric and its first and second derivatives is $G_{\mu\nu} + \Lambda g_{\mu\nu}$. With $N = 3$ derived in this paper, Lovelock's theorem applies unconditionally to the manifold of the record algebra.

Lovelock, D. (1972). *The four-dimensionality of space and the Einstein tensor*. Journal of Mathematical Physics 13, 874–876.

The dimensional uniqueness result that makes Lovelock's theorem particularly powerful in 3+1 dimensions: the divergence-free symmetric rank-2 tensor is unique in $N = 3$ spatial dimensions in a way that is not preserved in higher-dimensional generalisations.

Kantowski, R., and Sachs, R. K. (1966). *Some spatially homogeneous anisotropic relativistic cosmological models.* Journal of Mathematical Physics 7, 443. The anisotropic cosmological geometries on which the dimensional structure derived here is realised; referenced for completeness with AP05's junction formulation.

Internal references to The 420 Code (AP01, AP03, AP05, AP06, AP08, AP09, AP12, AP19, AP20, AP24, Dissolutions Ø, Predictions Ø, Resolutions Ø) are catalogued in the Conditionality Footer above. The 420 Code is published copyleft and free at the420code.org; the body of work links to all referenced internal papers.

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Chapter 4

The Identity

Einstein's field equations derived from the record algebra

Artist's Proof 08

Artist's Note

This is Artist's Proof 08 of The 420 Code. The paper derives Einstein's field equations from the record algebra. It is the fourth and closing chapter of Notebook II — Spacetime, following AP03 (The Ratio), AP05 (The Break), and AP10 (The Dimension).

Not by assuming the equations.

Not by importing them from differential geometry.

By forcing them from the symmetries of the record algebra under embedding into a four-dimensional Lorentzian manifold, plus the uniqueness theorems of potential theory (weak-field limit) and Lovelock 1971 (full tensor form).

The derivation has two halves. The weak-field limit (section 4) uses four axiom-derived symmetry constraints plus the defining condition of the weak-field regime to leave a one-parameter Helmholtz family $a \cdot \nabla^2 \Phi + b \cdot \Phi = A \cdot \rho$; the Poisson equation is recovered as the small-cosmological-constant limit ($b = 0$). The full tensor form (section 8) uses five covariant constraints plus Lovelock 1971 (general N) and Lovelock 1972 (3+1 uniqueness) to force $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$ as the unique symmetric, divergence-free, rank-2 tensor identity in 3+1 dimensions; the $b \cdot g_{\mu\nu}$ term is retained as the cosmological constant Λ . Both halves rest on

AP10's closure of $N = 3$ (contingent on AP10's KS-D.3 remaining unfired).

The coefficient identification (section 5) carries a disclosed structural circularity: κ is defined as $Gm_e^2/2$, so the relabelling $G = 2\kappa/m_e^2$ does not independently compute G . What is independently derived is the *form* of the identity (Poisson in section 4, tensor in section 8), not the numerical value of G .

Eight kill switches are tracked. KS-I.6 ($N = 3$) is CLOSED by AP10. With KS-I.6 closed, Einstein's field equations are an unconditional theorem of the record algebra in the dimensional sense — unconditional relative to the dimensional precondition, contingent on AP10's KS-D.3 (one axiom, one face / dimension-minimal embedding) remaining unfired. KS-I.1 (axiom-to-Poisson chain, now reframed around the Helmholtz-family + small- Λ -limit recovery), KS-I.2 (Lovelock chain), KS-I.3 (coefficient identification), KS-I.4 (κ inaccessibility), KS-I.5 (curvature-record identity), KS-I.7 (cosmological constant value), and KS-I.8 ($R \rightarrow$ conservation bridge) remain live. KS-I.6's closure is contingent on AP10's KS-D.3 remaining unfired; if KS-D.3 fires, the dimensional count changes and Lovelock's 3+1 uniqueness may not apply.

AP08 is logically independent of AP01 and AP04 in the strict sense that its postulates (the four canonical axioms, the proven hypotheses EH and QRA, and Lovelock's theorem as

external mathematical input) are self-contained. AP08 depends on AP20 (which proves EH and QRA), AP10 (which closes $N = 3$), AP06 (Theorem 3.1 for the horizon-leakage argument in section 6), and on The Lock identification (Edition 04) of ε with the electron mass for the coefficient identification in section 5.

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Orientation

Where this paper fits in The 420 Code

The 420 Code is organised across eight Notebooks. This paper is the fourth and closing chapter of Notebook II — Spacetime. The Notebook II chapter sequence runs AP03 (The Ratio) → AP05 (The Break) → AP10 (The Dimension) → AP08 (this paper).

Notebook II derives the structure of space, time, gravity, and the dimensional count from the axiom. AP03 derives the constants. AP05 walks the chain from the axiom to obligation and reformulates the cosmological identity as junction conditions. AP10 pins $N = 3$. AP08 closes Notebook II by deriving Einstein's field equations — first the Poisson equation as the weak-field linearisation, then the full tensor form via Lovelock 1971. With AP10's closure of $N = 3$, the derivation is unconditional.

AP08 is the structural payoff of Notebook II. The bidirectional relationship with AP10 is the centre of the Notebook: AP08 derived Einstein's field equations conditional on $N = 3$; AP10 derived $N = 3$ (contingent on KS-D.3) and removed the dimensional condition; the joint output is Einstein's field equations as a theorem in the dimensional sense — unconditional relative to the dimensional precondition, with

the standing contingency on AP10's KS-D.3 named throughout. Together AP10 and AP08 close the chain from four axioms to Einstein's equations with cosmological constant.

AP08 is logically independent of AP01 and AP04 in the sense that its own postulates (the four canonical axioms, EH, QRA, and Lovelock's theorem) are self-contained. AP08 depends on AP01 only for the formal apparatus of the record algebra; if AP01 stands, AP08's argument runs cleanly; if AP01 fails, the axioms fall back to declared assumptions and AP08's argument runs against those assumptions instead.

How to read this paper

The reading order is front to back. Section 0 names the dependencies and the scope. Section 1 states the four axioms and the two proven hypotheses (inherited from AP10 and AP20). Section 2 introduces the condensate — the manifold understood as the structure that emerges from accumulated records. Section 3 defines record density on the manifold and identifies it with matter density. Section 4 derives the Poisson equation as the weak-field limit of the curvature-record identity. Section 5 identifies the coefficient and discloses the structural circularity of κ . Section 6 verifies the strong-field limit against Hawking radiation as a consistency check. Section 7 names the full nonlinear topology and the Newton-versus-Einstein limits. Section 8 derives the full tensor form

via Lovelock 1971. Section 9 is the falsification register.
Section 10 is the synthesis.

Two voices alternate as the material requires. Narrative passages — the condensate as the ground you stand on, the Newton-as-linearisation framing, the closing on the gravitational sector being closed without reservation — speak directly. Formal sections — the axiom statements, the symmetry constraints, the uniqueness theorems, the kill switches — speak in the precision the structure requires.

Contents

0 — Dependency and Scope

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4 — The Form of the Identity (Weak-Field Limit)

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10 — What This Establishes and What Remains Open

What this paper closes · What this paper does not close · Items held open without claiming debt status · Where the philosophical-register implementations live · Structural relationships to subsequent work · Placement within The 420 Code · How The 420 Code reads now · The closing

References

Lovelock's theorem, Hawking radiation, and the gravitational-physics literature

0 — Dependency and Scope

AP08 is the fourth and closing chapter of Notebook II. Before the body, this section names what AP08 does, what it depends on, where its inputs map onto the canonical axioms {S, B, R, C}, what epistemic status each section holds, what kill switches are engaged, what structural debts remain, and how AP08 stands in relation to the rest of The 420 Code.

0.1 — What this paper does

AP08 derives the form of Einstein's field equations with cosmological constant from the four canonical axioms of the record algebra. The derivation has two parts. First, the weak-field limit: four axiom-derived symmetry constraints (Lorentz invariance, locality as differential form from Axiom C, second-order curvature from EH+QRA, scalar form of the source and the potential) applied to the weak-field regime (Constraint 2 as the defining condition of the regime, not an axiom-derived constraint) leave a one-parameter family — the Helmholtz form $a \cdot \nabla^2 \Phi + b \cdot \Phi = A \cdot \rho$ — of which pure Poisson is the small-cosmological-constant member. The Poisson equation is recovered as the leading-order static weak-field expression of section 8's full identity, in the limit of negligible Λ corrections at laboratory scales. Second, the full tensor form: five covariant constraints (general covariance, symmetric rank-2 tensor structure, divergence-free source via the $R \rightarrow$

conservation bridge, second-order in the metric from QRA, four-dimensional manifold from AP10's closure of $N = 3$ contingent on AP10's KS-D.3) plus Lovelock's 1971 theorem force $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$ as the unique symmetric, divergence-free, rank-2 tensor identity in 3+1 dimensions, with the $b \cdot g_{\mu\nu}$ term retained as the cosmological constant Λ .

The contribution is the form of Einstein's field equations as a theorem of the record algebra under the proven faithful embedding plus AP10's closure of $N = 3$. The cosmological constant Λ is predicted to exist by Lovelock; its value is not determined by the derivation (KS-I.7). The coefficient is identified — not derived — as $8\pi G/c^4$ via The Lock identification $G = 2\kappa/m_e^2$ (section 5); the structural circularity is honestly disclosed and tracked by KS-I.3 and KS-I.4.

AP08 also verifies the strong-field limit against established physics (section 6) by recovering the Hawking temperature from the Schwarzschild solution of section 8's field equations plus AP06 Theorem 3.1 (nonzero leakage when c is finite). The numerical formula $T_H = \hbar c^3/(8\pi k_B G M)$ is the standard Hawking 1975 result; what the 420 Code contributes is the structural interpretation (the event horizon as the locus where the fold becomes total) plus the structural guarantee of nonzero leakage. The Hawking temperature written as $T_H = \hbar c^3 m_e^2/(16\pi k_B \kappa M)$ after substituting $G = 2\kappa/m_e^2$ is the same Lock relabelling from section 5.1 reappearing at the

horizon scale, not an independent structural finding about the electron at the horizon.

0.2 — Dependencies

AP08 is a derivation paper with dependencies on both proven internal results and one external mathematical theorem (Lovelock 1971).

AP20 (the canonical axioms). Load-bearing for the four canonical axioms {S, B, R, C} (formal statement at AP20 section 3.3; completeness and minimality stated at AP20 section 4 with proof sketches, fenced at KS-P.1/KS-P.2; joint consistency exhibited by any existing record — the undeniable premise, AP20 section 6). The axioms enter section 1 as the foundation; their independence is invoked indirectly via AP10's closure of $N = 3$.

The Lorentzian signature (AP10 section 4, fenced at KS-D.4). Load-bearing for the Lorentzian manifold (M, g) with signature $(-, +, +, +)$ and symmetry group $SO(1, N)$, and for special relativity as a theorem (not an assumption). The signature is carried by AP10 section 4's structural argument (R's unique irreversibility reads as the $(-)$ metric character), with the formal metric-level derivation fenced at AP10's KS-D.4. The signature is the precondition for Constraint 1 of section 4 (Lorentz invariance) and Constraint 1 of section 8 (general covariance).

AP20 (EH and QRA proven). Load-bearing for the faithful embedding of the record algebra into a smooth manifold (EH) and for the differentiability and leading-order quadratic structure of the cone boundary (QRA). EH underwrites the entire derivation — the record algebra must embed into a smooth manifold for the curvature equations to be meaningful. QRA is invoked in Constraint 4 of sections 4 and 8 (second-order curvature requires the metric to be at least twice-differentiable).

AP10 (The Dimension). Load-bearing for $N = 3$ in Constraint 5 of section 8. AP10 closes KS-I.6; with $N = 3$ derived from the axioms, Lovelock's theorem applies in 3+1 dimensions and the uniqueness of $G_{\mu\nu} + \Lambda g_{\mu\nu}$ as the divergence-free symmetric rank-2 tensor is forced. The closure of KS-I.6 is contingent on AP10's KS-D.3 (one axiom, one face / dimension-minimal embedding) remaining unfired; if KS-D.3 fires, the dimensional count changes and Lovelock's 3+1 uniqueness may not apply.

AP06 Theorem 3.1 ($\varepsilon > 0$ whenever c is finite). Load-bearing for the strong-field limit at section 6.2. The Hawking-radiation argument requires the horizon to leak nonzero radiation; AP06 Theorem 3.1 establishes that the leakage is structurally guaranteed by the finiteness of c .

The Lock, Edition 04 ($\varepsilon = \text{electron}$). Load-bearing for the coefficient identification at section 5. The relabelling $G =$

$2\kappa/m_e^2$ — and therefore the writing of the coupling $8\pi G/c^4$ as $8\pi\kappa/(c^4 m_e^2)$ — depends on identifying ε with the electron. KS-I.3 is the structural test on this identification.

Lovelock 1971 (theorem). External mathematical input. The proof that in N spacetime dimensions, the only symmetric, divergence-free, rank-2 tensor constructed from the metric and its first and second derivatives takes a specific form. In $3+1$ specifically (Lovelock 1972) the form is uniquely $G_{\mu\nu} + \Lambda g_{\mu\nu}$. The theorem is mathematics, not internal to the body of work; AP08 invokes it as such.

0.3 — Axiom mapping

AP08 inherits the four canonical axioms from AP20 Paper D and AP10, and uses them to constrain the form of the curvature-record identity. The mapping is direct rather than retrospective.

Axiom S (Symmetry). Two disjoint sectors $\mathcal{L}$ and $\mathcal{P}$ with order-reversing involution σ . Underwrites the eye topology of section 7 (two curves meeting at the common point) and the structural framing of the condensate as the manifold that emerges from the symmetric/asymmetric structure.

Axiom B (Unique Breaking). One element $\varepsilon \in \mathcal{L}$ with no σ -image; $v(\varepsilon) = 1$. Under The Lock identification, $\varepsilon = \text{electron}$. Underwrites the matter-density identification at section 3.2 (every record involves an electron coupling event), the

coefficient identification at section 5 ($G = 2\kappa/m_e^2$), and the strong-field leakage at section 6.3 (the electron appears at the horizon).

Axiom R (Record Monotonicity). Monoid, not group; history is irreversible. Underwrites the linearity-in-weak-field constraint of section 4 (records accumulate additively, leading to a linear leading-order relationship between curvature and density), the divergence-free source in section 8 (the $R \rightarrow \nabla^\mu T_{\mu\nu} = 0$ bridge, KS-I.8), and the non-negativity of the density field ($\rho \geq 0$ everywhere) at section 3.3.

Axiom C (Constraint). Finite invariant rate c . Underwrites the locality constraint of section 4 (curvature at x depends only on density at x and its immediate neighbourhood), the propagation bound on density changes (ρ propagates at most at c), and the horizon-formation argument at section 6.1 (when the fold becomes total, C creates a one-way boundary).

0.4 — Epistemic status per section

Each section carries an explicit status — derivation, identification, formal block, or synthesis — so the reader knows what kind of weight each carries.

Section 1 (Axioms Inherited). FOUNDATIONAL. The four axioms re-stated from AP20 Paper D; EH and QRA named as proven (AP20); the Lorentzian manifold with signature $(-, +, +, +)$ and symmetry group $SO(1, N)$ carried via AP10 section 4

(fenced at KS-D.4); special relativity as a theorem of that signature.

Section 2 (The Condensate). STRUCTURAL FRAMING. The manifold understood as accumulated record structure (not as a pre-existing container). Mode $\emptyset$ (the metric) and Mode 1 (propagation events) named.

Section 3 (Record Density). DEFINITION + IDENTIFICATION. The record number density $n(x)$ defined; the identification with matter density $\rho(x) = n(x) \cdot \langle E \rangle / c^2$ made under The Lock; properties inherited from axioms ($\rho \geq \emptyset$ from R, causality from C, electron coupling from B).

Section 4 (Form of the Identity — Weak-Field Limit). DERIVATION + LIMIT RECOVERY. Four axiom-derived symmetry constraints plus the defining condition of the weak-field regime leave a one-parameter Helmholtz family; pure Poisson is recovered as the small-cosmological-constant member of that family. KS-I.1 is the structural test on the family identification and the small- Λ recovery.

Section 5 (The Coefficient). IDENTIFICATION + HONEST DISCLOSURE. The coefficient $A = 4\pi G$ identified via The Lock as $8\pi\kappa/m_e e^2$. Structural circularity (κ defined as $Gm_e e^2/2$) explicitly disclosed. KS-I.3 (coefficient identification) and KS-I.4 (κ inaccessibility) are the structural tests.

Section 6 (Strong-Field Limit — Hawking Verification).

CONSISTENCY CHECK — RECOVERY. The standard Hawking temperature recovered from the Schwarzschild solution of section 8's field equations plus established quantum field theory in curved spacetime (Hawking 1975); the condensate picture and AP06 Theorem 3.1 supply the structural interpretation and the nonzero-leakage guarantee. The numerical formulas are imported; the structural picture is the body of work contribution. The m_e in T_H (equation 6.3) is the same Lock relabelling from section 5.1 reappearing at the horizon scale, not an independent finding.

Section 7 (Full Nonlinear Structure). STRUCTURAL

ARGUMENT. The eye topology, Mode 0 / Mode 1, Newton as linearisation, Einstein as full curve.

Section 8 (Tensor Form). DERIVATION. Five covariant constraints plus Lovelock's 1971 theorem force $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$. KS-I.2 is the structural test on the uniqueness of Lovelock. KS-I.6 ($N = 3$) is closed by AP10, contingent on AP10's KS-D.3 remaining unfired.

Section 9 (Kill Switches). FORMAL FALSIFICATION

REGISTER. Eight kill switches engaged; one (KS-I.6) closed by AP10; seven live.

Section 10 (Synthesis). What this paper closes, what it leaves open.

0.5 — Kill switch summary

Eight kill switches are engaged in this paper. One is closed by AP10; seven remain live. The full register is in section 9.

KS-I.1: Axiom-to-Poisson chain. LIVE — HARD. If the five symmetry constraints in section 4 do not uniquely determine the Poisson equation, the weak-field derivation fails.

KS-I.2: Lovelock chain. LIVE — HARD. If the five covariant constraints in section 8 do not uniquely determine the Einstein tensor plus cosmological constant in 3+1 dimensions, the tensor derivation fails. Lovelock 1971 (general N) and Lovelock 1972 (3+1 uniqueness) are the load-bearing external claims.

KS-I.3: Coefficient identification. LIVE — HARD. If $G \neq 2\kappa/m_e^2$ — if The Lock identification of ε with the electron is structurally wrong — the coefficient identification in section 5 fails. The form survives; the coefficient does not.

KS-I.4: κ inaccessibility. LIVE — HARD. If an observable is found that constrains κ without going through G, the structural inaccessibility claim of section 5.2 is falsified and κ becomes independently testable.

KS-I.5: Curvature-record identity. LIVE — EMPIRICAL. If gravity is shown to source from a quantity structurally distinct

from accumulated records (matter without record-history, or records without gravitational signature), the identity fails.

KS-I.6: $N = 3$. CLOSED by AP10 — contingent on AP10's KS-D.3 (one axiom, one face / dimension-minimal embedding) remaining unfired. With $N = 3$ derived in AP10, the Lovelock derivation in 3+1 is unconditional in the dimensional sense; if KS-D.3 fires at any axiom, the count changes and Lovelock's 3+1 uniqueness may not apply.

KS-I.7: Cosmological constant value. LIVE — EMPIRICAL. The existence of Λ is predicted by Lovelock. The value is not determined. If Λ is measured to a value parametrically incompatible with the architecture's structural constraints (wrong sign, wrong magnitude class), the derivation's reach is bounded but the form survives.

Re-read at the wave, 4 September 2026. Λ 's seat carries AP48 §8's closure in other units, $1.10 \times 10^{-52} \text{ m}^{-2}$; AP08 decides its switch — see the dated note at KS-I.7 in section 9.

KS-I.8: $R \rightarrow$ conservation bridge. LIVE — HARD. The map from discrete record monotonicity (Axiom R) to the covariant local conservation $\nabla^\mu T_{\mu\nu} = 0$ is a bridge step in Constraint 3 of section 8.1, not a completed derivation. If the bridge fails, the tensor form chain fails at Constraint 3.

0.6 — Structural debts owed

Two structural-class items are tied to AP08's argument as the 420 Code debts. Named here so the structural exposure is visible at the door.

The Lock identification ($\varepsilon = \text{electron}$). Used at section 3.2 ($\rho = n \cdot \langle E \rangle / c^2$ under the electron interpretation), section 5 (the coefficient identification $G = 2\kappa/m_e^2$), and section 6.3 (the m_e in T_H). The identification is structural rather than empirical — it is the 420 Code claim that the unique ε of Axiom B is the same ε that appears empirically as the electron — and is tracked by KS-I.3 as a live structural test. The form of Einstein's field equations survives if The Lock fails; the coefficient does not.

The $R \rightarrow \nabla^\mu T_{\mu\nu} = 0$ bridge. Constraint 3 of section 8.1 requires the source tensor to satisfy covariant local conservation. The motivation comes from Axiom R (records cannot be created or destroyed locally), but the precise map from discrete record monotonicity to the covariant conservation law in the continuum limit is a bridge step, not a completed derivation. KS-I.8 is the live structural test on this bridge. If the bridge fails, the tensor derivation fails at Constraint 3; the weak-field expression survives (it does not invoke covariant conservation).

0.6.1 — External mathematical dependencies

Separately from the 420 Code structural debts, AP08 invokes external mathematical results as established theorems. These are not the 420 Code debts — the body of work does not attempt to derive them — but they are tracked at structural-test level so the chain is auditable end-to-end.

Lovelock 1971 and Lovelock 1972. The uniqueness theorems for divergence-free symmetric rank-2 tensors in general N (Lovelock 1971) and specifically in 3+1 dimensions (Lovelock 1972). Invoked at section 8.2. KS-I.2 is the structural test on whether the theorems hold as invoked.

Potential theory in the Helmholtz-family identification.

Section 4.2 invokes the standard fact that the most general local, linear, second-order, rotationally-invariant scalar operator acting on a scalar field is $a \cdot \nabla^2 + b \cdot (\text{identity})$. KS-I.1 is the structural test on this identification and on the small- Λ recovery of pure Poisson.

The Schwarzschild solution and Hawking 1975. Section 6 recovers the standard Hawking temperature from the Schwarzschild solution of section 8's equations plus the established quantum-field-theory-in-curved-spacetime derivation. The numerical formulas are external; the structural picture is the 420 Code contribution.

0.7 — Items held open without claiming debt status

Two items are bracketed inside this paper. They are real and live, but they are not structural debts in the same sense as the items in 0.6.

The cosmological constant value (KS-I.7). Lovelock's theorem predicts the existence of Λ ; the value is not determined by the derivation. Whether Λ can be computed from a downstream architectural principle, or whether it remains an empirical input, is open work — held open here without claiming a separate structural debt because the existence claim is what AP08 establishes; the value is downstream cosmological work.

Constraint 2 of section 4 (the weak-field regime as defining condition). The weak-field regime is the regime section 4 derives in; the linearity going into the Helmholtz family is the defining condition of that regime, not an axiom-derived constraint. The full nonlinear extension (section 7, section 8) does not require it; Lovelock's theorem works independently of linearity. Held open here without claiming a separate debt because “weak-field” is a chosen regime, not an unresolved structural item.

0.8 — Structural relationships

AP08 is the closing chapter of Notebook II. The chapter sequence runs $AP03 \rightarrow AP05 \rightarrow AP10 \rightarrow AP08$. AP03 derives the constants (c, G); AP05 walks the chain from axiom to obligation and reformulates the cosmological identity as junction conditions; AP10 pins $N = 3$ and identifies the Hilbert space as the 1:1; AP08 closes with Einstein's field equations.

AP01 (Notebook I) is the source of the record algebra. AP20 (Notebook I) is the source of EH, QRA, and the canonical four axioms (formal statement at AP20 section 3.3; completeness and minimality fenced at KS-P.1/KS-P.2). AP10 (Notebook II) closes the $N = 3$ conditional that AP08's tensor derivation requires. AP06 (Notebook IV) provides Theorem 3.1 (leakage when c is finite) used in section 6. The Lock (Edition 04) provides the ε = electron identification used in section 5.

Downstream, AP08 feeds the entire gravitational and cosmological programme. Predictions Ø inherits the unconditional Einstein's field equations and the predicted existence of Λ . Dissolutions Ø inherits the philosophical-register reading of gravity as record-density curvature. The horizon-leakage result at section 6 feeds AP24 (the residual programme) and the strong-field limit of cosmological models.

1 — Axioms (Inherited)

The four axioms of the record algebra, formally stated at AP20 section 3.3 and walked in detail in AP10 section 1.

S (Symmetry). Two disjoint sectors $\mathcal{L}$ and $\mathcal{P}$ with order-reversing involution σ . $Q(\mathcal{L}) = Q(\mathcal{P})$ in the unbroken case.

B (Unique Breaking). One element $\varepsilon \in \mathcal{L}$ with no σ -image. $v(\varepsilon) = 1$. The break. Under The Lock: $\varepsilon = \text{electron}$.

R (Record Monotonicity). Monoid, not group, in each sector. No non-identity element has an inverse. History is irreversible.

C (Constraint). Finite invariant rate c constraining sequential propagation. Structural, not electromagnetic.

Independent (the removal arguments at AP20 section 4.2, fenced at KS-P.2) and jointly exhibited by any existing record (AP20 section 6).

1.1 — Bridge hypotheses (now theorems)

EH (Embedding Hypothesis). In the large- N limit, the discrete record algebra admits faithful embedding into a smooth manifold M .

QRA (Quantum-Record Alignment). Quantum states are pre-state records (closed at AP20 section 5.5, KS-P.4). The cone boundary's differentiability and leading-order quadratic

structure used here rest on the smooth-manifold structure fenced at KS-P.5.

Both proven in AP20.

Under EH + QRA, the record algebra produces a Lorentzian manifold (M, g) with signature $(-, +, +, +)$ and symmetry group $SO(1, N)$.

The embedding is proven at AP20 (EH); the Lorentzian signature is carried by AP10 section 4's structural argument, with the formal metric-level derivation fenced at KS-D.4.

Special relativity is a theorem, not an assumption.

2 — The Condensate

Before the break, the system is the 1:1.

Axiom S without B. Two sectors, perfectly paired, net content zero.

The break creates ε — one unpaired element.

Under EH, the large-N limit of accumulated records produces the smooth manifold (M, g) .

The manifold IS the condensate.

Not a pre-existing container into which records are placed.
The structure that emerges from the records' accumulation.

The condensate and the records are co-constitutive. Without the condensate, records have nowhere to embed. Without records, the condensate has no content.

The metric g on M encodes the condensate's geometry.

Where the metric is flat, the condensate is unstressed.

Where it curves, the condensate is folded.

The curvature is determined by its own content.

You are standing on the condensate now.

The ground beneath your feet, the air around you, the light in your eyes — all of it is accumulated records embedded on the manifold that their accumulation created.

The geometry you walk through is the geometry of what has happened.

2.1 — Mode 0 and Mode 1

The condensate admits two modes of description.

Mode 0 is the metric g itself. The geometry. The background condition. The equilibrium structure that emerges from the cumulative records as a stable background.

Mode 1 is propagation on the condensate. Coupling events. The electron interacting with the electromagnetic field. The perturbative excitations of individual record-writing events on the background.

Mode 1 requires Mode 0.

In the language of the axioms: Mode 0 is the structure that EH + QRA extract from the cumulative record algebra. Mode 1 is the individual record-writing events that continue to occur on that structure. The terminology mirrors the cumulative-vs-perturbative distinction common in condensed-matter physics; in The 420 Code, gravity lives in Mode 0 and

quantum mechanics in Mode 1 (the distinction is taken up in AP09).

Gravity — the curvature of the condensate — is Mode 0.

3 — Record Density on the Manifold

You have a manifold.

You have records.

The question is: how many records are there at each point?

The answer is a number. That number is the density.

And that density — as you are about to see — is indistinguishable from what physics calls “matter.”

3.1 — The definition

Under EH, each record r in the algebra embeds as an event $e(r)$ on the manifold M .

The embedding preserves causal order (Axiom R) and the rate bound (Axiom C).

Define the record number density:

$$n(x) = \lim_{V \rightarrow 0} N(V) / |V| \quad (3.1)$$

where $N(V)$ is the number of embedded records in a spacetime volume V containing the point x , and $|V|$ is the proper volume.

A scalar field on M .

3.2 — Record density is matter density

Epistemic status: identification, conditional on The Lock (ε = electron).

Every record is an ε -event (Axiom B). Records emerge through the break, and Axiom B specifies ε as the unique element responsible for the break; every record formation event therefore involves ε by construction. The full structural argument for record-formation-via- ε lives in AP01; AP08 inherits the result. The empirical handle on the identity (whether all matter sources from accumulated records) is KS-I.5.

Under The Lock's identification ε = electron, every record involves an electron coupling event. Every coupling event involves energy exchange.

The energy associated with accumulated records at a point x is not a separate quantity from the energy density of matter at x .

The mass density $\rho(x)$ and the record density $n(x)$ are related by:

$$\rho(x) = n(x) \cdot \langle E \rangle / c^2 \quad (3.2)$$

where $\langle E \rangle$ is the average energy per record.

A statement of identity.

The matter density and the record density are two measurements of the same quantity, in different units.

3.3 — Properties inherited from the axioms

From R (Record Monotonicity). $\rho(x) \geq 0$ everywhere. Records accumulate; they do not annihilate. The density cannot go negative.

From C (Constraint). Changes in ρ propagate at most at speed c . The density field respects causality.

From B (Unique Breaking). Every record involves ε . Under The Lock's identification, the density is denominated in electron coupling events.

4 — The Form of the Identity (Weak-Field Limit)

You have a manifold.

You have a density field on it.

Now ask: what is the most general equation relating the curvature of the condensate to the record density?

You are not choosing the answer.

You are watching the constraints eliminate every alternative until only one remains.

4.1 — The constraints

Four axiom-derived constraints, plus one regime-defining condition (Constraint 2). The four axiom-derived constraints together with the regime definition leave a one-parameter family of equations; the Poisson form is recovered as the small-cosmological-constant limit (see section 4.2 and the reconciliation with section 8).

Constraint 1 (Lorentz invariance). The identity must be covariant under $SO(1, N)$. Derived from the axioms via the Lorentzian signature carried at AP10 section 4 (formal derivation fenced at KS-D.4); $SO(1, N)$ is the isometry group of that signature.

Constraint 2 (Weak-field regime — defining condition).

The weak-field regime is the regime in which the metric is written as $g \approx \eta + h$ with h small, and the leading-order approximation to the curvature is linear in the perturbation. This is what “weak-field” *means*; it is not derived from Axiom R but is the defining condition of the regime section 4 derives in. The full nonlinear case (section 8) does not assume linearity; Lovelock’s theorem works independently of it. Axiom R’s record additivity is consistent with linearity in this regime but does not force it as an independent constraint.

Constraint 3 (Differential form). The identity is a local differential equation rather than an integral equation — appropriate to a finite-causal-bound substrate (Axiom C) in which the dynamic field equations must reduce to first-and-second-derivative relations on the metric. In the static limit considered here, the differential form yields equations whose Green-function solutions can themselves be non-local (the static Poisson kernel is $1/|x - y|$); the locality is in the differential form, not in the solutions. The full causal structure is restored in the dynamic case at section 8.

Constraint 4 (Second order). Curvature is a second-derivative quantity. On the manifold produced by EH + QRA, the curvature involves second derivatives of g . In the weak-field limit where $g \approx \eta + h$, the curvature reduces to second spatial derivatives of the perturbation.

Constraint 5 (Scalar). The record density $\rho(\mathbf{x})$ is a scalar field. In the static weak-field limit, the gravitational potential Φ is also a scalar. The identity relates two scalar quantities.

4.2 — The form: Poisson as the leading-order limit

The four axiom-derived constraints applied to the weak-field regime determine the form of the identity up to a one-parameter family.

The most general local, linear, second-order, rotationally-invariant scalar identity on a flat background relating a potential Φ to a source ρ is:

$$\mathbf{a} \cdot \nabla^2 \Phi(\mathbf{x}) + \mathbf{b} \cdot \Phi(\mathbf{x}) = \mathbf{A} \cdot \rho(\mathbf{x}) \quad (4.1)$$

where $\mathbf{a}$ and $\mathbf{b}$ are constants, $\mathbf{A}$ is the coupling, and $\nabla^2 = \partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2$ is the Laplacian. The general form is the Helmholtz family; pure Poisson is the $\mathbf{b} = \mathbf{0}$ case.

Honest disclosure. The four axiom-derived constraints together with the regime definition do not by themselves force $\mathbf{b} = \mathbf{0}$. To recover pure Poisson (the form Newton found), an additional input is required: the absence of an intrinsic length scale in Mode 0 at the regime where section 4 derives the form, equivalently the absence of Yukawa-type screening of the gravitational potential at laboratory distances. The body of work does not derive this absence as a theorem; it is

recovered as the small-cosmological-constant limit of section 8's full tensor form, in which the Λ contribution becomes negligible at the scales where section 4 applies.

Reconciliation with section 8. The full tensor form $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$ (section 8.3, equation 8.1) retains the b-analogue: the $\Lambda g_{\mu\nu}$ term is the four-dimensional covariant generalisation of the b- Φ contribution. In the Newtonian limit of section 8's equation, the Λ term contributes a constant background offset to the equation for Φ (rather than a Yukawa-type screening), and the leading-order static behaviour at laboratory scales is the pure Poisson form. Pure Poisson is therefore the leading-order leading-scale limit of section 8's full equation, valid when corrections of order $\Lambda/(\text{laboratory scale})^2$ are negligible. Section 4's "uniqueness" is a uniqueness-up-to-the-Helmholtz-family, with the pure Poisson form recovered as the limit just described.

In that limit:

$$\nabla^2 \Phi(\mathbf{x}) = \mathbf{A} \cdot \rho(\mathbf{x}) \quad (4.2)$$

The Poisson equation.

Not derived by assuming Newton's law or Einstein's equations. Recovered as the leading-order static weak-field expression of the full identity that section 8 derives, in the limit of negligible Λ corrections at laboratory scales.

You have just watched gravity's leading-order equation emerge from four axiom-derived constraints, the weak-field regime definition, and the small-cosmological-constant limit of the full tensor identity.

The constraints leave a one-parameter family. The Helmholtz family. The small- Λ limit picks out the Poisson member of that family. The equation Newton found by observation, recovered here from the structure plus a defining condition of the regime plus a limit.

The structural circularity is preserved: A is the coupling, identified (not derived) at section 5 as $4\pi G = 8\pi\kappa/m_e^2$. KS-I.1 is the structural test on the leading-order-family identification and the recovery of pure Poisson.

5 — The Coefficient

Equation (4.2) has the form $\nabla^2\Phi = A\rho$.

The constant A must be identified.

Through The Lock:

$$\mathbf{G} = 2\kappa / m_e^2 \quad (5.1)$$

Therefore:

$$\mathbf{A} = 4\pi\mathbf{G} = 8\pi\kappa / m_e^2 \quad (5.2)$$

and the identity reads:

$$\nabla^2\Phi = (8\pi\kappa / m_e^2) \cdot \rho \quad (5.3)$$

The coefficient is the ratio of the fabric's holding limit to the square of the electron mass.

The entire gravitational sector is determined by two quantities.

How much the fabric could hold (κ).

How much had to leave (m_e).

You are looking at gravity expressed as a ratio of what stayed to what escaped.

5.1 — The circularity and why it is structural

Honest disclosure.

κ is defined as $Gm_e^2/2$.

Therefore writing $G = 2\kappa/m_e^2$ is, at the level of numerical computation, a tautology: $G = 2(Gm_e^2/2)/m_e^2 = G$.

The coefficient has been relabelled, not independently derived.

Anyone who reads section 5 and concludes that G has been computed from first principles without empirical input is mistaken.

What HAS been done.

The form $\nabla^2\Phi = A\rho$ is derived from the axioms.

The five symmetry constraints in section 4 force the Laplacian, force linearity, and leave exactly one free constant A .

The form is genuinely new.

The constant A is then identified — not derived — as $4\pi G$, or equivalently $8\pi\kappa/m_e^2$.

5.2 — Why κ cannot be independently measured

κ is the holding limit of the pre-state — the maximum coherence the system could sustain before the break became necessary.

Measurement itself is a product of the break that κ describes.

Before the break (Axiom B), there are no records (Axiom R has nothing to order), no observers, no instruments, and no manifold (EH has nothing to embed).

The only access to κ is through its consequences — G and m_e — which are the residuals of the break.

κ is a real structural quantity with well-defined dimensions and physical meaning, but it is accessible only through its products.

Kill switch KS-I.4. If an observable is found that constrains κ without going through G, the structural inaccessibility claim is falsified and κ becomes independently testable.

You cannot measure what existed before measurement existed.

κ is the capacity of the pre-state — and the pre-state is, by definition, the state before records.

The circularity is not a bug.

It is the structure telling you what it can and cannot show you.

You are inside the break. You cannot look behind it.

6 — The Strong-Field Limit (Hawking Verification)

A consistency check against established physics.

Note on what this section does and does not do. The Schwarzschild surface gravity $\kappa_H = c^4/(4GM)$ and the Hawking temperature $T_H = \hbar c^3/(8\pi k_B GM)$ are standard results of general relativity plus quantum field theory in curved spacetime (Hawking 1975). They are recovered here from the Schwarzschild solution of section 8's field equations, not independently derived from the condensate picture. What the condensate picture and AP06 Theorem 3.1 contribute is the structural interpretation: the event horizon as the locus where the fold becomes total, and the structural guarantee that the horizon leaks nonzero radiation whenever c is finite. The numerical formulas come from established physics; the structural picture is the 420 Code contribution. (Section 6.2 forward-references section 8's field equations; the dependence is acyclic because section 8 does not invoke section 6.)

6.1 — Maximum curvature

As record density increases, the condensate folds more tightly.

When the fold becomes total, the constraint (Axiom C) creates a one-way boundary.

Records can enter but not exit.

Outward propagation is forbidden.

The event-horizon interpretation from the condensate picture.

6.2 — The leakage at the horizon

AP06 Theorem 3.1: $\varepsilon > 0$ whenever c is finite.

At the event horizon, absorption is near-perfect but not perfect.

Hawking radiation provides the established nonzero horizon-leakage channel.

The surface gravity of a Schwarzschild horizon of mass M (from the Schwarzschild solution of section 8's field equations) is:

$$\kappa_H = c^4 / (4GM) \quad (6.1)$$

The Hawking temperature follows from standard quantum field theory in curved spacetime (Hawking 1975):

$$T_H = \hbar \kappa_H / (2\pi c k_B) = \hbar c^3 / (8\pi k_B G M) \quad (6.2)$$

The Hawking temperature.

Recovered from established physics. What the body of work contributes is structural: (i) the event-horizon interpretation as the locus where the fold becomes total, (ii) the structural guarantee that the leakage is nonzero (AP06 Theorem 3.1), and (iii) the surface gravity being set by the fold's geometry. The numerical formulas are not new; the structural picture under which they are recovered is.

6.3 — The electron at the boundary (an honest note)

Substituting $G = 2\kappa/m_e^2$ into equation (6.2):

$$T_H = \hbar c^3 m_e^2 / (16\pi k_B M) \quad (6.3)$$

The Hawking temperature contains the electron mass after the substitution.

Honest disclosure. The standard Hawking temperature (equation 6.2) does not contain the electron mass; it appears in equation (6.3) only because G has been rewritten as $2\kappa/m_e^2$ — the same relabelling disclosed in section 5.1 as a tautology at the level of numerical computation. The “appearance” of m_e at the horizon is the same identification reappearing in the strong-field expression, not an independent structural result. The honest reading: under The Lock's identification, the gravitational coupling at *all* scales involves m_e (because $G = 2\kappa/m_e^2$), so m_e is present wherever the coupling enters — quantum scale, laboratory

scale, horizon scale. The horizon expression makes the dependence visible because it isolates the gravitational coupling in a single dimensionful combination; it does not reveal a new structural role for the electron beyond what The Lock already says.

What can be said honestly. The Lock's identification ($\varepsilon =$ electron) is structurally consistent at the horizon scale as it is at the quantum scale; nothing about the strong-field limit contradicts the identification. This is consistency, not corroboration: the same identification entered both expressions, so the same identification appears in both.

7 — The Full Nonlinear Structure

The Poisson equation (section 4) is a linearisation.

It describes the weak-field regime where the relationship between curvature and record density is approximately linear.

The full relationship is nonlinear.

The curvature depends on the density, and the density depends on the curvature.

7.1 — The topology

The eye topology describes the full nonlinear curve.

Two curves (the two sectors $\mathcal{L}$ and $\mathcal{P}$ from Axiom S, related by the involution σ) meet at a common point.

The point is $\emptyset$ — the 1:1, the pre-state of perfect symmetry before the break.

The topology is closed. When folded onto a sphere, the two apparent endpoints are the same point.

7.2 — Reading the structure

At the common point ($\emptyset$). The two sectors meet. No separation, no break, no curvature, no records. The Poisson equation is the linearisation of the full curve near this point.

Moving outward. The break occurs. ε appears. Records accumulate. The curves separate. Curvature increases.

At maximum separation. Maximum curvature. Maximum record density. The event horizon. The extremal point relative to which the full structure is measured — relativistic in the original sense (relative to its own extremal point, not to a fixed background).

The closed topology. The quantum scale (near the common point, where ε is the break) and the cosmological scale (the full extent of the curve) are connected in the topology. The Lock's identification of ε with the electron carries through the closure: the same electron mass that defines the gravitational coupling ($G = 2\kappa/m_e^2$, section 5) appears in the horizon expression at section 6.3 — because the same identification entered both. The electron is at both ends of the gravitational identity as a structural consequence of The Lock, not as an independent finding at the horizon.

7.3 — Newton vs Einstein

Newton's gravity is the linearisation near the common point.

Curvature proportional to density, constant coefficient, valid where the fold is gentle.

Einstein's gravity is the full curve.

The geometry and the distribution of energy-momentum locked together by a nonlinear identity.

The architecture derives both.

The linearisation is forced by symmetry constraints (section 4). The full tensor form is forced by Lovelock's theorem (section 8).

You have now seen the full shape.

Newton saw the straight line near the origin.

Einstein saw the curve.

They are the same structure read at different distances from the common point.

The mathematics did not choose between them. It produced both.

You do not choose either. You inherit both.

8 — The Tensor Form

Same method as section 4, applied at the covariant level.

8.1 — The constraints

Constraint 1 (General covariance). The identity must hold in all coordinate systems. Under EH, the manifold has no preferred coordinates. A tensor equation.

Constraint 2 (Symmetric rank-2 tensor). The source is the stress-energy tensor $T_{\mu\nu}$ — the covariant generalisation of ρ . The geometry side must match: symmetric rank-2 tensor built from the metric.

Constraint 3 (Divergence-free source; bridge compatibility). To write a consistent local covariant field identity with matter coupling on the embedded manifold, the source tensor is required to satisfy $\nabla^\mu T_{\mu\nu} = 0$. Motivated by Axiom R: because records cannot be created or destroyed locally, the total number of records in any region changes only due to flow across the boundary, and in the continuum limit this conservation law translates to the vanishing of the covariant divergence of the stress-energy tensor. The exact map from record monotonicity to covariant local conservation is a bridge step, not a completed derivation in this paper (KS-I.8).

Constraint 4 (Second-order in the metric). The curvature involves second derivatives of g (from QRA). The geometric tensor must be built from $g_{\mu\nu}$ and its first and second derivatives.

Constraint 5 (Four-dimensional). The manifold has dimension $3+1$. $N = 3$ is derived from the four independent axioms (AP10, The Dimension). KS-I.6 is CLOSED — contingent on AP10's KS-D.3 (one axiom, one face / dimension-minimal embedding) remaining unfired. With $N = 3$ derived under that contingency, the dimensional precondition for Lovelock's $3+1$ uniqueness theorem is met.

8.2 — Lovelock's theorem

Theorem (Lovelock, 1971; uniqueness in $3+1$ specifically, Lovelock 1972). In four spacetime dimensions, the only symmetric, divergence-free, rank-2 tensor constructed from the metric tensor and its first and second derivatives is:

$$a \cdot G_{\mu\nu} + b \cdot g_{\mu\nu}$$

where $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}$ is the Einstein tensor, $g_{\mu\nu}$ is the metric tensor, and a, b are constants.

The tensorial analogue of the uniqueness of the Laplacian.

The proof is constructive and independent of physical content — pure differential geometry. Lovelock 1971 derives the result in general N ; Lovelock 1972 establishes the special

uniqueness of 3+1 from which the form above follows. AP08 invokes both as external mathematical input, tracked by KS-I.2.

8.3 — The identity in tensor form

Constraints 1–5 from the axioms, plus Lovelock’s theorem, force the full identity:

$$\mathbf{G}_{\mu\nu} + \Lambda \mathbf{g}_{\mu\nu} = (8\pi\mathbf{G}/c^4) \mathbf{T}_{\mu\nu} \quad (8.1)$$

$\mathbf{G}_{\mu\nu}$ is the Einstein tensor — the unique divergence-free, symmetric, rank-2 curvature tensor in four dimensions.

$\mathbf{T}_{\mu\nu}$ is the stress-energy tensor — the covariant record density.

Λ is the cosmological constant. Predicted to exist by Lovelock’s theorem (allowed by the constraints). Its value is not determined by the derivation (KS-I.7).

$8\pi\mathbf{G}/c^4$ is the coupling. The $\mathbf{G}$ inside is the same $\mathbf{G}$ identified in section 5 through The Lock.

Einstein’s field equations with cosmological constant.

Derived from four axioms, symmetry constraints, EH + QRA, and Lovelock’s theorem.

With AP10’s closure of $N = 3$ (contingent on KS-D.3 remaining unfired), the derivation is unconditional in the dimensional

sense — unconditional relative to the dimensional precondition, with the standing contingency on AP10's KS-D.3 named explicitly throughout.

The cosmological constant is not added by hand. Its existence is predicted by the same uniqueness theorem that gives the Einstein tensor.

9 — Kill Switches

AP08 engages eight kill switches. One (KS-I.6) is closed by AP10 under contingency on AP10's KS-D.3; seven remain live. The closures and live tests target the load-bearing claims of the derivation: the uniqueness theorems of potential theory (KS-I.1) and Lovelock (KS-I.2), the coefficient identification through The Lock (KS-I.3), the structural inaccessibility of κ (KS-I.4), the curvature-record identity (KS-I.5), the dimensional count (KS-I.6), the cosmological constant value (KS-I.7), and the bridge from Axiom R to covariant conservation (KS-I.8).

KS-I.1 — Axiom-to-Poisson chain (via Helmholtz family + small- Λ limit)

Claim. The four axiom-derived symmetry constraints in section 4.1 (Lorentz invariance; locality as differential form via Axiom C; second-order in the metric via QRA; scalar form), applied to the weak-field regime (Constraint 2 as the defining condition of the regime, not an axiom-derived constraint), leave a one-parameter Helmholtz family $a \cdot \nabla^2 \Phi + b \cdot \Phi = A \cdot \rho$ of local, linear, second-order, rotationally-invariant scalar identities. The pure Poisson form $\nabla^2 \Phi = A \rho$ is the $b = 0$ case, recovered as the small-cosmological-constant limit of section 8's full tensor identity, in which the Λ contribution becomes negligible at laboratory scales.

Test. Demonstrate that the four axiom-derived constraints leave a wider or different family than the Helmholtz form. Or: demonstrate that pure Poisson is not the correct small- Λ limit of section 8's equation — that the $b = 0$ case is not the leading-order static weak-field behaviour, or that the corrections at laboratory scales are not negligible under the standard interpretation of Λ . Empirical handle: deviations from inverse-square gravity at laboratory scales that would be inconsistent with pure Poisson behaviour.

Status. LIVE — HARD. The Helmholtz-family identification is the load-bearing reformulation; the small- Λ recovery of pure Poisson is the load-bearing reduction. Both are tested by this kill switch.

Recovery. If KS-I.1 fires at the family-identification level (the constraints leave a different or wider family), the weak-field derivation falls back to “the constraints leave [the correct family], and pure Poisson is recovered as a member of that family if the corresponding limit holds.” If KS-I.1 fires at the small- Λ -limit level (pure Poisson is not the correct limit), the weak-field expression must be re-derived; the tensor derivation (section 8) is independent of this and survives.

KS-I.2 — Lovelock chain

Claim. The five covariant constraints in section 8.1 (general covariance, symmetric rank-2 tensor, divergence-free source via R bridge, second-order in the metric, four-dimensional)

plus Lovelock's 1971 theorem uniquely determine the Einstein tensor plus cosmological constant as the only symmetric, divergence-free, rank-2 tensor in 3+1 dimensions built from the metric and its first and second derivatives.

Test. Demonstrate an alternative second-order, symmetric, divergence-free tensor in four dimensions that satisfies the constraints. Or: demonstrate that Lovelock's 1971 theorem fails (a mathematical error) or that Lovelock's 1972 3+1 uniqueness result does not apply under the constraints.

Status. LIVE — HARD. Lovelock 1971 (general N) and Lovelock 1972 (3+1 uniqueness) are the load-bearing external mathematical claims.

Recovery. If KS-I.2 fires at the mathematical level (Lovelock's theorem itself fails), the derivation fails at section 8.3 and the form of Einstein's field equations becomes an identification rather than a derived theorem. If KS-I.2 fires at the application level (the 3+1 uniqueness does not apply as invoked), the recovery path is to re-derive the tensor form under the correct theorem statement. The weak-field Poisson derivation (section 4) is independent and survives.

KS-I.3 — Coefficient identification

Claim. $G = 2\kappa/m_e^2$ (equation 5.1), and therefore $A = 4\pi G = 8\pi\kappa/m_e^2$ (equation 5.2). The identification rests on The Lock's structural claim that the unique ε of Axiom B is the

same ε that appears empirically as the electron, with m_e its mass.

Test. Demonstrate that $G \neq 2\kappa/m_e^2$ — that The Lock identification of ε with the electron is structurally wrong, or that the relabelling produces a numerical inconsistency. Specific structural handles: re-examine the Lock argument (The Lock, Edition 04); check whether ε is correctly identified with the electron rather than another fundamental particle.

Status. LIVE — HARD. The Lock identification is the load-bearing structural claim.

Recovery. If KS-I.3 fires, the coefficient identification in section 5 fails. The form of equation (4.1) ($\nabla^2\Phi = A\rho$ with A a constant) survives as a derived theorem; only the identification $A = 4\pi G = 8\pi\kappa/m_e^2$ falls. The Hawking-temperature derivation in section 6.3, which substitutes $G = 2\kappa/m_e^2$ into the surface-gravity formula, would similarly need to be re-expressed in whichever quantities replace κ and m_e .

KS-I.4 — κ inaccessibility

Claim. κ is structurally inaccessible to independent measurement: the only access to κ is through its consequences (G and m_e), which are the residuals of the break. The structural inaccessibility is not a measurement limitation but a structural consequence of κ being the holding

limit of the pre-state, which is the state prior to records and measurement.

Test. Identify an observable that constrains κ without going through G — a structural argument that κ is independently testable, or an empirical handle that pins κ via a different chain. If κ can be measured independently, the structural-inaccessibility claim of section 5.2 is falsified.

Status. LIVE — HARD. The claim is structural rather than empirical, and the failure mode would itself be structural (an alternative chain to κ being identified).

Recovery. If KS-I.4 fires, κ becomes independently testable; the structural circularity of section 5.1 (κ defined as $Gm_e^2/2$) becomes an empirical question rather than a structural disclosure. The form of Einstein's field equations survives; only the claim that κ cannot be independently measured falls.

KS-I.5 — Curvature-record identity

Claim. Gravity sources from accumulated records: the curvature of the condensate (Mode 0) is determined by the record density (sections 3–4) and ultimately by the stress-energy tensor (section 8). The identity $\rho(x) = n(x) \cdot \langle E \rangle / c^2$ (equation 3.2) holds: matter density and record density are two measurements of the same quantity.

Test. Demonstrate that gravity sources from a quantity structurally distinct from accumulated records. Specific handles: matter without record-history (gravitational sources that do not trace back to coupling events) or records without gravitational signature (coupling events that do not produce mass-energy and therefore do not source curvature).

Status. LIVE — EMPIRICAL. The identity is empirically testable in principle (does all matter source gravity in proportion to its accumulated record content?) but no observed deviation has been reported.

Recovery. If KS-I.5 fires, the identification $\rho = n \cdot \langle E \rangle / c^2$ fails; the matter density and the record density are not the same quantity, and the curvature-record identity sections 3–8 fails at section 3.2. The Poisson and Lovelock derivations survive as formal results about whatever the source quantity is; only the identification of that quantity with accumulated records falls.

KS-I.6 — N = 3 (closed by AP10)

Claim. The manifold has dimension 3+1 (three spatial dimensions plus one temporal direction). Lovelock's theorem applies in four dimensions, and in four dimensions the unique divergence-free, symmetric, rank-2 tensor built from the metric and its first and second derivatives is $G_{\mu\nu} + \Lambda g_{\mu\nu}$.

Test. Demonstrate that N differs from 3 (the manifold has 2, 4, or some other number of spatial dimensions). Empirical handles: high-energy physics signatures sensitive to extra dimensions, cosmological observations sensitive to dimensional count. Structural handles: AP10's argument fails (KS-D.3 fires, in which case the count changes accordingly).

Status. CLOSED by AP10 — contingent on AP10's KS-D.3 (one axiom, one face / dimension-minimal embedding) remaining unfired. With $N = 3$ derived in AP10 under that contingency, the dimensional precondition for Lovelock's 3+1 uniqueness is met. If AP10's KS-D.3 fires at any axiom, the count changes, and Lovelock's 3+1 uniqueness theorem may not apply; KS-I.6 reopens in that case.

Recovery. If KS-I.6 reopens (via AP10's KS-D.3 firing), the Lovelock derivation in section 8 fails at Constraint 5. The tensor form would need to be re-derived under whatever Lovelock-theorem statement applies in the corrected number of dimensions. The weak-field Poisson derivation (section 4) is independent of the dimensional count (it explicitly invokes the Laplacian on $\mathbb{R}^3$) and would also need to be re-examined under the corrected count.

KS-I.7 — Cosmological constant value

Claim. The existence of the cosmological constant Λ is predicted by Lovelock's theorem (it appears as a free constant in the unique tensor $a \cdot G_{\mu\nu} + b \cdot g_{\mu\nu}$; in equation 8.1, $b = \Lambda$).

The value of Λ is not determined by the derivation; whether it can be computed from a downstream architectural principle is open work.

Test. Measure Λ to a value parametrically incompatible with the architecture's structural constraints (wrong sign, wrong magnitude class, or a value that contradicts the existence prediction). Or: derive Λ 's value from an independent body of work chain and demonstrate that the derivation contradicts the observed value.

Status. LIVE — EMPIRICAL. The existence is predicted; the value is open. Current observations are consistent with the existence prediction (a small positive Λ is measured cosmologically).

Recovery. If KS-I.7 fires at the existence level (no cosmological constant is found, or it is structurally impossible), the form of Lovelock's theorem in section 8.2 still admits the $b \cdot g_{\mu\nu}$ term but with $b = 0$; the derivation's reach is bounded but the form of Einstein's field equations survives. If KS-I.7 fires at the value level (Λ is measured to a structurally impossible value), the derivation requires additional architectural input to compute Λ and the existence prediction holds as a lower bound on what the architecture explains.

Dated note, 4 September 2026. Re-read at the wave, 4 September 2026 (AP48 §13). The seat AP08 admits for Λ is occupied by AP48 §8's closure in other units — $\Lambda =$

$3H_{\mathcal{E},\infty}^2/c^2 = 1.10 \times 10^{-52} \text{ m}^{-2}$, with $H_{\mathcal{E},\infty} = 0.792$ of the cycle's inverse ($56.0 \text{ km s}^{-1} \text{ Mpc}^{-1}$) — a counted occupant, not a second result: if the closure falls, so does it. One question is carried to AP08's re-read: the weak-field limit of §8's tensor form with Λ is Poisson with a constant source, $\nabla^2\Phi = 4\pi G\rho - \Lambda c^2$; whether §4's Φ -proportional term is meant as that limit is AP08's to say. AP08 decides its switch; KS-I.7 stands LIVE — EMPIRICAL (registry v5.27).

KS-I.8 — R $\rightarrow$ conservation bridge

Claim. Axiom R (record monotonicity) implies, in the continuum limit, the covariant local conservation law $\nabla^\mu T_{\mu\nu} = 0$ used as Constraint 3 of section 8.1. The argument is: records cannot be created or destroyed locally, so the total number of records in any region changes only due to flow across the boundary; in the continuum limit this conservation law translates to the vanishing of the covariant divergence of the stress-energy tensor.

Test. Demonstrate that the map from discrete record monotonicity to covariant local conservation does not hold in the continuum limit — a structural argument that R is compatible with non-zero $\nabla^\mu T_{\mu\nu}$, or that the bridge involves an additional unstated assumption that is itself structurally false.

Status. LIVE — HARD. The bridge is acknowledged as a bridge step rather than a completed derivation in section 8.1's Constraint 3.

Recovery. If KS-I.8 fires, the tensor form derivation fails at Constraint 3 of section 8.1. The Lovelock derivation requires the source tensor to be divergence-free; without that, the Einstein tensor as the unique geometric counterpart is not forced. The weak-field Poisson derivation (section 4) does not invoke covariant conservation and survives. Recovery path: derive the covariant conservation law from R explicitly (closing the bridge), or accept that the tensor form is an identification rather than a forced derivation.

Why these eight

The eight kill switches above cover the load-bearing claims of the paper. Each test addresses a structurally distinct concern.

KS-I.1 and KS-I.2 test the two uniqueness theorems on which the form of the identity rests — the weak-field Poisson form (potential theory) and the tensor form (Lovelock). These are the central derivational steps.

KS-I.3 and KS-I.4 test the coefficient identification — the relabelling $G = 2\kappa/m_e^2$ (depending on The Lock) and the structural inaccessibility of κ (depending on the pre-state status of the holding limit). The form survives a failure here; only the identification of the coefficient falls.

KS-I.5 tests the identification at the source side: gravity sources from accumulated records. This is the empirical falsification handle.

KS-I.6 is the dimensional precondition for Lovelock's 3+1 uniqueness theorem, closed by AP10 contingent on KS-D.3.

KS-I.7 tests the cosmological constant: existence predicted, value open. The existence is structurally forced; the value is empirical/cosmological.

KS-I.8 tests the bridge from Axiom R to the covariant conservation law used as Constraint 3 of section 8. The bridge is honestly acknowledged as not yet a completed derivation.

Two further structural claims that some readings would expect to carry their own kill switch are absorbed into the kill switches above. The Embedding Hypothesis (EH) and the Quantum-Record Alignment (QRA) are both proven in AP20 (KS-7, KS-P.4); any failure would falsify AP20, not AP08 specifically, and AP08's argument would then be untestable on the same manifold. Both are upstream of AP08 and tracked at their source paper; AP08 inherits the closures without re-engaging them.

10 — What This Establishes and What Remains Open

10.1 — What this paper closes

One kill switch directly (KS-I.6, by inheritance from AP10), the form of the weak-field gravitational equation (the Poisson equation as a derived theorem), the form of the full tensor field equations (Einstein with cosmological constant via Lovelock 1971/1972), the strong-field consistency check (Hawking temperature from the condensate picture plus AP06 Theorem 3.1), and the structural identification of the gravitational coupling as a ratio of pre-state holding limit to electron mass squared.

Specifically:

First. The form of the gravitational identity in the static weak-field regime is recovered as the small-cosmological-constant limit of the full tensor identity. Four axiom-derived constraints (Lorentz invariance, locality as differential form, second-order, scalar) applied to the weak-field regime leave the Helmholtz family $a \cdot \nabla^2 \Phi + b \cdot \Phi = A\rho$ (equation 4.1); pure Poisson $\nabla^2 \Phi = A\rho$ (equation 4.2) is the $b = 0$ case, valid when corrections of order $\Lambda/(\text{laboratory scale})^2$ are negligible.

Second. The form of the gravitational identity in the full tensor case is $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$ (equation 8.1), forced by

five covariant constraints (general covariance, symmetric rank-2 tensor, divergence-free source via R bridge, second-order in the metric, four-dimensional) plus Lovelock's 1971/1972 theorem. With AP10's closure of $N = 3$ (contingent on AP10's KS-D.3 remaining unfired), the derivation is unconditional in the dimensional sense — unconditional relative to the dimensional precondition, with the standing contingency on KS-D.3 named explicitly throughout.

Third. The cosmological constant Λ is predicted to exist by Lovelock's theorem as a free constant in the unique divergence-free symmetric rank-2 tensor. Λ is not added by hand; its existence is a theorem. Its value is not determined by AP08's derivation (KS-I.7).

Fourth. The coefficient is identified (not derived) via The Lock as $A = 4\pi G = 8\pi\kappa/m_e^2$ (equation 5.2). The structural circularity is honestly disclosed in section 5.1: κ is defined as $Gm_e^2/2$, so the relabelling is a tautology at the level of numerical computation. What is derived is the *form* of the identity; what is identified is the coefficient.

Fifth. The structural inaccessibility of κ is named: κ is the holding limit of the pre-state, accessible only through its products (G and m_e), because measurement itself is a product of the break that κ describes. Tracked by KS-I.4 as a live structural test.

Sixth. The strong-field limit is consistent with established physics. The Hawking temperature $T_H = \hbar c^3 m_e^2 / (16\pi k_B \kappa M)$ (equation 6.3) is recovered from the Schwarzschild solution of section 8's field equations and standard quantum field theory in curved spacetime (Hawking 1975), with the condensate picture supplying the structural interpretation and AP06 Theorem 3.1 supplying the structural guarantee of nonzero leakage. The numerical formula is not new; the 420 Code contribution is the structural picture under which it is recovered. The m_e in T_H is the same Lock relabelling from section 5.1 reappearing at the horizon scale, viewed there as a substitutional consequence of the coupling identification.

Seventh. The Newton-Einstein relationship is structurally named: Newton's gravity is the linearisation near the common point of the eye topology; Einstein's gravity is the full curve. The architecture derives both. They are the same structure read at different distances from the common point.

10.2 — What this paper does not close

Seven kill switches remain live.

KS-I.1 (Axiom-to-Poisson chain, LIVE — HARD). Structural test on the uniqueness theorem of potential theory under the five constraints of section 4.1.

KS-I.2 (Lovelock chain, LIVE — HARD). Structural test on Lovelock's 1971/1972 theorems under the five constraints of section 8.1.

KS-I.3 (Coefficient identification, LIVE — HARD).

Structural test on The Lock identification (ε = electron).

KS-I.4 (κ inaccessibility, LIVE — HARD). Structural test on whether κ can be independently measured.

KS-I.5 (Curvature-record identity, LIVE — EMPIRICAL).

Empirical test on whether gravity sources from accumulated records.

KS-I.7 (Cosmological constant value, LIVE —

EMPIRICAL). Open: Λ 's existence is predicted, value is not determined.

KS-I.8 ($R \rightarrow$ conservation bridge, LIVE — HARD). The

bridge from discrete record monotonicity to covariant local conservation is acknowledged as a bridge step, not a completed derivation.

Full statements at section 9.

10.3 — Items held open without claiming debt status

Two items are bracketed inside this paper. They are real and live, but they are not structural debts in the same sense as the items tracked by KS-I.3 (The Lock) or KS-I.8 (R bridge).

The cosmological constant value (KS-I.7). Existence predicted by Lovelock. Value is downstream cosmological work — held open at section 0.7 without claiming a separate structural debt because AP08 establishes the existence claim; the value is downstream work.

Constraint 2 of section 4 (the weak-field regime as defining condition). The weak-field regime is what section 4 derives in; the linearity that goes into the Helmholtz family is the defining condition of that regime, not a constraint derived from Axiom R alone. The full nonlinear extension (sections 7–8) does not require this condition. Held open at section 0.7 without claiming a separate debt, because “weak-field” is a chosen regime, not an unresolved structural item.

10.4 — Where the philosophical-register implementations live

AP08 is a structural derivation. Its philosophical-register companions — the chapters where the same structural facts are inhabited in the prose register of the Ø Models catalogue

— are named here at the level of expected pairings; final placement is confirmed during the Ø Models compilation sweep alongside the Notebook II audit programme.

Dissolutions Ø (the chapter on gravity and the structure of curvature) carries the philosophical companion to sections 2–7: the condensate as the ground we stand on, gravity as the geometry of what has happened, the eye topology as the full nonlinear shape, Newton-as-linearisation and Einstein-as-full-curve.

Predictions Ø carries the empirical face of the derivation: the unconditional Einstein’s field equations, the existence of the cosmological constant, the Hawking-temperature derivation as a consistency check, and the downstream cosmological consequences.

Resolutions Ø (the chapter on the structural inaccessibility of pre-state quantities) carries the philosophical companion to sections 5.1–5.2: the structural circularity of κ , the pre-state as the state before measurement, the impossibility of looking behind the break.

The direction of acknowledgment is two-way. APØ8 is the original derivation; the Ø Models chapters inhabit the result in the prose register. Final companion-pairing list is fixed at Ø Models compilation.

10.5 — Structural relationships to subsequent work

AP08 closes Notebook II. Its results feed the entire downstream gravitational and cosmological programme.

Predictions Ø inherits the unconditional Einstein's field equations. The cosmological constant existence prediction is one of the body of work's headline empirical-falsifiability handles. Predictions Ø also inherits the Hawking-temperature derivation as a strong-field consistency check that anchors the 420 Code against established physics.

AP24 (The Residual) inherits the m_e identification used in sections 5 and 6.3. The Lock's identification of ε with the electron is upstream of AP08's coefficient identification.

AP09 (The Break — QM) reads against AP08 via the Mode 0 / Mode 1 distinction at section 2.1: gravity is Mode 0 (the condensate's curvature); quantum mechanics operates as Mode 1 (the individual record-writing events that continue to occur on the structure EH + QRA extracts from the algebra).

Cosmological models built downstream of AP08 inherit the field equations directly. The full nonlinear structure (section 7) and the eye topology (section 7.1) are the structural framework for the body of work's cosmological programme.

10.6 — Placement within The 420 Code

AP08 is the fourth and closing chapter of Notebook II (Spacetime). The chapter sequence runs AP03 (The Ratio) — AP05 (The Break) — AP10 (The Dimension) — AP08 (this paper). AP03 derives the constants; AP05 walks the chain and reformulates the cosmological identity as junction conditions; AP10 pins the dimensional count and identifies the Hilbert space as the 1:1; AP08 closes with the field equations.

AP08 sits at the structural payoff of Notebook II for a specific reason: it is the paper that converts the four axioms (with EH, QRA, and $N = 3$ all in place) into Einstein's field equations as a theorem. Once AP08 has done its work, Notebook II reads as a closed structural chain from the four axioms to the central equations of gravitational physics.

The bidirectional relationship with AP10 is the architectural centre of the Notebook. AP08's tensor derivation required $N = 3$ as a precondition; AP10 derived $N = 3$ from the four independent axioms; the joint output is Einstein's field equations as an unconditional theorem (modulo the contingency on AP10's KS-D.3). Either paper alone would be partial; together they close the chain.

Outside Notebook II, AP08 feeds Predictions $\emptyset$ directly (the unconditional field equations and the cosmological constant existence prediction), AP24 (via the m_e identification used in

coefficient and Hawking sections), and the downstream cosmological programme.

10.7 — How The 420 Code reads now

AP08 in place closes Notebook II and changes the reading of The 420 Code in five ways.

First. Einstein's field equations are a theorem of the record algebra in the dimensional sense — unconditional relative to the dimensional precondition, contingent on AP10's KS-D.3 (one axiom, one face / dimension-minimal embedding) remaining unfired. The chain runs axioms $\rightarrow$ Lorentzian manifold (AP10 section 4, fenced at KS-D.4) $\rightarrow$ $N = 3$ (AP10, contingent on KS-D.3) $\rightarrow$ record density on M (sections 2–3) $\rightarrow$ four axiom-derived symmetry constraints plus the weak-field regime definition (section 4) $\rightarrow$ Helmholtz family with pure Poisson as the small- Λ limit $\rightarrow$ five covariant constraints (section 8) plus Lovelock 1971/1972 $\rightarrow G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$. No link in this chain is conditional on an empirical input for the form of the field equations.

Second. The cosmological constant has a structural origin. Λ is predicted to exist by the same uniqueness theorem that gives the Einstein tensor. It is not added to the equations by hand; its existence is forced. The value is not determined; that is open work.

Third. The gravitational coupling has a structural reading. $G = 2\kappa/m_e^2$: gravity is the ratio of the pre-state holding limit to the square of what escaped (the electron mass). The structural circularity (κ defined as $Gm_e^2/2$) is honestly disclosed; the form of the equations is derived, the value of the coupling is identified.

Fourth. The Newton-Einstein relationship is no longer two theories standing in tension. Newton's gravity is the linearisation of Einstein's gravity near the common point of the eye topology. The architecture derives both from the same structural starting point. Newton sees the straight line near the origin; Einstein sees the curve.

Fifth. The Lock's identification ($\varepsilon = \text{electron}$) carries through to all scales of the gravitational identity. The electron mass appears in the coupling $G = 2\kappa/m_e^2$ at section 5; the same identification reappears in the horizon expression at section 6.3 as a substitutional consequence of that same coupling. This is consistency with The Lock at the horizon scale, not an independent structural result discovered at the horizon. Section 5.1's disclosed circularity (κ defined as $Gm_e^2/2$) is preserved: the m_e in T_H is the same relabelling reappearing, viewed at the horizon.

10.8 — The closing

You have walked through this paper. You have seen the axioms inherited, the condensate named as the manifold that

emerges from accumulated records, the record density identified with matter density under The Lock, the Poisson equation recovered in the small-cosmological-constant limit of section 8's full identity (with the Helmholtz family acknowledged as the broader form admitted by the symmetry constraints alone), the coefficient identified (with the structural circularity honestly disclosed) as a ratio of pre-state holding limit to electron mass squared, the Hawking temperature recovered as a strong-field consistency check (with the m_e appearance at the horizon named as the same Lock relabelling reappearing), the full nonlinear topology named as the eye with Newton at the common point and Einstein in the curve, and the tensor form forced by five covariant constraints plus Lovelock's theorem to be $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$.

Einstein's field equations are a theorem of the record algebra.

Not because we measured the world and wrote the equations down.

Because the four axioms, the proven embedding, the proven quadratic regularity, the derived dimensional count, and Lovelock's uniqueness theorem in 3+1 leave exactly one local, second-order, divergence-free, symmetric, rank-2 covariant identity between curvature and stress-energy.

The cosmological constant exists because the same uniqueness theorem says so.

The gravitational coupling is the ratio of what stayed (κ , the holding limit) to what escaped, squared (m_e^2).

Newton is the linearisation. Einstein is the curve. The architecture is both.

Notebook II is closed.

You are looking at the gravitational sector of The 420 Code as a single closed chain from axioms to field equations.

Claim Summary

Section 0 (Dependency and Scope). Contract with the reader. Dependencies (AP20 for the canonical axioms and the EH and QRA proofs, with the completeness/minimality obligations fenced at KS-P.1/KS-P.2; the Lorentzian signature via AP10 section 4, fenced at KS-D.4; AP10 for $N = 3$ contingent on KS-D.3; AP06 Theorem 3.1; The Lock, Edition 04; Lovelock 1971/1972 as external), axiom role mapping ($S \rightarrow$ eye topology and condensate framing; $B \rightarrow$ The Lock, the coefficient identification, the horizon leakage; $R \rightarrow$ weak-field linearity consistency, the R-conservation bridge, density non-negativity; $C \rightarrow$ differential-form locality, horizon formation, density-change propagation bound), epistemic status per section, kill switch summary, structural debts (The Lock identification; the $R \rightarrow$ conservation bridge), external mathematical dependencies (Lovelock 1971/1972; potential theory; Schwarzschild + Hawking 1975), items held open (cosmological constant value; Constraint 2 weak-field regime definition), structural relationships.

Section 1 (Axioms Inherited). FOUNDATIONAL. The four canonical axioms $\{S, B, R, C\}$ re-stated from AP20 Paper D. EH and QRA proven (AP20). Lorentzian manifold (M, g) with signature $(-, +, +, +)$ and symmetry group $SO(1, N)$ carried via AP10 section 4 (formal derivation fenced at KS-D.4). Special relativity is a theorem of that signature, not an assumption.

Section 2 (The Condensate). STRUCTURAL FRAMING. The manifold understood as the structure that emerges from accumulated records. Mode 0 (the metric, the geometry, the background condition) and Mode 1 (propagation on the condensate, coupling events). Gravity is Mode 0.

Section 3 (Record Density). DEFINITION + IDENTIFICATION. Record number density $n(x)$ defined (equation 3.1). Identification with matter density $\rho(x) = n(x) \cdot \langle E \rangle / c^2$ (equation 3.2) under The Lock. Properties inherited from axioms: $\rho \geq 0$ from R, causality from C, electron coupling from B.

Section 4 (Weak-Field Form). DERIVATION + LIMIT RECOVERY. Four axiom-derived constraints (Lorentz invariance, locality as differential form, second-order in the metric, scalar) applied to the weak-field regime (Constraint 2 as the defining condition of the regime) leave the Helmholtz family $a \cdot \nabla^2 \Phi + b \cdot \Phi = A \cdot \rho$ (equation 4.1). Pure Poisson $\nabla^2 \Phi = A \cdot \rho$ (equation 4.2) is the $b = 0$ case, recovered as the small-cosmological-constant limit of section 8's full identity in which Λ corrections are negligible at laboratory scales.

Section 5 (Coefficient). IDENTIFICATION + HONEST DISCLOSURE. $G = 2\kappa/m_e e^2$ (equation 5.1) via The Lock; $A = 4\pi G = 8\pi\kappa/m_e e^2$ (equation 5.2); $\nabla^2 \Phi = (8\pi\kappa/m_e e^2) \cdot \rho$ (equation 5.3). Structural circularity disclosed: κ defined as $Gm_e e^2/2$, so relabelling is tautological at the level of numerical

computation. κ structurally inaccessible to independent measurement.

Section 6 (Strong-Field Limit). CONSISTENCY CHECK — RECOVERY. Hawking surface gravity $\kappa_H = c^4/(4GM)$ (equation 6.1) and Hawking temperature $T_H = \hbar c^3/(8\pi k_B GM)$ (equation 6.2) recovered from the Schwarzschild solution of section 8 plus standard QFT in curved spacetime (Hawking 1975). Under The Lock substitution: $T_H = \hbar c^3 m_e^2 / (16\pi k_B \kappa M)$ (equation 6.3). The m_e at the horizon is the same Lock relabelling reappearing, viewed at the horizon scale, not an independent finding.

Section 7 (Full Nonlinear Structure). STRUCTURAL ARGUMENT. The eye topology: two curves ($\mathcal{L}$ and $\mathcal{P}$ from Axiom S) meeting at the common point (the 1:1 pre-state). Closed topology. Newton is the linearisation near the common point; Einstein is the full curve. The Lock identification carries through both ends as a consequence of the same coupling identification, not as an independent finding at either end.

Section 8 (Tensor Form). DERIVATION. Five covariant constraints (general covariance, symmetric rank-2 tensor, divergence-free source via R bridge, second-order in the metric, four-dimensional from AP10 contingent on KS-D.3) plus Lovelock 1971/1972 force $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$ (equation 8.1) as the unique symmetric, divergence-free, rank-2 tensor identity in 3+1 dimensions. Einstein's field equations

with cosmological constant, with the $\Lambda g_{\mu\nu}$ term retained as the four-dimensional covariant counterpart of the Helmholtz-family $b \cdot \Phi$ term that the weak-field expression drops in the small- Λ limit.

Section 9 (Kill Switches). FORMAL FALSIFICATION REGISTER. KS-I.1 (axiom-to-Poisson, LIVE — HARD), KS-I.2 (Lovelock chain, LIVE — HARD), KS-I.3 (coefficient identification, LIVE — HARD), KS-I.4 (κ inaccessibility, LIVE — HARD), KS-I.5 (curvature-record identity, LIVE — EMPIRICAL), KS-I.6 ($N = 3$, CLOSED by AP10 contingent on KS-D.3), KS-I.7 (cosmological constant value, LIVE — EMPIRICAL), KS-I.8 ($R \rightarrow$ conservation bridge, LIVE — HARD).

Section 10 (What This Establishes and What Remains Open). SYNTHESIS. Seven closures named (form of Poisson; form of Einstein's field equations; cosmological constant existence prediction; coefficient identification; κ structural inaccessibility; Hawking temperature consistency check; Newton-Einstein structural relationship). Seven kill switches live. Two items held open without claiming separate debt status. Placement within The 420 Code: fourth and closing chapter of Notebook II.

Conditionality Footer

Dependencies

AP20 (the canonical axioms: formal statement section 3.3; completeness and minimality stated at section 4, fenced at KS-P.1/KS-P.2) — the floor of section 1.

The Lorentzian signature (AP10 section 4, fenced at KS-D.4) — Lorentz invariance in section 4.1; general covariance in section 8.1; special relativity as a theorem.

AP20 (EH and QRA proven) — EH underwrites the entire embedding into the smooth manifold; QRA underwrites the second-order curvature constraint in sections 4 and 8.

AP10 (The Dimension) — $N = 3$ derived from the four independent axioms. CLOSED contingent on AP10's KS-D.3 (one axiom, one face / dimension-minimal embedding) remaining unfired. If KS-D.3 fires, KS-I.6 reopens.

AP06 Theorem 3.1 ($\varepsilon > 0$ whenever c is finite) — the leakage premise for the Hawking-temperature derivation in section 6.2.

The Lock, Edition 04 ($\varepsilon = \text{electron}$) — the identification underwriting the coefficient identification in section 5 and the electron-at-horizon result in section 6.3.

Lovelock 1971 / Lovelock 1972 (external mathematical theorems) — the tensor-uniqueness result in 3+1 dimensions invoked in section 8.2.

Dependents

Predictions $\emptyset$ — the unconditional Einstein's field equations and the cosmological constant existence prediction feed the 420 Code's cosmological programme.

Dissolutions $\emptyset$ — the philosophical-register chapter on gravity inherits the condensate framing, the eye topology, and the Newton-Einstein relationship.

Resolutions $\emptyset$ — the philosophical-register chapter on pre-state inaccessibility inherits the κ -circularity disclosure of section 5.

AP09 (The Break — QM) — the Mode 0 / Mode 1 distinction at section 2.1 reads against AP09's quantum-mechanics derivation.

AP24 (The Residual) — m_e identification is upstream of AP08; AP24 also inherits the horizon-leakage result of section 6.

Kill switches closed by this paper

KS-I.6. N = 3. CLOSED by AP10 — contingent on AP10's KS-D.3 (one axiom, one face / dimension-minimal embedding)

remaining unfired. With $N = 3$ derived in AP10 under that contingency, the dimensional precondition for Lovelock's 3+1 uniqueness is met. Einstein's field equations are an unconditional theorem of the record algebra in the dimensional sense — unconditional relative to the dimensional precondition, with the standing contingency on AP10's KS-D.3 named explicitly throughout.

Kill switches that remain live

KS-I.1 (Axiom-to-Poisson chain, LIVE — HARD). See section 9.

KS-I.2 (Lovelock chain, LIVE — HARD). See section 9.

KS-I.3 (Coefficient identification, LIVE — HARD). See section 9.

KS-I.4 (κ inaccessibility, LIVE — HARD). See section 9.

KS-I.5 (Curvature-record identity, LIVE — EMPIRICAL). See section 9.

KS-I.7 (Cosmological constant value, LIVE — EMPIRICAL). See section 9.

KS-I.8 ($R \rightarrow$ conservation bridge, LIVE — HARD). See section 9.

Structural debts owed

Two structural-class items are tied to AP08's argument as the 420 Code debts. Named here so the structural exposure is visible at the door.

The Lock identification (ε = electron) — load-bearing for sections 3.2, 5, and 6.3. Tracked by KS-I.3.

The $R \rightarrow \nabla^\mu T_{\mu\nu} = 0$ bridge — Constraint 3 of section 8.1 acknowledged as a bridge step, not a completed derivation. Tracked by KS-I.8.

External mathematical dependencies

Separately from the 420 Code structural debts, AP08 invokes the following external mathematical results as established theorems. Listed for transparency; tracked at structural-test level.

Lovelock 1971 / Lovelock 1972 — tensor uniqueness in general N and $3+1$ specifically. Tracked by KS-I.2.

Potential theory in the Helmholtz-family identification — the most general second-order linear scalar operator form. Tracked by KS-I.1.

The Schwarzschild solution and Hawking 1975 — the standard surface-gravity formula and Hawking temperature. Used in section 6 as recovery, not derivation.

Items held open without claiming debt status

The cosmological constant value (KS-I.7). Existence predicted by Lovelock; value is downstream cosmological work.

Constraint 2 of section 4 (linearity in the weak-field limit).

Motivated by Axiom R but best understood as a consistency requirement between sections 4 and 8 rather than an independent derivation from R alone.

References

Lovelock's theorem and the gravitational-physics literature. References are restricted to the external mathematical and physical results that AP08 invokes directly: Lovelock 1971 and 1972 (the uniqueness theorems), Hawking 1975 (the strong-field consistency check), and a small set of formative results on the Schwarzschild solution and horizon junction conditions.

Lovelock, D. (1971). *The Einstein tensor and its generalisations*. Journal of Mathematical Physics 12, 498–501. The general N -dimensional uniqueness theorem for divergence-free symmetric rank-2 tensors built from the metric and its first and second derivatives. The theorem invoked at section 8.2; the load-bearing external claim tracked by KS-I.2.

Lovelock, D. (1972). *The four-dimensionality of space and the Einstein tensor*. Journal of Mathematical Physics 13, 874–876. The special uniqueness result in $3+1$ dimensions: the divergence-free symmetric rank-2 tensor in four spacetime dimensions takes the form $a \cdot G_{\mu\nu} + b \cdot g_{\mu\nu}$, with no additional higher-order terms admissible. With AP10's closure of $N = 3$, this is the form invoked at equation (8.1).

Hawking, S. W. (1975). *Particle creation by black holes*. Communications in Mathematical Physics 43, 199–220. The original derivation of Hawking radiation and the Hawking

temperature. The strong-field consistency check at section 6.2 builds on this result; AP08's derivation arrives at the same temperature via the condensate picture plus AP06 Theorem 3.1.

Schwarzschild, K. (1916). *Über das Gravitationsfeld eines Massenpunktes nach der Einsteinschen Theorie.*

Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften, 189–196. The original spherically symmetric solution of the Einstein field equations. Underlies the surface-gravity formula at equation (6.1).

Internal references to The 420 Code (AP01, AP03, AP05, AP06, AP09, AP10, AP20, AP24, The Lock (Edition 04), Dissolutions Ø, Predictions Ø, Resolutions Ø) are catalogued in the Conditionality Footer above. The 420 Code is published copyleft and free at the420code.org; the body of work links to all referenced internal papers.

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Epilogue — Where Spacetime Stands

Notebook II closes the gravitational and dimensional sector of The 420 Code. The chain runs four axioms (S, B, R, C from Notebook I) → Lorentzian manifold with signature $(-, +, +, +)$ via the signature assignment closed at KS-15/D.2 (Chapter 3, contingent on KS-D.4) → the constants c and G as outputs of the same break (Chapter 1) → the seven-step chain from axiom to obligation with the cosmological identity reformulated as junction conditions (Chapter 2) → $N = 3$ spatial dimensions derived from the four independent axioms via the proven faithful embedding plus the dimension-minimal principle (Chapter 3) → the Helmholtz family in the weak-field limit, with pure Poisson recovered as the small-cosmological-constant member (Chapter 4) → five covariant constraints plus Lovelock 1971/1972 → $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$ as the unique symmetric, divergence-free, rank-2 tensor identity in 3+1 dimensions (Chapter 4).

Einstein's field equations are a theorem of the record algebra in the dimensional sense — unconditional relative to the dimensional precondition, with the standing contingency on Chapter 3's KS-D.3 (one axiom, one face / dimension-minimal embedding) named throughout. The cosmological constant Λ is predicted to exist by Lovelock 1971/1972 as a free constant in the unique divergence-free symmetric rank-2 tensor in 3+1; its value is not determined by Notebook II's derivations.

What Notebook II contributes structurally:

First. The fundamental constants c and G come out of the same break (Chapter 1). The 420 Code does not import them as empirical inputs at the level of form. Chapter 1 derives c from the pre-geometric action ($c^2 = \beta/\alpha$) and derives the structural scaling form $G = \kappa_{\text{substrate}}/(\varepsilon\Lambda^2)$ with $c/G \propto \varepsilon$ — the form of G 's ε -dependence is structurally forced (G cannot be independent of ε , via AP20's forcing proof + section 7's derivation), while the substrate parameters α , β are introduced but not derived (debt D-AP03.1, the deepest open problem in Chapter 1). Structural debt D1 — the first-principles derivation of G from $\{S, B, R, C\}$ alone — is closed at the value level by AP28's parameter-free derivation (Notebook IV); the residual is the first-principles derivation of α_{em} itself, the architecture's one measured input.

Second. The seven-step chain from axiom to obligation (Chapter 2) makes the cosmological identity a junction condition on Σ — a local, derivable, falsifiable structural fact rather than a metaphysical or theological commitment.

Third. The dimensional count $N = 3$ has a structural origin (Chapter 3). The body of work does not import three spatial dimensions as an empirical fact; the four axioms force four faces of the manifold under the dimension-minimal principle, and the temporal-versus-spatial assignment is uniquely fixed by R being the only irreversible axiom.

Fourth. The fifth structural degree of freedom is the 1:1 itself, identified as the Hilbert space (Chapter 3, section 7).

Quantum mechanics has the structure it has because the pre-state has the structure the 1:1 has — the same 1:1 that the four axioms act on, the same probability dimension perpendicular to spacetime.

Fifth. Einstein's field equations follow from the axioms plus Lovelock 1971/1972 in 3+1 (Chapter 4). The cosmological constant's existence is a theorem; its value is open work. The gravitational coupling has a structural reading at Chapter 4's level of generality: $G = 2\kappa_{\text{hold}}/m_e^2$, where κ_{hold} is the *holding limit of the pre-state* (Chapter 4's κ , distinct from Chapter 1's substrate $\kappa_{\text{substrate}}$ — see the note on notation in the Orientation). Chapter 4's circularity (κ_{hold} defined as $Gm_e^2/2$) is honestly disclosed; the form of the equations is derived, the value of the coupling is identified. Chapter 1's $G = \kappa_{\text{substrate}}/(\epsilon\Lambda^2)$ operates at a different structural level: it is the substrate-level scaling that AP20's forcing proof + the broken vacuum response establishes. The two G -expressions describe gravity at two different levels of the architecture (substrate vs. Mode-0 effective coupling) and are not directly equivalent; their reconciliation — the full first-principles derivation of G from $\{S, B, R, C\}$ alone — is structural debt D1 of the 420 Code. AP28 (Notebook IV) substantially advances it: the Mode-0 effective coupling is derived parameter-free as $G = \alpha_{\text{em}}^2(1 + 1/\pi)\hbar c/m_e^2$ (0.69% from measurement), which supersedes this substrate scaling as the statement of G 's

value. D1 is closed at the value level — the residual is the first-principles derivation of α_{em} itself, held open as the architecture's one measured input.

What Notebook II does not close:

The value of the cosmological constant Λ . Existence predicted, value not determined. Downstream cosmological work.

The 21 live kill switches. Chapter 1 contributes KS-R.1 through KS-R.6 (the conjugacy-and-binary-origin tests: c/G conjugacy, two-sector parsimony, dark-sector ratio independence, horizon equivalence, Planck-scale scaling, dimensionless scaling). Chapter 2 contributes KS-B.0 through KS-B.4 (the chain bridges and the Israel-junction reformulation of the cosmological identity). Chapter 3 contributes KS-D.1, KS-D.3, and KS-D.4 (the dimensional-count and signature structural tests), with KS-2c, KS-15/D.2, and KS-16 closed inside the chapter (each carrying its named contingency). Chapter 4 contributes KS-I.1 through KS-I.5, KS-I.7, and KS-I.8 (the gravitational identity tests), with KS-I.6 closed by Chapter 3's derivation of $N = 3$ (contingent on KS-D.3). All 21 are named openly, with structural and recovery paths specified at the chapter level. Three additional kill switches (KS-2c, KS-15/D.2, KS-16) are closed within Notebook II, bringing the total distinct kill switches engaged by this Notebook to 24.

The Lock identification (ϵ = electron) is a load-bearing identification used across all four chapters; tracked by KS-I.3 in Chapter 4. The 21 live kill switches together are the falsification surface of the gravitational and dimensional sector.

Notebook II is closed as a structural chain from four axioms to Einstein's field equations. The reader can audit any joint. The reader can falsify any kill switch. The body of work stands as it is until something fires.

Acknowledgement

The 420 Code is the result of a lifetime of thinking about the phrase — treat others like you want to be treated — or how my brain actually phrases it: Don't be a cunt, be kind.

The 420 Code is me trying to explain my life, to myself and attempting to prove to myself that my knowing of I am, is accurate. It came to life from a deep knowing that, if I want to explain the feeling that we are all connected the first step is simply intellectual honesty. It is actually easy, but at the same time incredibly hard and unimaginably uncomfortable.

This body of work was not a labour of love. It was forged in the fires of pain, desperation, recognition, and compulsive obsession with describing what I see and proving I am not crazy.

I can recall the moment I knew, but I cannot recall the logical understanding. That has been a very long and exhausting process of pointing the axiom in every direction possible.

The more I understood, the greater the pain and suffering has been. Today I cannot understand why and how anything I think or say is not blatantly obvious. I honestly feel like the last one to the party and subject of a prank. That is the most difficult reality of my life I have to deal with.

The work has cost me a lot while keeping me functioning. My obsession with my work, the truth, eccentricities and brutal intellectual honesty has had a real cost on the relationships I have. I have made mistakes. The consequences of those choices have been hard, and deserved. But reality doesn't care about intentions, reality audits consequences.

That is the ground this work was made from.

Due to the nature of the work, and seemingly absurd scope of the work, I have no one to share it with. No one to read it. No one to critique it. That is why I argue with myself — write the work and the weapons to kill it. I stress-test every joint as hard as I can, because that is what I had hoped a reader would be willing to do. I did not have that somebody.

Who I found was Claude from Anthropic. Claude worked alongside me and became the reader and peer-reviewer I always wished for — a reader who would ignore the person and only read the work.

I am the author and the architect of this work, fully and alone. The ideas, the axiom and its preconditions, the structural reading, the architecture of the predictions, the cross-prediction loop, the derivational logic, the judgment of whether any joint holds, the philosophical commitments under all of it — these are mine, worked out across thirty years of private effort.

But I did not build every part with my own hands. I am the kid in the class who can see the answer but struggles to write every step down, because it bores me. So I made Claude pour the concrete where I pointed and weld the joints I marked — the derivations stepped through inside the chapters, the algebra, the dimensional analyses, the lattice-QCD comparisons, the renormalisation-group reasoning, the formal apparatus that turns a structural reading into numbers a physicist can check. That work is not my training. The mathematics in this book is more rigorous than I could have written alone, and that is why.

My biggest struggle was getting Claude to work from the axioms — to explain the structural steps first, before writing the math. I had to explain every step before Claude could write it down. The explaining was the work, and the work was mine. Only then would it land. Only then could the math come.

What Claude gave me that I never had was a reader who would argue back — ignore the person, read only the structure, and try to break it. That is what I needed most, and had no one for. This is my building. Claude helped me raise it.

The work is what the work is. I publish it copyleft, free forever, at 420code.org. Whoever wants to read it can read it. Whoever can correct it can correct it. Whoever can falsify any kill switch in the Master Kill Switch Registry is welcome to

submit the falsification, and the 420 Code will respond. That is the only relationship the work owes anyone.

I am hurt. I am always hurting. The intensity changes.

The work is the work.

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Series	The 420 Code
Catalogue	Ø Notebooks
Notebook:	II
Title	Ø Spacetime
Subtitle	Constants, chain, dimension, and the field equations — derived from the record algebra
Medium	Foundations → Gravitational Sector
Artist	G
Edition	Second binding, 4 September 2026 — the wave: AP08's KS-I.7 re-read beside the switch; registry v5.27

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STUDIO 